The Erdős–Hajnal Property for All Induced Paths and Infinite Families of Sparse Graphs*

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Abstract

A graph is H-free if it contains no induced subgraph isomorphic to H. A central conjecture of Erdős and Hajnal (1989) asserts that for every finite graph H, there exists a constant $c = c(H) > 0$ such that every H-free graph G on $|G| \geq 1$ vertices contains a clique or stable set of size at least $|G|^c$. While the conjecture was recently verified for all graphs on at most five vertices via the resolution of the five-vertex path P_5 by Nguyen, Scott, and Seymour (2025), every longer path P_k ($k \geq 6$), every prime subdivided spider $S_{1,2,d}$ ($d \geq 2$), and every prime unicyclic long-tail bird Bird_d ($d \geq 2$) escapes all vertex-substitution and leaf/co-leaf attachment closures.

Starting directly from published results for graphs of order at most five, we prove the Erdős–Hajnal conjecture unconditionally for all finite paths P_k and co-paths $\overline{P_k}$ ($k \geq 1$), all linear and co-linear forests, all (s,d)-brooms $B(s,d)$ ($s,d \geq 1$), all double brooms $\mathrm{DB}(a,d,b)$ ($a,d,b \geq 1$), all prime subdivided spiders $S_{1,2,d}$ ($d \geq 2$) and leaf-substituted spiders, all prime unicyclic birds Bird_d ($d \geq 1$, with $\mathrm{Bird}_1 = \mathrm{Bull}$), all bicyclic sawtooths $M_{2,d}$ ($d \geq 1$), all trees of order at most 6, all forests of component order at most 6, 10 of the 11 non-isomorphic trees on 7 vertices, all opposite-chirality forest pairs $\{F_1, \overline{F_2}\}$, all hooked-claw pairs $\{P_t, \mathrm{Hook}_4(Q,S)\}$, and countably infinitely many pairwise non-isomorphic prime non-split viral extension towers. Our proof combines a recursive depth-r Gallai modular-core decomposition inside combs and co-combs with a universal recursive terminal-path pure-substitution witness construction that avoids exponential casework across arbitrary path length $k \geq 6$.

Note:

We would like to explicitly acknowledge that Dr. Tung H. Nguyen was the first, to our knowledge, resolve the P_k case, preceding any other group of humans or AI systems. Following discussions with Dr. Nguyen on how to present our respective P_k results, he kindly proposed that both parties announce their findings independently while mutually acknowledging each other's work.

These discussions were initiated after our Cogentic system obtained a solution for the P_6 case, which was verified by Prof. Joonkyung Lee. We initially contacted Dr. Nguyen to explore a potential collaboration on more general cases; he then informed us that he had recently solved the general P_k case and was in the process of drafting his manuscript. In parallel, we continued exploring

*This paper was entirely obtained by Cogentic [5], an agentic framework for mathematical discovery, using Gemini Argon as the base model. There was no human interaction other than the initial prompt for cogentic to generate a full proof. The Cogentic harness was built by Yang Cai, Vineet Gupta, Yanchen Jiang, Christopher Liaw, Aranyak Mehta, Grigoris Velegkas and Di Wang.

Junehyuk Jung and Joonkyung Lee proposed we attempt to solve the problem, and helped inspecting and revising the natural language manuscript.

several variants, including longer paths. During these attempts, one of our agent's runs successfully resolved P_k. However, we emphasize that this result was obtained strictly subsequent to our initial communication with Dr. Nguyen.

In the interest of full transparency, we are uploading the raw Cogentic outputs—including both the natural-language manuscript (the remainder of this paper) and the corresponding Lean code—on GitHub first. Following this, we will post a human-readable expository article on arXiv describing the core ideas and structure of the AI-generated proof to meet the standards of the mathematics community.

1 Introduction

All graphs in this paper are finite and simple. For a graph $G = (V(G), E(G))$, we write $|G| := |V(G)|$ for its order, $\overline{G}$ for its complement, $\omega(G)$ for its clique number, $\alpha(G)$ for its stability (independence) number, and $\kappa(G) := \max(\omega(G), \alpha(G))$ (with $\kappa(\emptyset) := 0$). Given a graph H (or a finite family $\mathcal{F}$ of graphs), a graph G is *H-free* (respectively, *$\mathcal{F}$-free*) if no induced subgraph of G is isomorphic to H (respectively, to any member of $\mathcal{F}$). We say that H (or $\mathcal{F}$) has the *Erdős–Hajnal property*, written $H \in \mathcal{EH}$ (respectively, $\mathcal{F} \in \mathcal{EH}$), if there exists a constant $c > 0$ such that $\kappa(G) \geq |G|^c$ for every H-free (respectively, $\mathcal{F}$-free) graph G with $|G| \geq 1$.

In 1989, Erdős and Hajnal [10] conjectured that every finite graph H has the Erdős–Hajnal property; see the surveys [6, 14]. Whereas general n-vertex H-free graphs are known unconditionally only to satisfy $\kappa(G) \geq 2^{c\sqrt{\log n \log\log n}}$ [4] (improving the classical $2^{c\sqrt{\log n}}$ bound of [10]), a sustained structural program has resolved the conjecture for several fundamental classes of small and structured graphs. Because complementation swaps cliques and stable sets while mapping H-free graphs to $\overline{H}$-free graphs, $H \in \mathcal{EH}$ holds if and only if $\overline{H} \in \mathcal{EH}$. Furthermore, Alon, Pach, and Solymosi [1] proved that $\mathcal{EH}$ is closed under vertex substitution $H_1[v_0 \leftarrow H_2]$, reducing the conjecture for disconnected or reducible graphs to prime induced building blocks. Since P_4-free graphs (cographs) are perfect [23] and therefore satisfy $\kappa(G) \geq |G|^{1/2}$, all graphs on at most four vertices lie in $\mathcal{EH}$, and the subsequent breakthroughs for the bull [7], the five-vertex cycle C_5 [8, 9], and finally the five-vertex path P_5 by Nguyen, Scott, and Seymour [21] established the Erdős–Hajnal property for all graphs of order at most five.

Central to modern progress on the conjecture is the interplay between local induced-copy density and polynomial density decrement. A graph G is *ε-sparse* if every vertex has degree at most $\varepsilon|G|$, and *ε-restricted* if either G or $\overline{G}$ is ε-sparse. Strengthening foundational work of Fox and Sudakov [12] and Bousquet, Lagoutte, and Thomassé [2], Bucić, Fox, and Pham [3] proved that a finite non-empty family $\mathcal{F}$ of non-empty graphs has the Erdős–Hajnal property if and only if it satisfies the *polynomial Rödl property* (every $\mathcal{F}$-free graph G contains an ε-restricted induced subgraph of order at least $\varepsilon^d|G|$ for all $\varepsilon \in (0, 1/2)$), and if and only if $\mathcal{F}$ is *viral* (the same conclusion holds whenever each $H \in \mathcal{F}$ has at most $(\varepsilon^d|G|)^{|H|}$ induced copies in G). Virality enjoys two fundamental closure operations: Fox, Nguyen, Scott, and Seymour [11] proved that viral families are closed under vertex substitution, and Nguyen, Scott, and Seymour [18] proved that if $H, \overline{J} \in \mathcal{F}$ satisfy $|H|, |J| \geq 3$ with $\deg_H(v_1) \leq 1$ and $\deg_J(v_2) \leq 1$ such that $(\mathcal{F} \setminus \{H\}) \cup \{H - v_1\}$ and $(\mathcal{F} \setminus \{\overline{J}\}) \cup \{\overline{J - v_2}\}$ are viral, then $\mathcal{F}$ is viral. In particular, letting $\mathcal{J}(\mathcal{B})$ denote the closure of an isomorphism- and complement-closed viral base class $\mathcal{B} \supseteq \{K_1, K_2, \overline{K_2}\}$ under vertex substitution and adding a vertex of degree at most 1, every opposite-chirality pair $\{H_1, \overline{H_2}\}$ $(H_1, H_2 \in \mathcal{J}(\mathcal{B}))$ is viral, and every single graph in $\mathcal{H}(\mathcal{B}) := \mathcal{J}(\mathcal{B}) \cap \overline{\mathcal{J}(\mathcal{B})}$ belongs to $\mathcal{EH}$.

Despite the power of $\mathcal{H}(\mathcal{B})$, every graph H of order $|H| \geq 6$ that is prime ($m_{\max}(H) = 1$)

and satisfies $\Delta(H) \le |H| - 3$ has minimum co-degree $\delta(\overline{H}) = |H| - 1 - \Delta(H) \ge 2$ and therefore escapes $\mathcal{H}(\mathcal{B})$ whenever $H \notin \mathcal{B}$ (Proposition 2.6). In particular, every path P_k ($k \ge 6$), every subdivided spider $S_{1,2,d}$ ($d \ge 2$), and every unicyclic long-tail bird Bird_d ($d \ge 2$) lies outside $\mathcal{H}(\{K_1, K_2, \overline{K_2}, C_5, P_5, \overline{P_5}, \mathrm{Bull}\})$. While Nguyen, Scott, and Seymour [19] established the near-polynomial bound $\kappa(G) \ge 2^{(\log |G|)^{1-o(1)}}$ for all P_k-free graphs, and Huang, Ju, and Zhou [16] recently axiomatized a two-stage sparsification framework to resolve two six-vertex graphs (E-graph and $\mathrm{Bird} = \mathrm{Bird}_2$), no infinite family of prime graphs with $\Delta(H) \le |H| - 3$ outside $\mathcal{H}$ was previously known to satisfy the single-graph Erdős–Hajnal conjecture.

The purpose of this paper is to prove the single-graph Erdős–Hajnal conjecture unconditionally for all finite paths P_k ($k \ge 1$) and for several infinite families of sparse trees, unicyclic graphs, bicyclic graphs, and prime non-split graphs. Our proof is completely self-contained starting directly from published theorems on graphs of order at most five ($|H| \le 4$, Bull, C_5, and P_5). To state our main results precisely, recall that for integers $s, d \ge 1$, the (s,d)-*broom* $B(s,d) := P_{d+2}[v_1 \leftarrow \overline{K_s}]$ is the $(s+d+1)$-vertex tree obtained from $P_{d+2} = (v_1 - \cdots - v_{d+2})$ by substituting an s-vertex stable set $\overline{K_s}$ for its endpoint v_1, and for integers $a, d, b \ge 1$, the *double broom* $\mathrm{DB}(a,d,b) := P_{d+3}[v_1 \leftarrow \overline{K_a}, v_{d+3} \leftarrow \overline{K_b}]$ substitutes $\overline{K_a}$ and $\overline{K_b}$ for the two endpoints of P_{d+3}. For every integer $d \ge 2$, the $(d+4)$-vertex *subdivided spider* $S_{1,2,d}$ is the tree on $\{c, w_1, w_2, x_2, u_1, \ldots, u_d\}$ with edge set $\{cw_1, cw_2, w_2x_2, cu_1\} \cup \{u_q u_{q+1} : 1 \le q \le d-1\}$. For every integer $d \ge 1$, the $(d+4)$-vertex *unicyclic bird* Bird_d on $\{b_0, b_1, b_2, b_3, u_1, \ldots, u_d\}$ has edge set $\{b_0b_1, b_1b_2, b_2b_3, b_1b_3, b_2u_1\} \cup \{u_q u_{q+1} : 1 \le q \le d-1\}$ (so $\mathrm{Bird}_1 = \mathrm{Bull}$ and $\mathrm{Bird}_2 = \mathrm{Bird}$), and the $(d+5)$-vertex *bicyclic sawtooth* $M_{2,d} := \mathrm{Bird}_d[b_0 \leftarrow K_2]$ substitutes K_2 for the leaf b_0.

Theorem 1.1 (Infinite Families of Paths, Spiders, Birds, Sawtooths, Brooms, and Small Trees)**.**

(1) (Paths, linear forests, brooms, and double brooms.) *For every integer $k \ge 1$, both the k-vertex path P_k and its complement $\overline{P_k}$ have the Erdős–Hajnal property, satisfy the polynomial Rödl property, and are viral. Consequently, every non-empty linear forest $F = \bigsqcup_{r=1}^{m} P_{k_r}$ ($m \ge 1$, $k_1, \ldots, k_m \ge 1$), every co-linear forest $\overline{F}$, every (s,d)-broom $B(s,d)$ ($s, d \ge 1$), every double broom $\mathrm{DB}(a,d,b)$ ($a, d, b \ge 1$), and all of their complements have the Erdős–Hajnal property, satisfy the polynomial Rödl property, and are viral.*

(2) (Prime subdivided spiders, prime unicyclic birds, and bicyclic sawtooths.) *For every integer $d \ge 2$, both $S_{1,2,d}$ and $\overline{S_{1,2,d}}$ (together with every leaf-substituted spider $S_{1,2,d}[w_1 \leftarrow \overline{K_a}, x_2 \leftarrow \overline{K_b}, u_d \leftarrow \overline{K_c}]$ for $a, b, c \ge 1$ and its complement) have the Erdős–Hajnal property, satisfy the polynomial Rödl property, and are viral. For every integer $d \ge 1$, both the unicyclic bird Bird_d and $\overline{\mathrm{Bird}_d}$ and the bicyclic sawtooth $M_{2,d}$ and $\overline{M_{2,d}}$ have the Erdős–Hajnal property, satisfy the polynomial Rödl property, and are viral.*

(3) (Small trees and forests.) *Every non-empty tree of order at most 6, every non-empty forest whose connected components each have order at most 6, and 10 of the 11 non-isomorphic trees on 7 vertices (every 7-vertex tree except the three-legged subdivided claw $S_{2,2,2}$, alongside all of their complements) belong to $\mathcal{H}(\mathcal{B}^*_{\mathrm{new}}) \subseteq \mathcal{EH}$ (defined before Theorem 1.3), satisfy the polynomial Rödl property, and are viral.*

Theorem 1.1 is deduced by strong induction on graph order from our second main result, which establishes all three hypotheses of the symmetric two-stage sparsification framework (Definition 3.1) uniformly across every infinite family $\{P_k\}$ ($k \ge 6$), $\{S_{1,2,d}\}$ ($d \ge 2$), and $\{\mathrm{Bird}_d\}$ ($d \ge 2$).

Theorem 1.2 (Uniform Leaf-Reducibility, Stage-1 Coupled Generalized Niceness, and Stage-2 Thresholded Wonderfulness)**.** *For every integer $r \ge 1$, the depth-r Stage-1 comb/co-comb dichotomy*

(Theorem 4.3) holds unconditionally. Furthermore, for every $H \in \{P_k : k \geq 6\} \cup \{S_{1,2,d} : d \geq 2\} \cup \{\mathrm{Bird}_d : d \geq 2\}$, $\{H\}$ *is unconditionally leaf-reducible (Definition 3.1*(1)*) and Stage-2 thresholded wonderful (Definition 3.1*(3)*), and both* $\{H\}$ *and* $\{\overline{H}\}$ *are unconditionally Stage-1 coupled generalized nice (Definition 3.1*(2)*).*

To state our third main theorem, let $\mathcal{B}_0^* := \{K_1, K_2, \overline{K_2}, P_5, \overline{P_5}, \mathrm{Bull}\}$ and $\mathcal{B}_0^+ := \mathcal{B}_0^* \cup \{C_5\}$ be the isomorphism- and complement-closed base classes of generators of order at most 5, set $\mathcal{B}_k^* := \{K_1, K_2, \overline{K_2}, \mathrm{Bird}_2, \overline{\mathrm{Bird}_2}\} \cup \{P_t, \overline{P_t} : 1 \leq t \leq k\}$ $(k \geq 6)$, set $\mathcal{B}_{\mathrm{new}}^* := \mathcal{B}_9^* \cup \{\mathrm{Bull}\} \cup \{S_{1,2,d}, \overline{S_{1,2,d}} : 2 \leq d \leq 5\} \cup \{\mathrm{Bird}_d, \overline{\mathrm{Bird}_d} : 3 \leq d \leq 6\}$, and set

$$\begin{aligned}\mathcal{B}_\infty^* \;:=\; & \mathcal{B}_{\mathrm{new}}^* \cup \{P_k, \overline{P_k} : k \geq 10\} \cup \{S_{1,2,d}, \overline{S_{1,2,d}} : d \geq 6\} \\ & \cup \{\mathrm{Bird}_d, \overline{\mathrm{Bird}_d} : d \geq 7\} \cup \{M_{2,d}, \overline{M_{2,d}} : d \geq 1\}.\end{aligned}$$

For any non-empty graph Q and subset $S \subseteq V(Q)$, let $\mathrm{Hook}_4(Q, S)$ denote the graph obtained from $Q \sqcup P_4$ ($P_4 = (u_1 - u_2 - u_3 - u_4)$) by joining u_1 to every vertex of S while leaving $\{u_2, u_3, u_4\}$ anticomplete to $V(Q)$. Finally, for any prime graph G_0 of order $n_0 := |G_0| \geq 6$ with $\Delta(G_0) \leq n_0 - 3$ and a leaf $x_0 \in V(G_0)$ ($\deg_{G_0}(x_0) = 1$), let $\{H_{n_0+r}^{(G_0)}\}_{r \geq 0}$ denote the canonical alternating co-leaf/leaf extension tower above G_0 (Definition 2.1(5)).

Theorem 1.3 (Strict Closure Hierarchy, Opposite-Chirality Forest Pairs, Hooked Claws, and Infinite Non-Split Prime Viral Towers)**.**

(1) (Strict closure hierarchy and opposite-chirality pairs.) *We have the strict inclusions*

$$\mathcal{H}(\mathcal{B}_0^*) \subsetneq \mathcal{H}(\mathcal{B}_9^*) \subsetneq \mathcal{H}(\mathcal{B}_{\mathrm{new}}^*) \subsetneq \mathcal{H}(\mathcal{B}_\infty^*) \subseteq \mathcal{EH},$$

witnessed explicitly by $P_6 \in \mathcal{H}(\mathcal{B}_9^*) \setminus \mathcal{H}(\mathcal{B}_0^+)$, $S_{1,2,2} \in \mathcal{H}(\mathcal{B}_{\mathrm{new}}^*) \setminus \mathcal{H}(\mathcal{B}_9^*)$, *and* $S_{1,2,6} \in \mathcal{H}(\mathcal{B}_\infty^*) \setminus \mathcal{H}(\mathcal{B}_{\mathrm{new}}^*)$. *For every* $H_1, H_2 \in \mathcal{J}(\mathcal{B}_\infty^*)$, *the opposite-chirality pair* $\{H_1, \overline{H_2}\}$ *belongs to* $\mathcal{EH}$, *satisfies the polynomial Rödl property, and is viral. In particular, for every pair of non-empty finite forests* F_1 *and* F_2 *(*$|F_1|, |F_2| \geq 1$*), the pair* $\{F_1, \overline{F_2}\}$ *belongs to* $\mathcal{EH}$, *satisfies the polynomial Rödl property, and is viral.*

(2) (Hooked-claw pairs.) *For every non-empty graph* $Q \in \mathcal{EH}$ *(*$|Q| \geq 1$*), every subset* $S \subseteq V(Q)$, *and every integer* $t \geq 1$, *both* $\{P_t, \mathrm{Hook}_4(Q, S)\}$ *and* $\{\overline{P_t}, \overline{\mathrm{Hook}_4(Q, S)}\}$ *belong to* $\mathcal{EH}$, *satisfy the polynomial Rödl property, and are viral.*

(3) (Countably infinitely many prime non-split viral towers.) *For every prime non-split graph* G_0 *of order* $n_0 := |G_0| \geq 6$ *with* $\Delta(G_0) \leq n_0 - 3$, *leaf* $x_0 \in V(G_0)$ *(*$\deg_{G_0}(x_0) = 1$*), and* $G_0, G_0 - x_0 \in \mathcal{H}(\mathcal{B}_\infty^*)$ *— in particular, for* $(G_0, x_0) \in \{(P_k, v_k) : k \geq 6\} \cup \{(S_{1,2,d}, u_d) : d \geq 2\} \cup \{(\mathrm{Bird}_d, u_d) : d \geq 2\}$ *(where* $S_{1,2,2} - u_2 \cong B(2,2)$ *and* $\mathrm{Bird}_2 - u_2 \cong \mathrm{Bull}$*) — the graphs* $\{H_{n_0+r}^{(G_0)}\}_{r \geq 0}$ *are pairwise non-isomorphic (*$H_{n_0+r_1}^{(G_0)} \cong H_{n_0+r_2}^{(G_0)} \iff r_1 = r_2$*), prime (*$m_{\max} = 1$*), non-split, satisfy* $\overline{H}_{n_0+r}^{(G_0)} \notin \mathcal{J}(\mathcal{B}_0^+)$ *(and in particular* $\overline{H}_{n_0+r}^{(G_0)} \notin \mathcal{J}(\{K_1, K_2, \overline{K_2}, C_5\})$*), and belong to* $\mathcal{H}(\mathcal{B}_\infty^*) \subseteq \mathcal{EH}$ *(so every* $H_{n_0+r}^{(G_0)}$ *and* $\overline{H}_{n_0+r}^{(G_0)}$ *has the Erdős–Hajnal property, satisfies the polynomial Rödl property, and is viral).*

1.1 Overview of the Proof Architecture

Passing from five-vertex graphs [21] to paths P_k $(k \geq 6)$, subdivided spiders $S_{1,2,d}$ $(d \geq 2)$, and unicyclic birds Bird_d $(d \geq 2)$ of arbitrary length encounters two fundamental structural obstructions in the two-stage iterative sparsification paradigm, together with a coupling barrier at the interface between Stage 1 and Stage 2. Remarkably, both obstructions admit uniform, induction-friendly

resolutions starting directly from $P_5 \in \mathcal{EH}$ [21] and Bull $\in \mathcal{EH}$ [7], requiring no finite casework for intermediate graph orders.

1. Stage 1: Arbitrary-Depth Comb/Co-Comb Dichotomy, Modular Collapse, and Decoupled Interface (Sections 3 and 4). When a $\overline{P_k}$-free induced subgraph F is sparse, upside-down comb extraction (Lemma 3.2(3)) either passes to a sparser induced subgraph, extracts a one-sided sparse trigger pair (X, Y), or produces an (ℓ, w)-comb $((a_i, B_i) : i \in [\ell])$ with an external apex v complete to $\bigcup_{i=1}^{\ell} B_i$ and anticomplete to $\{a_1, \dots, a_\ell\}$ in F. For $\overline{P_5}$ [21], no external vertex $u \in B_j$ $(j \neq i)$ can be mixed on an anticonnected subset $M \subseteq B_i$, because v, a_i, u and an edge of $\overline{F}[M]$ mixed with u immediately induce P_5 in $\overline{F}$. For $k \geq 6$, however, an external vertex $u = z_0 \in B_j$ and a non-neighbor $z_1 \in B_i \setminus N_F(z_0)$ form an induced four-vertex path $(v - a_i - z_0 - z_1)$ in $\overline{F}$, leaving room for an induced path of length $k - 4 \geq 2$ inside $N_F(z_0) \cap B_i$. In particular, when $k \geq 7$, vertices of $B_i \setminus N_F(z_0)$ can be mixed on large anticonnected components of $F[N_F(z_0) \cap B_i]$, so single-level modularity breaks down completely.

In Section 4, we resolve this obstruction uniformly across all depths $r = k - 5 \geq 1$ via three ingredients:

(i) *Per-handle s-vertex external apex stems and nested cutter chains.* Rather than tracking a single external apex v $(s = 1)$, we equip both (ℓ, w)-combs and (ℓ, w)-co-combs with per-handle s-vertex external apex stems $\{\mathbf{v}^{(i)} = (v_1^{(i)}, \dots, v_s^{(i)}) : i \in [\ell]\}$ (Definition 2.2). In either a comb or a co-comb, any ordered pair (B_i, B_j) $(i \neq j)$ admits a canonical stem-handle path $(\hat{v}_1^{(i,j)} - \cdots - \hat{v}_s^{(i,j)} - \hat{a}_{i,j}) \cong P_{s+1}$ in $\overline{F}$ that is complete in F to B_i while $\hat{a}_{i,j}$ is anticomplete in F to B_j. Along any nested cutter chain $M_0 \supseteq M_1 \supseteq \cdots \supseteq M_{r-1}$ inside B_i where $M_q \subseteq N_F(z_{q-1}) \cap M_{q-1}$ and $z_q \in M_{q-1} \setminus N_F(z_{q-1})$ $(1 \leq q \leq r-1)$, the $(s+1+r)$-tuple $(\hat{v}_1^{(i,j)}, \dots, \hat{v}_s^{(i,j)}, \hat{a}_{i,j}, z_0, z_1, \dots, z_{r-1})$ induces P_{s+1+r} in $\overline{F}$ with its first $s + r$ vertices complete in F to M_{r-1}. At terminal depth $d = r - 1 \geq 0$, if z_d is mixed on M_d, forbidding $\overline{P_{s+4+r}}$ forces every anticonnected component of $F[N_F(z_d) \cap M_d]$ to be a *proper module* of $F[M_d]$; symmetrically, whenever $N_F(z_d) \cap M_d \neq \emptyset$, forbidding $\overline{S_{1,2,s+r}}$ (respectively, $\overline{\mathrm{Bird}_{s+r}}$) forces every connected component (respectively, anticonnected component) of the non-neighborhood $F[M_d \setminus N_F(z_d)]$ to be a proper module of $F[M_d]$.

(ii) *Gallai modular-core split trees and Regime A vs. Regime B arity capping.* On each tooth B_i, we maintain an active-leaf split tree $\mathcal{T}_i$ governed by the subadditive potential $\Phi_{D_r}(\mathcal{T}_i) = \sum_{L \in \mathcal{L}(\mathcal{T}_i)} |L|^{1/D_r}$ up to a parameterized stopping threshold $L_{\max} \in [2, \ell]$ and global arity cap p_{cap}. At any earlier depth $0 \leq d \leq r - 2$ where no balanced cutter exists, dyadic bucket extraction either yields a sparse/co-sparse blockade or split of the active leaf, or promotes extreme cutters into a $(k_{\mathrm{comb}}, w_{\mathrm{sub}})$-comb or co-comb equipped with $(s + 1 + d)$-vertex apex stems, reducing the path slack from r to $r' = r - 1 - d \in [1, r - 1]$. Passing the stopping threshold $L'_{\max} := k' = \min(k_{\mathrm{comb}}, p_{\mathrm{cap}})$ to the depth-r' induction hypothesis cleanly separates two regimes: in *Regime A* $(k_{\mathrm{comb}} \leq p_{\mathrm{cap}})$, dyadic bucket sizing guarantees $w_{\mathrm{sub}} > \frac{|M_i|}{16(k')^2}$ independently of the target cross-block sparsity η_{out} and p_{cap}, so $(k')^{-2} \geq k^{-2}$ absorbs into $k^{-E_{D_r}}$ and advances $\Phi_{D_r}(\mathcal{T}_i)$; in *Regime B* $(k_{\mathrm{comb}} > p_{\mathrm{cap}})$, $L'_{\max} = p_{\mathrm{cap}}$ forces the recursive call to return a blockade of *exact length* p_{cap}, exiting immediately without splitting $\mathcal{T}_i$.

(iii) *Decoupled Stage 1–2 interface via one-shot viral purification.* Because recursive leaf splits produce weakly x_{sp}-sparse-or-co-sparse blockades (rather than 100% complete pairs), Stage 1 outputs blockades whose width $\lfloor \varepsilon^{c_1} |G| \rfloor$ is completely independent of $x_{\mathrm{sp}} \in [\varepsilon^{3c_2}, \varepsilon^{c_7}]$. In Stage 2 (Theorem 3.5), we evaluate Stage 1 at $\varepsilon_{\mathrm{GN}} = y^a$ and $x_{\mathrm{sp}} = y^{3ac_2}$ $(c_2 \geq 4d'|H_*| + 8)$

and invoke virality of the leaf-reduced family $\overline{\mathcal{F}^-}$ inside each block to upgrade every edge of the pattern graph J simultaneously to 100% completeness (0 non-edges in G).

2. Stage 2: Recursive Terminal-Path Pure-Substitution Arms (Sections 3 and 5). Once Stage 2 purifies a blockade $\mathcal{B} = (B_1, \dots, B_\ell)$ in a $\overline{H_*}$-free graph G so that every edge $ij \in E(J)$ is 100% complete in G and every non-edge $ij \notin E(J)$ is two-sided δ_{cross}-sparse in G, we must show that if an external vertex $v_0 \notin V(\mathcal{B})$ has open-interval attachment $0 < |N_G(v_0) \cap B_i| < \frac{1}{2}|B_i|$ on at least $y\ell$ blocks $i \in I_0(v_0)$, then $J[I_0(v_0)]$ forbids a fixed witness graph $J_0 \in \mathcal{EH}$ (Proposition 3.4). When mapping v_0 to an interior vertex of H_* in $\overline{G}$, representatives of $\overline{G}$-neighbors of v_0 are chosen sequentially inside $B_j \setminus N_G(v_0)$ ($> \frac{1}{2}w$), avoiding earlier δ_{cross}-sparse neighborhoods so all non-edges of J_0 outside the special set S_{J_0} become edges in $\overline{G}$ ((E-Edge)), while all edges of J_0 are 100% non-edges in $\overline{G}$ ((E-Non)). However, representatives of $\overline{G}$-non-neighbors of v_0 ($u \in S_{J_0}$) must be chosen inside $B_j \cap N_G(v_0)$, which is only guaranteed to be non-empty. Across the non-edges $E_{\text{non}}(S_{J_0})$ inside S_{J_0}, the adjacency pattern in $\overline{G}$ is therefore an arbitrary Boolean vector in $\{0,1\}^{|E_{\text{non}}(S_{J_0})|}$.

Rather than constructing finite ad-hoc witnesses whose Boolean case analysis blows up exponentially with k, Section 5 introduces a single recursive pure-substitution operator $(\text{Arm}_r(B_0), S_r(B_0)) \in \mathcal{J}(\{B_0, \overline{K_2}\})$ along any terminal induced path $(w_1 - \cdots - w_L)$ of a base graph $G_0 = \overline{B_0}$ ($L \geq 2$). Starting from $(\text{Arm}_1(B_0), S_1(B_0)) := (\overline{K_2}, \{x_1\})$, we define $(\text{Arm}_r(B_0), S_r(B_0))$ ($2 \leq r \leq L-1$) by substituting disjoint copies of $\text{Arm}_{r-1}(B_0)$ into the first r terminal-path vertices $(w_1, \dots, w_r)$ of B_0 while keeping $(w_{r+1}, \dots, w_L) \sqcup V_{\text{head}}$ outside the special set. A clean backward induction across $q = r, r-1, \dots, 1$ (Lemma 5.1) shows that under any open-interval realization satisfying (E-Non) and (E-Edge), either some slot fails to link at an interior special vertex and immediately closes an induced copy of the one-leaf extension $G_0^+ = G_0 + (w_0 - w_1)$ inside $\overline{G}[\text{Arm}_r(B_0)]$, or all slots link across $(1, \dots, r)$ to output an induced $(r+1)$-vertex path $(z_1 - \cdots - z_r - y)$ with $z_1, \dots, z_r \in S_r(B_0)$ and $y \notin S_r(B_0)$. Joining two such arms via K_2-substitution bridges across v_0 whenever neither arm already embeds G_0^+, establishing thresholded wonderfulness in one uniform stroke for all P_k ($k \geq 6$), all $S_{1,2,d}$ ($d \geq 2$), and all Bird_d ($d \geq 2$).

2 Notation, Viral Closures, and Prime Extension Towers

For a graph $G = (V(G), E(G))$ and $v \in V(G)$, we write $N_G(v) := \{u \in V(G) : uv \in E(G)\}$ for the open neighborhood of v, $N_G[v] := \{v\} \cup N_G(v)$ for its closed neighborhood, $\deg_G(v) := |N_G(v)|$ for its degree, and $\overline{\deg}_G(v) := \deg_{\overline{G}}(v) = |G| - 1 - \deg_G(v)$ for its co-degree. We set $\delta(G) := \min_{v \in V(G)} \deg_G(v)$ and $\Delta(G) := \max_{v \in V(G)} \deg_G(v)$ (with $\delta(\emptyset) = \Delta(\emptyset) = 0$). For $X \subseteq V(G)$ and $v \in V(G)$, $G[X]$ denotes the subgraph induced by X, $G \setminus X := G[V(G) \setminus X]$, $G - v := G \setminus \{v\}$, and $\overline{G}[X] := \overline{G[X]}$. Write $G_1 \sqcup G_2$ for vertex-disjoint union, $H \subseteq_{\text{ind}} G$ if H is isomorphic to an induced subgraph of G, $[k] := \{1, \dots, k\}$ for $k \geq 1$, and $\mathbf{1}[\mathcal{E}] \in \{0,1\}$ for the indicator of an event $\mathcal{E}$.

For graphs H_1, H_2 and $v_0 \in V(H_1)$, $H_1[v_0 \leftarrow H_2]$ denotes the graph obtained from H_1 by *substituting* H_2 for v_0 (replacing v_0 with a disjoint copy of $V(H_2)$ and joining every $u \in V(H_2)$ to $N_{H_1}(v_0)$); simultaneous substitution at distinct vertices $v_1, \dots, v_m \in V(H_1)$ is written $H_1[v_1 \leftarrow G_1, \dots, v_m \leftarrow G_m]$, and we write $\{u, u'\}_{K_2}$ (respectively, $\{u, u^*\}_{\overline{K_2}}$) for a labeled 2-vertex clique (respectively, 2-vertex stable set). A graph G is *anticonnected* if $\overline{G}$ is connected, and the *anticonnected components* of G are the induced subgraphs corresponding to the connected components of $\overline{G}$. A subset $M \subseteq V(G)$ is a *module* of G if every $z \in V(G) \setminus M$ is either complete ($M \subseteq N_G(z)$) or anticomplete ($M \cap N_G(z) = \emptyset$) to M, and a *proper module* if $1 \leq |M| < |G|$. We write $m_{\max}(G)$ for

the largest cardinality of a proper module of G ($m_{\max}(G) := 0$ when $|G| \le 1$); G is *prime* if $|G| \ge 4$ and $m_{\max}(G) = 1$, a *cograph* if G is P_4-free, and *split* if $V(G)$ partitions into a clique and a stable set. For any graph H and subset $S \subseteq V(H)$, we write $E_{\text{non}}(S) := \{\{u, v\} \subseteq S : u \neq v,\ uv \notin E(H)\}$ for the set of non-edges of H inside S.

Definition 2.1 (Graph Families, Closures $\mathcal{J}(\mathcal{B})$ and $\mathcal{H}(\mathcal{B})$, and Canonical Extension Towers)**.**

(1) *(Paths, linear forests, brooms, and double brooms.)* For each integer $k \ge 1$, P_k denotes the k-vertex path on $\{v_1, \dots, v_k\}$ with edge set $\{v_i v_{i+1} : 1 \le i \le k-1\}$, and K_k denotes the k-vertex complete graph. A *linear forest* is a non-empty disjoint union $\bigsqcup_{r=1}^{m} P_{k_r}$ with $m \ge 1$ and $k_1, \dots, k_m \ge 1$, and a *co-linear forest* is the complement of a linear forest. For integers $s, d \ge 1$, the (s,d)-*broom* is $B(s,d) := P_{d+2}[v_1 \leftarrow \overline{K_s}]$ ($s+d+1$ vertices), and for integers $a, d, b \ge 1$, the *double broom* is $\mathrm{DB}(a,d,b) := P_{d+3}[v_1 \leftarrow \overline{K_a}, v_{d+3} \leftarrow \overline{K_b}]$ ($a+d+b+1$ vertices).

(2) *(Subdivided spiders, leaf-substituted spiders, and $S_{2,2,2}$.)* For integers $1 \le a \le b \le c$, the *spider* $S_{a,b,c}$ is the $(a+b+c+1)$-vertex tree obtained by identifying one endpoint each of three disjoint paths of edge-lengths a, b, c into a unique degree-3 center c. For every integer $d \ge 2$, $S_{1,2,d}$ has vertex set $V(S_{1,2,d}) := \{c, w_1, w_2, x_2, u_1, \dots, u_d\}$ and edge set $E(S_{1,2,d}) := \{cw_1, cw_2, w_2x_2, cu_1\} \cup \{u_q u_{q+1} : 1 \le q \le d-1\}$; for integers $a, b, c \ge 1$, the corresponding *leaf-substituted spider* is $S_{1,2,d}[w_1 \leftarrow \overline{K_a}, x_2 \leftarrow \overline{K_b}, u_d \leftarrow \overline{K_c}]$. The 7-vertex *subdivided claw* $S_{2,2,2}$ has vertex set $\{c, a_1, b_1, a_2, b_2, a_3, b_3\}$ and edge set $\{ca_1, a_1b_1, ca_2, a_2b_2, ca_3, a_3b_3\}$.

(3) *(Unicyclic birds,* Bull*, and bicyclic sawtooths.)* For every integer $d \ge 1$, the $(d+4)$-vertex *unicyclic bird* Bird_d has vertex set $V(\mathrm{Bird}_d) := \{b_0, b_1, b_2, b_3, u_1, \dots, u_d\}$ and edge set $E(\mathrm{Bird}_d) := \{b_0b_1, b_1b_2, b_2b_3, b_1b_3, b_2u_1\} \cup \{u_q u_{q+1} : 1 \le q \le d-1\}$; we write $\mathrm{Bull} := \mathrm{Bird}_1$ (5 vertices) and $\mathrm{Bird} := \mathrm{Bird}_2$ (6 vertices). For every integer $d \ge 1$, the $(d+5)$-vertex *bicyclic sawtooth* is $M_{2,d} := \mathrm{Bird}_d[b_0 \leftarrow \{b_0, b_0'\}_{K_2}]$.

(4) *(Hooked claws and closure operators $\mathcal{J}(\mathcal{B}), \mathcal{H}(\mathcal{B})$.)* For any non-empty graph Q ($|Q| \ge 1$) and subset $S \subseteq V(Q)$, $\mathrm{Hook}_4(Q, S)$ is obtained from $Q \sqcup P_4$ ($P_4 = (u_1 - u_2 - u_3 - u_4)$) by adding all edges between u_1 and S while leaving $\{u_2, u_3, u_4\}$ anticomplete to $V(Q)$; when $S = V(Q)$, we abbreviate $\mathrm{Hook}_4(Q) := \mathrm{Hook}_4(Q, V(Q))$. For any class $\mathcal{B}$ of non-empty finite graphs containing $\{K_1, K_2, \overline{K_2}\}$ and closed under isomorphism and complementation, $\mathcal{J}(\mathcal{B})$ is the smallest class of non-empty finite graphs containing $\mathcal{B}$ closed under isomorphism, vertex substitution $H_1[v_0 \leftarrow H_2]$, and adding a vertex of degree at most 1, and $\mathcal{H}(\mathcal{B}) := \mathcal{J}(\mathcal{B}) \cap \overline{\mathcal{J}(\mathcal{B})}$.

(5) *(Canonical (x_r, y_r) alternating extension towers.)* Let G_0 be a graph of order $n_0 := |G_0| \ge 6$ with a designated leaf $x_0 \in V(G_0)$ ($\deg_{G_0}(x_0) = 1$). Set $H_{n_0}^{(G_0)} := G_0$, and recursively for integers $r \ge 1$ form $H_{n_0+r}^{(G_0)}$ on $V(H_{n_0+r-1}^{(G_0)}) \sqcup \{u_{n_0+r}\}$ with pointers (x_r, y_r) by setting:
- for *odd* $r \ge 1$ (co-leaf step): $N_{H_{n_0+r}^{(G_0)}}(u_{n_0+r}) := V(H_{n_0+r-1}^{(G_0)}) \setminus \{x_{r-1}\}$ and $(x_r, y_r) := (x_{r-1}, u_{n_0+r})$;
- for *even* $r \ge 2$ (leaf step): $N_{H_{n_0+r}^{(G_0)}}(u_{n_0+r}) := \{y_{r-1}\}$ and $(x_r, y_r) := (u_{n_0+r}, y_{r-1})$.

Definition 2.2 (Combs, Co-Combs, and Per-Handle s-Vertex External Apex Stems)**.** A *blockade* in a graph F is a sequence $\mathcal{B} = (B_1, \dots, B_\ell)$ of pairwise disjoint non-empty subsets of $V(F)$, called *blocks*; it is an (ℓ, w)-*blockade* if its length is ℓ and $\min_{i \in [\ell]} |B_i| \ge w$, where $V(\mathcal{B}) := \bigcup_{i=1}^{\ell} B_i$. An (ℓ, w)-*comb* (respectively, (ℓ, w)-*co-comb*) in F is a sequence of pairs $((a_i, B_i) : i \in [\ell])$ where $(B_1, \dots, B_\ell)$ is an (ℓ, w)-blockade in F, $a_1, \dots, a_\ell \in V(F) \setminus V(\mathcal{B})$ are distinct *handles*, and for all $i, j \in [\ell]$: in a *comb*, a_i is complete in F to B_i and anticomplete in F to B_j for $j \neq i$; in a *co-comb*, a_i is anticomplete in F to B_i and complete in F to B_j for $j \neq i$.

For an integer $s \geq 1$, an (ℓ, w)-comb or (ℓ, w)-co-comb $((a_i, B_i) : i \in [\ell])$ in F is *equipped with per-handle s-vertex external apex stems* $\{\mathbf{v}^{(i)} = (v_1^{(i)}, \dots, v_s^{(i)}) : i \in [\ell]\}$ if each $\mathbf{v}^{(i)}$ consists of s distinct vertices in $V(F) \setminus (\{a_1, \dots, a_\ell\} \cup V(\mathcal{B}))$ such that:
(Stem-1) $\overline{F}[\{v_1^{(i)}, \dots, v_s^{(i)}\}]$ is the induced path $(v_1^{(i)} - v_2^{(i)} - \cdots - v_s^{(i)}) \cong P_s$;
(Stem-2) $v_s^{(i)}$ is non-adjacent in F (adjacent in $\overline{F}$) to a_i, whereas every $v_r^{(i)}$ $(1 \leq r \leq s-1)$ is adjacent in F to a_i; and
(Stem-3) every stem vertex $v_r^{(i)}$ $(1 \leq r \leq s,\ i \in [\ell])$ is complete in F (anticomplete in $\overline{F}$) to $\bigcup_{q=1}^{\ell} B_q$.
When $\mathbf{v}^{(1)} = \cdots = \mathbf{v}^{(\ell)} = (v_1, \dots, v_s)$, we call $(v_1, \dots, v_s)$ a *shared s-vertex external apex stem* (and for $s = 1$, $v := v_1$ an *external apex*). For any distinct $i, j \in [\ell]$, setting $(\hat{\mathbf{v}}^{(i,j)}, \hat{a}_{i,j}) := (\mathbf{v}^{(i)}, a_i)$ in a comb and $(\hat{\mathbf{v}}^{(i,j)}, \hat{a}_{i,j}) := (\mathbf{v}^{(j)}, a_j)$ in a co-comb yields the *unified handle-stem pair* satisfying $\overline{F}[\{\hat{v}_1^{(i,j)}, \dots, \hat{v}_s^{(i,j)}, \hat{a}_{i,j}\}] \cong P_{s+1}$ with $\hat{\mathbf{v}}^{(i,j)}$ complete in F to $B_i \cup B_j$ and $\hat{a}_{i,j}$ complete in F to B_i and anticomplete in F to B_j.

We next prove from first principles the equivalence of the Erdős–Hajnal property, the polynomial Rödl property, and virality (Theorem 2.3, following Bucić, Fox, and Pham [3, Theorem 4] and Kleitman and Winston [17]), together with Rödl's theorem [22]. Note that if G is x-restricted with $|G| \geq 1$ and $x > 0$, greedily picking a vertex and restricting to its non-neighborhood in G (when G is x-sparse) or in $\overline{G}$ (when G is x-co-sparse) removes at most $1 + x|G|$ vertices per step and produces a stable set or clique of size at least $\frac{|G|}{1+x|G|}$ (hence $\geq \frac{1}{2x}$ when $|G| \geq x^{-1}$).

Theorem 2.3 (Rödl [22] and Equivalence of $\mathcal{EH}$, Polynomial Rödl, and Virality; Bucić, Fox, and Pham [3, Theorem 4])**.**

(a) (Rödl's theorem.) *For every non-empty finite family $\mathcal{F} = \{H_1, \dots, H_m\}$ of non-empty graphs and every $\xi \in (0, 1/2)$, there exists $\delta = \delta(\mathcal{F}, \xi) > 0$ such that every $\mathcal{F}$-free graph G contains a ξ-restricted induced subgraph F with $|F| \geq \delta|G|$.*

(b) (Equivalence with virality.) *Let $\mathcal{F} = \{H_1, \dots, H_{m_\mathcal{F}}\}$ be a non-empty finite family of non-empty graphs with $h := \max_{H \in \mathcal{F}} \max(|H|, 2)$, and write $\mathrm{ind}_H(G)$ for the number of injective maps $\phi : V(H) \hookrightarrow V(G)$ that are graph isomorphisms onto $G[\phi(V(H))]$. The following three assertions are equivalent:*

- (i) *$\mathcal{F} \in \mathcal{EH}$: there exists $c_\mathcal{F} \in (0, 1/2]$ such that $\kappa(G) \geq |G|^{c_\mathcal{F}}$ for every $\mathcal{F}$-free graph G with $|G| \geq 1$;*
- (ii) (Polynomial Rödl property.) *there exists $d_\mathcal{F} > 0$ such that for every $\varepsilon \in (0, 1/2)$, every $\mathcal{F}$-free graph G contains an ε-restricted induced subgraph $S \subseteq G$ with $|S| \geq \varepsilon^{d_\mathcal{F}}|G|$;*
- (iii) (Virality.) *there exists $d_\mathcal{F} > 0$ such that for every $\varepsilon \in (0, 1/2)$ and every graph G with $\mathrm{ind}_H(G) \leq (\varepsilon^{d_\mathcal{F}}|G|)^{|H|}$ for all $H \in \mathcal{F}$, G contains an ε-restricted induced subgraph $S \subseteq G$ with $|S| \geq \varepsilon^{d_\mathcal{F}}|G|$.*

Specifically, (i) *implies* (iii) *with* $d_\mathcal{F} := \frac{22h}{c_\mathcal{F}^2}\left(\frac{10}{c_\mathcal{F}} + \log_2(m_\mathcal{F} + 1) + 1\right)$.

Proof. Write $H := H_1$ and $h_0 := |H|$; if $h_0 = 1$, every H-free graph is empty. For $h_0 \geq 2$ with $V(H) = \{v_1, \dots, v_{h_0}\}$, set $h := \max(h_0, 4)$, $x_0 := \xi/8$, $c_1 := (x_0/(2h))^h$, $\ell := \lceil 8/\xi \rceil$, and $\delta := \frac{1}{4}c_1^{2\ell-1}$. If $1 \leq |G| < 2h\, c_1^{-(2\ell-2)}$, then $\delta|G| < \frac{h}{2}c_1 < 1$, so any singleton $\{v\} \subseteq V(G)$ suffices; thus assume $|G| \geq 2h\, c_1^{-(2\ell-2)}$ and set $m := \lfloor c_1^{2\ell-2}|G| \rfloor \geq 2h$. Every $Y \subseteq V(G)$ with $|Y| \geq 2h$ contains disjoint subsets $L, R \subseteq Y$ of size $\geq c_1|Y|$ with L pointwise x_0-sparse to R in G or $\overline{G}$, else greedy selection across h_0 parts of Y embeds H into $G[Y]$. Induction on $p + q \geq 2$ $(p, q \geq 1)$ then shows that every $Y \subseteq V(G)$ of size $\geq c_1^{-(p+q-2)}m$ contains either $U \subseteq Y$ of size pm with $|E(G[U])| \leq \frac{1}{2}(p + p^2x_0)m^2$ or $U \subseteq Y$ of size qm with $|E(\overline{G}[U])| \leq \frac{1}{2}(q + q^2x_0)m^2$. Taking $Y = V(G)$ and $p = q = \ell$ gives U of

size ℓm with $|E(G^*[U])| \le \frac{\xi}{8}|U|^2$ ($G^* \in \{G, \overline{G}\}$), and deleting vertices of degree $> \frac{\xi}{2}|U|$ in $G^*[U]$ yields a ξ-restricted subgraph of order $\ge \frac{1}{2}|U| \ge \delta|G|$, proving (a).

In (b), (iii) $\implies$ (ii) is immediate. For (ii) $\implies$ (i) with $c := \frac{1}{2(d_{\mathcal{F}}+1)}$, $\kappa(G) = 1 = |G|^c$ if $|G| = 1$, $\kappa(G) \ge 2 \ge |G|^c$ if $2 \le |G| \le 2^{2(d_{\mathcal{F}}+1)}$, and for $|G| > 2^{2(d_{\mathcal{F}}+1)}$ applying (ii) at $\varepsilon := \frac{1}{2}|G|^{-c} \in (0, 1/4)$ gives an ε-restricted $S \subseteq G$ with $|S| \ge \varepsilon^{d_{\mathcal{F}}}|G| \ge \varepsilon^{-1}$, so $\kappa(G) \ge \frac{1}{2\varepsilon} = |G|^c$. To deduce (iii) from (i), note first by induction on $n + \ell$ that if $0 \le \ell \le k$, $u \ge 0$, $(1-\varepsilon)^\ell n \le u$, and every subset $S \subseteq V(G)$ of size $> u$ satisfies $\Delta(G[S]) \ge \varepsilon|S| - 1$, then G has at most $\binom{n}{\ell}\binom{u}{k-\ell}$ independent k-sets (Kleitman and Winston [17]). Second, if every t-vertex $\mathcal{F}$-free graph has $\kappa \ge k$ and G ($|G| = n \ge 2t$) satisfies $\mathrm{ind}_H(G) \le \frac{n^{|H|}}{m_{\mathcal{F}} 2^{h+1} t^{|H|-1}}$ for all $H \in \mathcal{F}$, a uniformly random $2t$-subset $S \subseteq V(G)$ expects at most $\frac{t}{2}$ induced copies from $\mathcal{F}$, so double counting homogeneous k-sets across $\binom{V(G)}{2t}$ yields at least $\frac{1}{2}\left(\frac{n}{2t}\right)^k$ homogeneous k-sets in G. Setting $f := 1/c_{\mathcal{F}} \ge 2$, $\varepsilon_0 := 2^{-10f}(m_{\mathcal{F}}+1)^{-1}$, $k := 400\lceil \frac{f^2}{\varepsilon}\log^2\frac{1}{\varepsilon}\rceil$, $t := \lceil k^f \rceil$, $\delta := \frac{1}{32k^{f-1}}$, and $\ell := \lceil \frac{1}{\varepsilon}\log\frac{2}{\delta}\rceil$ for $\varepsilon \in (0, \varepsilon_0)$ (with $\varepsilon^{d_{\mathcal{F}}} n > 1$, as otherwise any singleton $\{v\} \subseteq V(G)$ suffices; hence $n \ge 2t$), comparing $\frac{1}{4}\left(\frac{n}{2t}\right)^k$ against $\binom{n}{\ell}\binom{\lfloor \delta n \rfloor}{k-\ell}$ gives $1 \le \frac{24\delta t}{k} \le \frac{4}{5}$ unless G has an ε-restricted subgraph of order $> \lfloor \delta n \rfloor \ge \varepsilon^{d_{\mathcal{F}}} n$; rescaling $\varepsilon' := \varepsilon_0 \varepsilon$ handles $\varepsilon \in [\varepsilon_0, 1/2)$. □

We now establish viral closure under vertex substitution and simultaneous degree-≤ 1 / co-degree-≤ 1 attachments (adapting Fox, Nguyen, Scott, and Seymour [11] and Nguyen, Scott, and Seymour [18, Theorems 1.10 and 7.8] to allow both isolated and pendant vertices across arbitrary complement-closed viral base classes).

Proposition 2.4 (Substitution and Degree-≤ 1 Attachment Virality and Pair Inheritance)**.**

(1) (Viral substitution closure.) *Let $\mathcal{F}$ be a non-empty finite family of non-empty graphs, let $H = H_1[v_0 \leftarrow H_2] \in \mathcal{F}$, and suppose $\mathcal{F}_1 := (\mathcal{F} \setminus \{H\}) \cup \{H_1\}$ and $\mathcal{F}_2 := (\mathcal{F} \setminus \{H\}) \cup \{H_2\}$ are viral with common exponent $d \ge 1$. Then $\mathcal{F}$ is viral with exponent $d' := d + |H_1|d + |H_1|$ (and $\mathcal{F} \in \mathcal{EH}$).*

(2) (Simultaneous degree-≤ 1 and co-degree-≤ 1 attachment closure.) *Let $\mathcal{F}$ be a finite family of graphs, and suppose $H, \overline{J} \in \mathcal{F}$ ($|H|, |J| \ge 3$) contain vertices $v_0 \in V(H)$ and $w_0 \in V(J)$ with $\deg_H(v_0) \le 1$ and $\deg_J(w_0) \le 1$. If $(\mathcal{F} \setminus \{H\}) \cup \{H - v_0\}$ and $(\mathcal{F} \setminus \{\overline{J}\}) \cup \{\overline{J - w_0}\}$ are both viral, then $\mathcal{F}$ is viral.*

(3) (Opposite-chirality pair inheritance and $\mathcal{H}(\mathcal{B})$.) *Let $\mathcal{B}$ be any isomorphism- and complement-closed class of non-empty viral graphs containing $\{K_1, K_2, \overline{K_2}\}$. Then for every $H_1, H_2 \in \mathcal{J}(\mathcal{B})$, the pair $\{H_1, \overline{H_2}\}$ is viral, satisfies the polynomial Rödl property, and belongs to $\mathcal{EH}$; in particular, every single graph $H \in \mathcal{H}(\mathcal{B}) := \mathcal{J}(\mathcal{B}) \cap \overline{\mathcal{J}(\mathcal{B})}$ (and $\overline{H}$) is viral, satisfies the polynomial Rödl property, and belongs to $\mathcal{EH}$.*

Proof. In (1) ($h_1 := |H_1|$, $h_2 := |H_2|$, $h := h_1 + h_2 - 1$), if $h_1 = 1$ then $H \cong H_2$, and if $\mathrm{ind}_K(G) \le (\varepsilon^{d'}|G|)^{|K|}$ ($K \in \mathcal{F}$) with $\mathrm{ind}_{H_1}(G) \le (\varepsilon^d|G|)^{h_1}$, virality of $\mathcal{F}_1$ on G directly yields an ε-restricted subgraph of order $\ge \varepsilon^d|G| \ge \varepsilon^{d'}|G|$. When $h_1 \ge 2$ and $\mathrm{ind}_{H_1}(G) > (\varepsilon^d|G|)^{h_1}$, more than $\frac{1}{2}\varepsilon^{h_1 d}|G|^{h_1-1}$ embeddings $\psi : V(H_1) \setminus \{v_0\} \hookrightarrow V(G)$ have extension sets X_ψ to H_1 of size $\ge \varepsilon^{h_1 d+1}|G|$. Since $\sum_\psi \mathrm{ind}_{H_2}(G[X_\psi]) \le \mathrm{ind}_H(G) \le (\varepsilon^{d'}|G|)^h$, averaging finds ψ^* with $\mathrm{ind}_{H_2}(G[X_{\psi^*}]) \le (\varepsilon^d|X_{\psi^*}|)^{h_2}$ (while $\mathrm{ind}_K(G[X_{\psi^*}]) \le (\varepsilon^{d'}|G|)^{|K|} \le (\varepsilon^d|X_{\psi^*}|)^{|K|}$ for all $K \in \mathcal{F}\setminus\{H\}$), and virality of $\mathcal{F}_2$ on $G[X_{\psi^*}]$ yields an ε-restricted subgraph of order $\ge \varepsilon^d|X_{\psi^*}| \ge \varepsilon^{d'}|G|$.

In (2), degree-0 vertices reduce immediately to (1) via $\overline{K_2}$ or K_2 substitution, so assume $\deg_H(v_0) = \deg_J(w_0) = 1$ with unique neighbors $u_0 \in V(H - v_0)$ and $z_0 \in V(J - w_0)$, common viral exponent $d \ge 4$, $h := \max(|H|, |J|, 4)$, $c := 4^{-h}$, and $a := 4dh + 2$. In any y^2-sparse $\mathcal{F}$-free graph F_0 ($0 < y \le c$) with $y^{a+1}|F_0| \ge 1$ and no y^4-restricted subgraph of order $\ge y^{4d+2}|F_0|$, greedy extraction

peels pairwise anticomplete blocks $B_0, \dots, B_{k-1}$ of size $\geq y^{a+1}|F_0|$ ($k := \lceil y^{-1} \rceil$): while residual $|R| \geq (1-hy)^k |F_0| \geq 4^{-h}|F_0|$, applying virality of $\mathcal{F}_1$ at sparsity y^4 inside a $\lceil y|R| \rceil$-subset $S \subseteq R$ finds an induced copy $\Phi \subseteq S$ of $H \setminus \{u_0, v_0\}$ with $\geq (y^4)^{d(|H|-1)}|S| \geq y^{4dh}|R| \geq y^{a+1}|F_0|$ extensions to $H - v_0$ in S, anticomplete to the $\geq (1-hy)|R|$ non-neighbors of Φ in $R \setminus S$. Dually, when $\overline{F_0}$ is y^2-sparse, applying the same argument to $\overline{F_0}$, J, and $\overline{\mathcal{F}_2}$ peels a complete $(\lceil y^{-1} \rceil, y^{a+1}|F_0|)$-blockade in F_0. Iterating from Theorem 2.3(a) across dyadic scales $y_i = c^{2^{i-1}}$ produces a pure $(r, |F|/r^C)$-blockade ($r \geq 2$) whenever $\kappa(F) < |F|^{1/C}$, proving $\mathcal{F} \in \mathcal{EH}$ by induction on $|F|$ and virality by Theorem 2.3(b). In (3), every graph of order ≤ 2 lies in $\{K_1, K_2, \overline{K_2}\} \subseteq \mathcal{B}$ and $\overline{J_a[w_0 \leftarrow J_b]} = \overline{J_a}[w_0 \leftarrow \overline{J_b}]$, so strong induction on $|H_1| + |H_2| \geq 2$ splits on non-trivial substitutions ($|F_a|, |F_b| \geq 2$) via (1) and on pairs of degree-≤ 1 extensions of order ≥ 3 via (2). □

We next prove the hooked-claw theorem for $\{P_t, \mathrm{Hook}_4(Q,S)\}$, recording both its direct single-level comb neighborhood modularity at $S = V(Q)$ and its unconditional inheritance from Theorem 1.1(1).

Proposition 2.5 (Hooked-Claw Defective Modularity and Pair Inheritance)**.** *Let $Q \in \mathcal{EH}$ be a non-empty graph ($|Q| \geq 1$) with Erdős–Hajnal exponent $c_Q \in (0,1]$.*

(1) *For every integer $m \geq 0$, every $\{\overline{P_{6+m}}, \overline{\mathrm{Hook}_4(Q, V(Q))}\}$-free graph F containing an (ℓ, w)-comb $((a_i, B_i) : i \in [\ell])$ ($\ell \geq 2$) with external apex $v \in V(F)$, every $i \neq j \in [\ell]$, every non-empty $M \subseteq B_i$, every $u \in B_j$ not complete to M in F, and every $w_0 \geq 1$, if $\kappa(F) < w_0^{c_Q/(m+1)}$, then every anticonnected component C_s of $F[N_F(u) \cap M]$ satisfying $|C_s| \geq 2^{m+1} w_0$ is a proper module of $F[M]$.*

(2) *Whenever $P_t \in \mathcal{EH}$ ($t \geq 1$) — and hence unconditionally for every integer $t \geq 1$ by Theorem 1.1(1) — both $\{P_t, \mathrm{Hook}_4(Q,S)\}$ and $\{\overline{P_t}, \overline{\mathrm{Hook}_4(Q,S)}\}$ belong to $\mathcal{EH}$, satisfy the polynomial Rödl property, and are viral for every subset $S \subseteq V(Q)$.*

Proof. Suppose in (1) that $\kappa(F) < w_0^{c_Q/(m+1)}$ and some anticonnected component C_s of $F[N_F(u) \cap M]$ satisfies $|C_s| \geq 2^{m+1} w_0$. Since $C_s \subseteq N_F(u) \subsetneq M$ and every vertex of $(N_F(u) \cap M) \setminus C_s$ is complete in F to C_s, suppose for contradiction that some $z \in M \setminus N_F(u)$ is mixed on C_s. Set $\tau_r := (2\kappa(F))^r \kappa(F)^{1/c_Q}$ ($0 \leq r \leq m$); since $c_Q \in (0,1]$ and $\kappa(F) \geq 1$, $\kappa(F)^{m+1/c_Q} \leq \kappa(F)^{(m+1)/c_Q} < w_0$, so $2\tau_m < 2^{m+1} w_0 \leq |C_s|$. Starting from $(y_0, y_1, y_2, y_3) := (v, a_i, u, z)$ and $D^{(0)} := C_s$, inductively for $q = 1, \dots, m$, partition $D^{(q-1)} = W^{(q)} \sqcup Z^{(q)}$ where $W^{(q)} := N_{\overline{F}}(y_{q+2}) \cap D^{(q-1)} \neq \emptyset$ and $Z^{(q)} := N_F(y_{q+2}) \cap D^{(q-1)} \neq \emptyset$. Since $(y_{q-1} - y_q - y_{q+1} - y_{q+2}) \cong P_4$ in $\overline{F}$ has $\{y_{q-1}, y_q, y_{q+1}\}$ anticomplete and y_{q+2} complete in $\overline{F}$ to $W^{(q)}$, $\overline{F}[W^{(q)}]$ is Q-free, giving $|W^{(q)}| \leq \tau_0 \leq \tau_{m-q+1}$ and $|Z^{(q)}| > 2\kappa(F)\tau_{m-q}$. Since $F[Z^{(q)}]$ has at most $\kappa(F)$ anticonnected components, a largest anticonnected component $D^{(q)} \subseteq Z^{(q)}$ satisfies $|D^{(q)}| > 2\tau_{m-q} > \kappa(F)^{1/c_Q}$ and has an $\overline{F}$-neighbor $y_{q+3} \in W^{(q)}$, which must also have an F-neighbor in $D^{(q)}$ (else the shifted suffix $(y_q - y_{q+1} - y_{q+2} - y_{q+3}) \cong P_4$ in $\overline{F}$ makes $\{y_q, y_{q+1}, y_{q+2}\}$ complete in F and y_{q+3} complete in $\overline{F}$ to $D^{(q)}$, forcing $\overline{F}[D^{(q)}]$ to be Q-free). At $q = m$, picking $x_1 x_2 \in E(\overline{F}[D^{(m)}])$ mixed with y_{m+3} induces $(y_0 - \cdots - y_{m+3} - x_1 - x_2) \cong P_{6+m}$ in $\overline{F}$, a contradiction. Part (2) follows since every $\{P_t, \mathrm{Hook}_4(Q,S)\}$-free graph is P_t-free, combined with Theorem 2.3(b). □

We close Section 2 by proving the general escape criterion from $\mathcal{H}(\mathcal{B})$ and the canonical prime non-split alternating extension tower theorem.

Proposition 2.6 (Escape Criterion from $\mathcal{H}(\mathcal{B})$)**.** *Let $\mathcal{B}$ be any isomorphism- and complement-closed class of non-empty finite graphs containing $\{K_1, K_2, \overline{K_2}\}$, and let H be a prime graph ($m_{\max}(H) = 1$)*

of order $|H| \geq 6$ *with* $\Delta(H) \leq |H| - 3$*. If* $H \notin \mathcal{B}$*, then* $\delta(\overline{H}) = |H| - 1 - \Delta(H) \geq 2$*,* $\overline{H} \notin \mathcal{J}(\mathcal{B})$*, and hence* $H, \overline{H} \notin \mathcal{H}(\mathcal{B}) := \mathcal{J}(\mathcal{B}) \cap \overline{\mathcal{J}(\mathcal{B})}$*. In particular:*

(1) *every path* P_k *(*$k \geq 6$*) is prime, non-split, and has* $\Delta(P_k) = 2 \leq k - 3$*, so* $P_k, \overline{P_k} \notin \mathcal{H}(\mathcal{B}^*_{k-1})$ *(and* $P_6 \notin \mathcal{H}(\mathcal{B}^+_0)$*);*

(2) *every subdivided spider* $S_{1,2,d}$ *(*$d \geq 2$*) is prime, non-split (containing induced* $2K_2 = \{w_2x_2, u_1u_2\}$*), and of order* $d+4 \geq 6$ *with* $\Delta(S_{1,2,d}) = 3 \leq |S_{1,2,d}| - 3$*, so* $S_{1,2,d}, \overline{S_{1,2,d}} \notin \mathcal{H}(\mathcal{B}^*_k)$ *for all* $k \geq 6$*;*

(3) *every unicyclic bird* Bird_d *(*$d \geq 2$*) is prime, non-split (containing induced* $2K_2 = \{b_0b_1, u_1u_2\}$*), and of order* $d + 4 \geq 6$ *with* $\Delta(\mathrm{Bird}_d) = 3 \leq |\mathrm{Bird}_d| - 3$*, so for all* $d \geq 3$ *and* $k \geq 6$*,* $\mathrm{Bird}_d \notin \mathcal{H}(\mathcal{B}^*_k \cup \{S_{1,2,p}, \overline{S_{1,2,p}} : p \geq 2\} \cup \{\mathrm{Bird}_q, \overline{\mathrm{Bird}_q} : 3 \leq q < d\})$*.*

Proof. Since H is prime, $m_{\max}(\overline{H}) = m_{\max}(H) = 1$, and $\Delta(H) \leq |H| - 3$ implies $\delta(\overline{H}) = |H| - 1 - \Delta(H) \geq 2$. Every graph $G_0 \in \mathcal{J}(\mathcal{B}) \setminus \mathcal{B}$ satisfies either $\delta(G_0) \leq 1$ (if formed by adding a vertex of degree at most 1) or $m_{\max}(G_0) \geq 2$ (if $G_0 \cong F_1[v_0 \leftarrow F_2]$ with $|F_1|, |F_2| \geq 2$). Because $\mathcal{B}$ is closed under isomorphism and complementation, $H \notin \mathcal{B}$ implies $\overline{H} \notin \mathcal{B}$, hence $\overline{H} \notin \mathcal{J}(\mathcal{B})$ and $H, \overline{H} \notin \mathcal{H}(\mathcal{B})$.

For primality in (1)–(3): in P_k ($k \geq 4$), any proper module M of size ≥ 2 with extreme indices $a < b$ forces $a = 1$ and $b = k$ (else v_{a-1} or v_{b+1} is mixed on M), and then the first $c \in [k] \setminus M$ cannot be adjacent to both v_1 and v_k; in $S_{1,2,d}$ ($d \geq 2$), connectivity forces some $z \notin M$ to be complete to any proper module M of size ≥ 2, so M cannot contain an edge (as $S_{1,2,d}$ is triangle-free) and cannot be a stable set (as $S_{1,2,d}$ is C_4-free and has no twin leaves); in Bird_d ($d \geq 2$), connectivity forces any proper module M of size ≥ 2 to satisfy $M \subseteq N_{\mathrm{Bird}_d}(z)$ for some $z \notin M$ (hence $|M| \leq \Delta(\mathrm{Bird}_d) = 3$), which is impossible since b_0 separates b_1 from $\{b_2, b_3, u_1\}$, u_1 separates b_2 from $\{b_0, b_3\}$, b_1 separates b_3 from u_1, and for $M = N_{\mathrm{Bird}_d}(u_q)$ ($1 \leq q \leq d-1$) the predecessor of M along $(b_3 - b_2 - u_1 - \cdots - u_d)$ is mixed on M. Non-isomorphism among $(d+4)$-vertex generators follows from edge counts and maximum degrees ($|E(\mathrm{Bird}_d)| = d + 4 > d + 3 = |E(S_{1,2,d})| = |E(P_{d+4})|$, $\Delta(\mathrm{Bird}_d) = 3$, and $|E(\overline{P_{d+4}})|, |E(\overline{S_{1,2,d}})| > d + 4$ for $d \geq 2$). Finally, no graph containing induced $2K_2$ can be partitioned into a clique and a stable set, proving non-splitness. □

Theorem 2.7 (Canonical Alternating Prime Non-Split Extension Tower Theorem)**.** *Let* G_0 *be any prime graph of order* $n_0 := |G_0| \geq 6$ *satisfying* $\Delta(G_0) \leq n_0 - 3$ *and possessing a vertex* $x_0 \in V(G_0)$ *of degree* $\deg_{G_0}(x_0) = 1$*, and set* $\mathcal{B}(G_0) := \{G_0, \overline{G_0}, G_0 - x_0, \overline{G_0 - x_0}, K_1, K_2, \overline{K_2}\}$*. Then for every integer* $r \geq 0$*, the* $(n_0 + r)$*-vertex tower graph* $H^{(G_0)}_{n_0+r}$ *(Definition 2.1(5)) satisfies:*

(1) $H^{(G_0)}_{n_0+r}$ *contains* G_0 *as an induced subgraph, is prime (*$m_{\max}(H^{(G_0)}_{n_0+r}) = 1$*), is non-split whenever* G_0 *is non-split, and satisfies* $\overline{H^{(G_0)}_{n_0+r}} \notin \mathcal{J}(\mathcal{B}^+_0)$ *(hence* $H^{(G_0)}_{n_0+r} \notin \mathcal{H}(\mathcal{B}^+_0)$*);*

(2) $H^{(G_0)}_{n_0+r}$ *is uniquely step-by-step reconstructible up to isomorphism from its degree-*1 *and co-degree-*1 *vertex sets* $A_1(H) := \{v \in V(H) : \deg_H(v) = 1\}$ *and* $B_1(H) := \{v \in V(H) : \overline{\deg}_H(v) = 1\}$*; and*

(3) $H^{(G_0)}_{n_0+r} \in \mathcal{H}(\mathcal{B}(G_0))$*, so whenever* $G_0, G_0 - x_0 \in \mathcal{EH}$*, both* $H^{(G_0)}_{n_0+r}$ *and* $\overline{H^{(G_0)}_{n_0+r}}$ *have the Erdős–Hajnal property, satisfy the polynomial Rödl property, and are viral.*

Proof. Observe first that if G_* is prime on $m_0 \geq 4$ vertices and $w_0 \in V(G_*)$ satisfies $\deg_{G_*}(w_0) \geq 1$ and $N_{G_*}(z) \neq \{w_0\}$ for all $z \in V(G_*) \setminus \{w_0\}$, then adding u with $N_{G'_*}(u) = \{w_0\}$ preserves primality: any proper module M of G'_* with $|M| \geq 2$ would have $u \in M$ (else $M = V(G_*)$ contradicts $\deg_{G'_*}(u) = 1 < 4 \leq m_0$, while if $u \notin M$ and $M \subsetneq V(G_*)$ then M is a proper module of G_* of size ≥ 2) and hence $M = \{u, z\}$ with $N_{G_*}(z) = \{w_0\} \setminus \{z\}$, contradicting $\deg_{G_*}(w_0) \geq 1$ if

$z = w_0$ and $N_{G_*}(z) \neq \{w_0\}$ if $z \neq w_0$. By complementation, adding a co-leaf u with $N_{\overline{G_*'}}(u) = \{w_0\}$ preserves primality whenever $\overline{\deg}_{G_*}(w_0) \geq 1$ and $N_{\overline{G_*}}(z) \neq \{w_0\}$ for all $z \in V(G_*) \setminus \{w_0\}$.

Write $H_r := H_{n_0+r}^{(G_0)}$; clearly $G_0 = H_0 \subseteq_{\text{ind}} H_r$, so H_r is non-split whenever G_0 is non-split, and pairwise non-isomorphism across $r \geq 0$ holds since $|H_r| = n_0 + r$. Because G_0 is prime on $n_0 \geq 6$ vertices, G_0 has no isolated vertices ($\delta(G_0) = 1$), while $\delta(\overline{G_0}) = n_0 - 1 - \Delta(G_0) \geq 2$ gives $B_1(H_0) = \emptyset$ and $x_0 \in A_1(H_0)$. By induction on $r \geq 1$, every H_r is prime and has disjoint $(|A_1(H_r)|, |B_1(H_r)|)$ signatures:

- At $r = 1$ (co-leaf u_{n_0+1} with $\overline{H_1}$-neighbor x_0): $\overline{\deg}_{G_0}(z) \geq 2$ for all $z \in V(G_0)$ implies H_1 is prime with $A_1(H_1) = \{x_0\}$, $B_1(H_1) = \{u_{n_0+1}\}$, and $x_0 u_{n_0+1} \notin E(H_1)$.
- At even $r \geq 2$ (leaf u_{n_0+r} with H_r-neighbor y_{r-1}): in H_{r-1}, $\deg_{H_{r-1}}(z) \geq 2$ for all $z \neq x_{r-1}$ while $x_{r-1}y_{r-1} \notin E(H_{r-1})$ (so $N_{H_{r-1}}(x_{r-1}) \neq \{y_{r-1}\}$), implying H_r is prime; moreover, every $z \in V(H_{r-1}) \setminus \{y_{r-1}\}$ has $\overline{\deg}_{H_r}(z) = \overline{\deg}_{H_{r-1}}(z) + 1 \geq 2$, yielding $B_1(H_r) = \{y_{r-1}\}$ and $A_1(H_r) = \{x_{r-1}, u_{n_0+r}\}$ with $x_{r-1}y_{r-1} \notin E(H_r)$ and $u_{n_0+r}y_{r-1} \in E(H_r)$, so u_{n_0+r} is the unique vertex in $A_1(H_r)$ adjacent to y_{r-1}.
- At odd $r \geq 3$ (co-leaf u_{n_0+r} with $\overline{H_r}$-neighbor x_{r-1}): in H_{r-1}, $\overline{\deg}_{H_{r-1}}(z) \geq 2$ for all $z \neq y_{r-2}$ while $y_{r-2}x_{r-1} \in E(H_{r-1})$ (so $N_{\overline{H_{r-1}}}(y_{r-2}) \neq \{x_{r-1}\}$), implying H_r is prime; moreover, every $z \in V(H_{r-1}) \setminus \{x_{r-1}\}$ has $\deg_{H_r}(z) = \deg_{H_{r-1}}(z) + 1 \geq 2$, yielding $A_1(H_r) = \{x_{r-1}\}$ and $B_1(H_r) = \{y_{r-2}, u_{n_0+r}\}$ with $y_{r-2}x_{r-1} \in E(H_r)$ and $u_{n_0+r}x_{r-1} \notin E(H_r)$, so u_{n_0+r} is the unique vertex in $B_1(H_r)$ non-adjacent to x_{r-1}.

Thus $(|A_1(H_r)|, |B_1(H_r)|) = (|A_1(G_0)|, 0), (1,1), (2,1), (1,2)$ for $r = 0$, $r = 1$, even $r \geq 2$, and odd $r \geq 3$, reconstructing $H_{r-1} = H_r - u_{n_0+r}$ uniquely at every step $r \geq 1$. Since $\overline{G_0} \subseteq_{\text{ind}} \overline{H_r}$ is prime of order $n_0 \geq 6$ with $\delta(\overline{G_0}) \geq 2$, whereas every prime induced subgraph of order ≥ 6 in any member of $\mathcal{J}(\mathcal{B}_0^+)$ has $\delta \leq 1$, we have $\overline{H_r} \notin \mathcal{J}(\mathcal{B}_0^+)$ and $H_r \notin \mathcal{H}(\mathcal{B}_0^+)$. Finally, setting $H_{-1} := G_0 - x_0$, strong induction on $r \geq 0$ shows $H_r, \overline{H_r} \in \mathcal{J}(\mathcal{B}(G_0))$: $H_0, \overline{H_0} \in \mathcal{B}(G_0)$ holds directly; for odd $r \geq 1$, $\overline{\deg}_{H_r}(u_{n_0+r}) = 1$ with $\overline{H_r} - u_{n_0+r} \cong \overline{H_{r-1}}$ gives $\overline{H_r} \in \mathcal{J}(\mathcal{B}(G_0))$, while $\deg_{H_r}(x_{r-1}) = 1$ with $H_r - x_{r-1} \cong K_2[1 \leftarrow H_{r-2}, 2 \leftarrow \{u_{n_0+r}\}_{K_1}]$ ($r - 2 \geq -1$) gives $H_r \in \mathcal{J}(\mathcal{B}(G_0))$; and for even $r \geq 2$, $\deg_{H_r}(u_{n_0+r}) = 1$ with $H_r - u_{n_0+r} \cong H_{r-1}$ gives $H_r \in \mathcal{J}(\mathcal{B}(G_0))$, while $\overline{\deg}_{H_r}(y_{r-1}) = 1$ with $\overline{H_r} - y_{r-1} \cong K_2[1 \leftarrow \overline{H_{r-2}}, 2 \leftarrow \{u_{n_0+r}\}_{K_1}]$ ($r - 2 \geq 0$) gives $\overline{H_r} \in \mathcal{J}(\mathcal{B}(G_0))$. Proposition 2.4(3) then proves (3). □

3 Blockade Regularization and the Symmetric Two-Stage Reduction

For disjoint non-empty subsets $A, B \subseteq V(G)$ and $x \in [0,1]$, write $E_G(A,B) := \{ab \in E(G) : a \in A, b \in B\}$. We say B is *pointwise x-sparse to A* in G if $|N_G(b) \cap A| \leq x|A|$ for every $b \in B$, (A,B) is *two-sided x-sparse* in G if A is pointwise x-sparse to B and B is pointwise x-sparse to A, (A,B) is *complete-or-x-sparse* if (A,B) is complete or two-sided x-sparse in G, (A,B) is *weakly x-sparse* if $|E_G(A,B)| \leq x|A||B|$, (A,B) is *weakly x-co-sparse* if $|E_{\overline{G}}(A,B)| \leq x|A||B|$, and (A,B) is *weakly x-sparse-or-co-sparse* if it is weakly x-sparse or weakly x-co-sparse in G. A blockade $\mathcal{B} = (B_1, \ldots, B_\ell)$ in G is *equal-sized* if $|B_1| = \cdots = |B_\ell|$, *anticonnected* if $G[B_i]$ is anticonnected for every $i \in [\ell]$, and *pairwise-pure*, *complete*, *anticomplete*, *complete-or-x-sparse*, or *weakly x-sparse-or-co-sparse* if every pair (B_i, B_j) $(1 \leq i < j \leq \ell)$ has that property; it is *pure* if it is complete or anticomplete. Given a graph J on $[\ell]$ (a *pattern graph* of $\mathcal{B}$) and subsets $X \subseteq B_i, Y \subseteq B_j$ $(i \neq j)$, we write $Q_J(X,Y) := E_G(X,Y)$ when $ij \notin E(J)$ and $Q_J(X,Y) := E_{\overline{G}}(X,Y)$ when $ij \in E(J)$ for the set of *wrong cross-pairs*, calling $\{v \in Y : \{u,v\} \in Q_J(X,Y)\}$ the *wrong neighbors* of $u \in X$ in Y, and

saying $\mathcal{B}$ has *pairwise defect at most* x relative to J if $|Q_J(B_i, B_j)| \le x|B_i||B_j|$ for all $1 \le i < j \le \ell$.

Definition 3.1 (Coupled Two-Stage Framework and Open-Interval Realizations)**.** Let $\mathcal{F}$ be a finite non-empty family of non-empty graphs.

(1) *(Leaf-reducibility.)* $\mathcal{F}$ is *leaf-reducible* if there exist $H_* \in \mathcal{F}$ with $|H_*| \ge 2$ and a vertex $v_0 \in V(H_*)$ with $\deg_{H_*}(v_0) \le 1$ such that $(\mathcal{F} \setminus \{H_*\}) \cup \{H_* - v_0\} \in \mathcal{EH}$.

(2) *(Coupled generalized niceness.)* $\mathcal{F}$ is *coupled generalized nice* if there exist constants $c_1, \dots, c_9 > 0$ such that for every $\varepsilon \in (0, 1/2)$, every coupled cross-block sparsity parameter $x_{\mathrm{sp}} \in [\varepsilon^{3c_2}, \varepsilon^{c_7}]$, and every $\overline{\mathcal{F}}$-free graph G, at least one of the following holds:
  (GN1) G contains a weakly x_{sp}-sparse-or-co-sparse $(\lceil \varepsilon^{-1} \rceil, \lfloor \varepsilon^{c_1}|G| \rfloor)$-blockade (whose block width depends only on ε and is independent of x_{sp});
  (GN2) $\kappa(G) \ge (\varepsilon^{c_3} x_{\mathrm{sp}}^{c_3}|G|)^{c_4}$;
  (GN3) G contains a pairwise-pure $(k, |G|/k^{c_5})$-blockade with a cograph (P_4-free) pattern graph for some integer $k \in [\lceil \varepsilon^{-c_6} \rceil, \lceil \varepsilon^{-c_9} \rceil]$; or
  (GN4) G contains a $2\varepsilon^{c_7}$-restricted induced subgraph of order at least $\varepsilon^{c_8}|G|$.

(3) *(Thresholded wonderfulness.)* $\mathcal{F}$ is *thresholded wonderful* if there exist an integer $a \ge 7$ and a constant $y_{\mathrm{thr}} \in (0, 1/4]$ such that for every $y \in (0, y_{\mathrm{thr}}]$, every $\overline{\mathcal{F}}$-free graph G, and every equal-sized anticonnected complete-or-(two-sided)-δ_{cross}-sparse (ℓ, w)-blockade $\mathcal{B} = (B_1, \dots, B_\ell)$ in G with $\ell \ge y^{-a}$, $\delta_{\mathrm{cross}} \in (0, \min(y^a, \frac{1}{8\ell^2})]$, and $w = |B_1| = \cdots = |B_\ell| \ge 1$ (where every pair (B_i, B_j) with $ij \in E(J)$ is 100% complete in G and every pair with $ij \notin E(J)$ is two-sided δ_{cross}-sparse in G), either:
  (W1) G contains a y^4-restricted induced subgraph of order at least w; or
  (W2) there exists $i^* \in [\ell]$ such that at most $y|G|$ vertices $v \in V(G) \setminus V(\mathcal{B})$ satisfy $0 < |N_G(v) \cap B_{i^*}| < \frac{1}{2}|B_{i^*}|$.

(4) *(Open-interval realization of a witness pair (J_0, S_{J_0}).)* Given a finite graph J_0 and a special vertex subset $S_{J_0} \subseteq V(J_0)$, an *open-interval realization* of (J_0, S_{J_0}) in a graph G around an external vertex $v_0 \in V(G)$ is a family of distinct vertices $\{v_u : u \in V(J_0)\} \subseteq V(G) \setminus \{v_0\}$ such that v_0 is anticomplete in $\overline{G}$ to $\{v_u : u \in S_{J_0}\}$ and complete in $\overline{G}$ to $\{v_u : u \in V(J_0) \setminus S_{J_0}\}$, and for all distinct $x, u \in V(J_0)$:
  (E-Non) if $xu \in E(J_0)$, then $v_x v_u \notin E(\overline{G})$ ($v_x v_u \in E(G)$);
  (E-Edge) if $xu \notin E(J_0)$ and $\{x, u\} \not\subseteq S_{J_0}$, then $v_x v_u \in E(\overline{G})$ ($v_x v_u \notin E(G)$).

(In [16, Sections 2.1 and 3], generalized niceness fixes a single defect ε^{c_2} on complete-or-weakly-sparse blockades; allowing any coupled parameter $x_{\mathrm{sp}} \in [\varepsilon^{3c_2}, \varepsilon^{c_7}]$ with x_{sp}-independent block width $\lfloor \varepsilon^{c_1}|G| \rfloor$ in (GN1) enables Theorem 3.5 to purify weakly co-sparse pairs to 100% completeness via virality before invoking (3).)

We next establish from first principles the blockade regularization, cograph refinement, symmetric layout refinement, upside-down comb extraction, and multi-scale iteration tools underlying Sections 3 and 4.

Lemma 3.2 (Blockade Regularization, Layout Refinement, Comb Extraction, and Stage-1 Reduction; adapted from [16, 19, 20, 21])**.**

(1) (Anticonnected blockade dichotomy and equal-sized cograph trigger peeling.) *For every non-empty graph G and integer $k \ge 2$, either G has an anticonnected component of size $\ge |G|/k$ or G contains a complete $(k, |G|/k^2)$-blockade. Furthermore, for $a, b \ge 1$, $c := 2^{-4b}$, $x \ge 0$, $0 < y \le c$, and any graph F_0 with $w := \lfloor y^{a+2}|F_0| \rfloor \ge 1$, if every $F \subseteq F_0$ with $|F| \ge c|F_0|$ contains disjoint $X, Y \subseteq V(F)$ with $|Y| \ge (1 - by)|F|$ where either* (a) *$|X| = w$ and Y is*

complete or pointwise x-sparse to X in F, or (b) $|X| \ge y^a|F|$ *and (X,Y) is pure in F, then F_0 contains an equal-sized $(\lceil y^{-1}\rceil, w)$-blockade with a cograph (P_4-free) pattern graph J whose edges are complete in F_0 and whose non-edges ($i<j$) are pointwise x-sparse from B_j to B_i (hence pairwise-pure when $x=0$).*

(2) (Defective cograph refinement and x_{sp}-independent symmetric layout refinement.)

(a) *Let $C \ge 1$, $\varepsilon \in (0, 1/8]$, $x_{\mathrm{def}} \in [0, \varepsilon^{12C}]$, and $|G| \ge 4\varepsilon^{-5C}$. If every $F \subseteq G$ with $|F| \ge \frac{1}{2}\varepsilon^{2C}|G|$ contains either a 2ε-restricted subgraph of order $\ge \varepsilon^{5C}|G|$, a cograph $(k, |F|/k^C)$-blockade ($2 \le k < \lceil\varepsilon^{-2}\rceil$) with pairwise defect $\le x_{\mathrm{def}}$, or an equal-sized cograph (k, w_0)-blockade ($k \ge \lceil\varepsilon^{-2}\rceil$, $w_0 \ge \varepsilon^{5C}|G|$) with pairwise defect $\le \frac{1}{8}\varepsilon^4$, then G contains a 2ε-restricted induced subgraph of order $\ge \varepsilon^{5C}|G|$.*

(b) *Let $\varepsilon \in (0, 1/2)$, $d \ge 1$, $C_{\mathrm{cap}} \ge 2$, $K_{\max} := \lceil\varepsilon^{-20dC_{\mathrm{cap}}}\rceil$, $x_{\mathrm{sp}} \ge 0$, and $x_{\mathrm{in}} := x_{\mathrm{sp}}\varepsilon^{4d}$. For any non-empty graph G, if every induced subgraph $F \subseteq G$ with $|F| \ge \varepsilon^d|G|$ contains a weakly x_{in}-sparse-or-co-sparse $(k, |F|/k^d)$-blockade for some $k \in [2, K_{\max}]$, then G contains a weakly x_{sp}-sparse-or-co-sparse $(\lceil\varepsilon^{-1}\rceil, \lfloor\varepsilon^{(20C_{\mathrm{cap}}+4)d^2}|G|\rfloor)$-blockade whose block width is completely independent of x_{sp}.*

(3) (Upside-down comb extraction and multi-scale iteration.) *For $0 < x \le y \le 2^{-8}$ and any y^3-sparse graph F with $|F| \ge y^{-4}$, either* (i) *F is $2y^4$-sparse,* (ii) *F contains disjoint $X, Y \subseteq V(F)$ with $|X| \ge y^4|F|$, $|Y| \ge (1-4y)|F|$, and Y pointwise x-sparse to X in F, or* (iii) *F contains an $(\ell, y^4|F|/\ell^2)$-comb with an external apex $v \in V(F)$ for some integer $\ell \in [\lceil\frac{1}{2}y^{-1}\rceil, \lfloor x^{-1}\rfloor]$. Moreover, if $0 < x < \lambda_0 < 1/2$, $b_1 > 1$, $(b_1-1)b_2 \ge b_3 > 0$, F_0 is λ_0-restricted, and every λ-restricted $F \subseteq F_0$ ($\lambda \in [x, \lambda_0]$, $|F| \ge \lambda^{b_2}|F_0|$) failing property $\mathcal{O}$ contains a λ^{b_1}-restricted subgraph of order $\ge \lambda^{b_3}|F|$, then F_0 either has an induced subgraph with $\mathcal{O}$ or contains an x-restricted subgraph of order $\ge x^{b_1 b_2}|F_0|$.*

(4) (Stage-1 comb-dichotomy reduction to coupled generalized niceness.) *Let $\mathcal{F}$ be a finite non-empty family of non-empty graphs that either contains P_t ($t \ge 2$, setting $d' := 0$) or is leaf-reducible with $|H_*| \ge 3$, $\deg_{H_*}(v_0) = 1$, and leaf-reduced viral exponent $d' \ge 4$. Suppose there exist constants $C_{\mathrm{comb}} \ge 18$, $C_{\mathrm{cap}} \ge 2$, and $c_{\mathrm{A}}, c_{\mathrm{B}} > 0$ such that for every $0 < x \le y \le 2^{-16}$, every $\overline{\mathcal{F}}$-free graph F, and every (ℓ, w)-comb in F with an external apex v ($\ell \in [\lceil\frac{1}{2}y^{-1}\rceil, \lfloor x^{-1}\rfloor]$, $w \ge y^4|F|/\ell^2$), either* (CD1) *$\kappa(F) \ge w^{c_{\mathrm{A}}}$,* (CD2) *$F$ contains a pairwise-pure $(k, |F|/k^{C_{\mathrm{comb}}})$-blockade for some $k \in [\max(2, \lceil\frac{1}{2}\ell^{c_{\mathrm{B}}}\rceil), p_{\mathrm{cap}}]$ ($p_{\mathrm{cap}} := \lceil x^{-C_{\mathrm{cap}}}\rceil$), or* (CD3) *$F$ contains a weakly x-sparse-or-co-sparse $(k, |F|/k^{C_{\mathrm{comb}}})$-blockade for some $k \in [\max(2, \lceil\frac{1}{2}\ell^{c_{\mathrm{B}}}\rceil), p_{\mathrm{cap}}]$. Then $\mathcal{F}$ is coupled generalized nice (Definition 3.1(2)) with $c_2 = d \ge \max(120, 4d'|H_*| + 8)$ and $3c_2 \ge c_7 = 6$.*

Proof. In (1), if every anticonnected component C_s of G satisfies $|C_s| < |G|/k$, grouping consecutive components greedily into $k-1$ minimal prefix unions of size $\ge |G|/k^2$ (each $< (\frac{1}{k} + \frac{1}{k^2})|G|$) leaves a k-th union of size $> |G|/k^2$, forming a complete $(k, |G|/k^2)$-blockade. For cograph peeling, since $(1-by)^{\lceil y^{-1}\rceil - 1} \ge 2^{-4b} = c$, repeatedly peeling (X_i, Y_i) from R_{i-1} ($R_0 = V(F_0)$, $R_i = Y_i$) for $i = 1, \dots, \lceil y^{-1}\rceil$ preserves $|B_i| = w$ with zero sub-sampling loss in case (a); each vertex $i \in [\lceil y^{-1}\rceil]$ is pure in J to $\{i+1, \dots, \lceil y^{-1}\rceil\}$, so J is P_4-free.

In (2), Seinsche's theorem [23] gives $\omega(J)\alpha(J) \ge |V(J)|$ for every cograph J. Maintaining $\mathcal{L} = (J, (A_j : j \in V(J)))$ from $(K_1, (V(G)))$ with $|A_j| \ge \varepsilon^{4C}|G|$, $\sum_j |A_j|^{1/C} \ge |G|^{1/C}$, and $\sum_{i \ne j} |Q_J(A_i, A_j)| \le x_{\mathrm{def}}(|G|^2 - \sum_j |A_j|^2)$, splitting a largest block A_{j_0} ($|A_{j_0}| \ge \varepsilon^{2C}|G|$) into $2 \le k < \lceil\varepsilon^{-2}\rceil$ blocks preserves $|B_p| \ge |A_{j_0}|/k^C \ge \varepsilon^{4C}|G|$ and P_4-freeness of J until $|V(J)| \ge \lceil\varepsilon^{-2}\rceil$. Choosing a clique or stable set $I \subseteq V(J)$ of size $r := \lceil\varepsilon^{-1}\rceil$, sub-sampling w_0-subsets $X_i \subseteq A_i$ ($w_0 := \lceil\varepsilon^{5C}|G|\rceil \le \varepsilon^{4C}|G|$), deleting vertices with $> \varepsilon w_0$ wrong neighbors across $\bigsqcup_{j \in I \setminus \{i\}} X_j$, and

equalizing each surviving block to exact size $w' := \lceil \frac{1}{2} w_0 \rceil \geq 1$ yields S of order $rw' \geq \varepsilon^{5C}|G|$ with wrong degree $\leq (w'-1) + \varepsilon w_0 \leq 2\varepsilon|S|$, proving (a). Maximizing $|V(J)| < \lceil \varepsilon^{-1} \rceil$ under $|A_j| \geq \varepsilon^{2d}|G|$, $\sum_j |A_j|^{1/d} \geq |G|^{1/d}$, and $\sum_{i \neq j} |Q_J(A_i, A_j)| \leq x_{\mathrm{sp}} \varepsilon^{4d}(|G|^2 - \sum_j |A_j|^2)$ cancels ε^{4d} against $|A_i||A_j| \geq \varepsilon^{4d}|G|^2$ once $|V(J)| \geq \lceil \varepsilon^{-1} \rceil$ (and if a split has arity $k \in [\lceil \varepsilon^{-1} \rceil, K_{\max}]$, $k \leq \varepsilon^{-(20dC_{\mathrm{cap}}+1)}$ gives $|A_{j_0}|/k^d \geq \varepsilon^{(20C_{\mathrm{cap}}+4)d^2}|G|$), proving (b).

In (3), if (i) and (ii) fail on a y^3-sparse graph F, some $v \in V(F)$ has $|A_0| > 2y^4|F|$ ($A_0 := N_F(v)$), and $\geq (1-4y)|F|$ non-neighbors $R_1 \subseteq V(F) \setminus N_F[v]$ satisfy $|N_F(u) \cap A_0| \leq \frac{1}{2} y^2 |A_0|$. Greedily removing maximum increments $D_i := N_F(a_i) \cap A^{(i)}$ ($a_i \in R_1$, $|D_i| > x|A^{(i)}|$, $|A^{(i)}| \geq y^4|F|$) removes $> \frac{1}{2}|A_0|$ vertices; picking a dyadic bucket J_t ($2^{-(t+1)}W < |D_i| \leq 2^{-t}W$, $W = \frac{1}{2}y^2|A_0|$) with $\sum_{i \in J_t} |D_i| > \frac{1}{4(t+1)^2}|A_0|$, pruning J_t in reverse order into L ($|B_q| \geq \frac{1}{2}|D_q|$, $|L| > \frac{1}{9}y^{-2} > 2y^{-1}$), and truncating to $\ell := \min(|L|, \lfloor x^{-1} \rfloor) \in [\lceil \frac{1}{2} y^{-1} \rceil, \lfloor x^{-1} \rfloor]$ gives $|B_q| \geq \frac{y^4|F|}{\ell^2}$ both when $\ell = |L|$ ($4 \cdot 2^t \leq y^2\ell^2$) and when $\ell = \lfloor x^{-1} \rfloor$ ($\frac{1}{2}xy^4|F| \geq \frac{y^4|F|}{\ell^2}$). Multi-scale iteration follows along $\lambda_t = \lambda_0^{b_1^t}$ since $(b_1 - 1)b_2 \geq b_3$ preserves $|F_{t+1}| \geq \lambda_{t+1}^{b_2}|F_0|$.

In (4), when $\overline{F'}$ is y^3-sparse and $\overline{\mathcal{F}}$-free, either minimal trigger-set expansion in $\overline{F'}$ (when $P_t \in \mathcal{F}$, [19, 3.1]) or virality of $\overline{\mathcal{F}^-}$ at scale y^4 produces either a y^4-restricted subgraph or a complete pair (X, Y) with $|X| \geq y^{a_{\mathrm{leaf}}}|F'|$ ($a_{\mathrm{leaf}} := 4d'|H_*| + 6$) and $|Y| \geq (1 - b_{\mathrm{leaf}} y)|F'|$. Running (3) from $\lambda_0 = c^{10}$ down to $c\varepsilon^6$ on any $F \subseteq G$ ($|F| \geq \varepsilon^d|G|$) at comb defect $x_3 := (x_{\mathrm{sp}}\varepsilon^{4d})^2$ and cap $p_{\mathrm{cap}} := \lceil x_3^{-C_{\mathrm{cap}}} \rceil \leq K_{\max}(\varepsilon)$ (averaging w_1-subsets $X \subseteq X_0$ of exact size $w_1 := \lfloor y^{a_{\mathrm{leaf}}+2}|F_t| \rfloor$ and Markov-pruning Y_0 to restore pointwise x_{in}-sparsity when (3)(ii) occurs) yields a weakly x_{in}-sparse-or-co-sparse $(k, |F|/k^d)$-blockade ($k \in [2, K_{\max}(\varepsilon)]$) for sufficiently large d, and (2b) gives (GN1). □

We now prove the Sequential Open-Interval Realization Lemma (covering both zero-cut pure-substitution witnesses and three-stage isolated-block cut-edge witnesses, together with two-polarity pendant-P_3 lifting) and deduce the Master Open-Interval Witness Reduction (Proposition 3.4).

Lemma 3.3 (Sequential Open-Interval Realization and Pendant-P_3 Lifting)**.** *Let G be a graph, let $\mathcal{B} = (B_1, \ldots, B_\ell)$ be an equal-sized anticonnected complete-or-(two-sided)-δ_{cross}-sparse (ℓ, w)-blockade in G ($w \geq 1$) with pattern graph J on $[\ell]$, and for $v_0 \in V(G) \setminus V(\mathcal{B})$ set $I_0(v_0) := \{i \in [\ell] : 0 < |N_G(v_0) \cap B_i| < \frac{1}{2}|B_i|\}$.*

(1) (Sequential open-interval realization and isolated-block cut-edge extension.) *Suppose J_0 is a non-empty finite graph with $\delta_{\mathrm{cross}} \leq \frac{1}{4|V(J_0)|}$, vertex partition $V(J_0) = S \sqcup U_{\mathrm{cut}} \sqcup W_{\mathrm{seq}}$ where U_{cut} is a clique in J_0 complete in J_0 to S, and $\{j_u : u \in V(J_0)\} \subseteq I_0(v_0)$ induces J_0 in $J[I_0(v_0)]$. Then G contains distinct vertices $\{a_s : s \in S\} \cup \{c_p, a_p : p \in U_{\mathrm{cut}}\} \cup \{c_w : w \in W_{\mathrm{seq}}\}$ ($a_s \in N_G(v_0) \cap B_{j_s}$, $c_p a_p \in E(\overline{G}[B_{j_p}])$ with $c_p \in B_{j_p} \setminus N_G(v_0)$ and $a_p \in N_G(v_0) \cap B_{j_p}$, and $c_w \in B_{j_w} \setminus N_G(v_0)$) such that on $\mathcal{X}_{\mathrm{all}} := \{v_0\} \cup \{a_s\} \cup \{c_p, a_p\} \cup \{c_w\}$, v_0 is anticomplete in $\overline{G}$ to $\{a_s : s \in S\} \cup \{a_p : p \in U_{\mathrm{cut}}\}$ and complete in $\overline{G}$ to $\{c_p : p \in U_{\mathrm{cut}}\} \cup \{c_w : w \in W_{\mathrm{seq}}\}$, and for all distinct $x, y \in \mathcal{X}_{\mathrm{all}} \setminus \{v_0\}$, $xy \in E(\overline{G}) \iff \mathrm{blk}(x)\,\mathrm{blk}(y) \notin E(J_0)$ except possibly inside $\{a_s a_{s'} : \{s, s'\} \in E_{\mathrm{non}}(S)\}$. In particular, when $U_{\mathrm{cut}} = \emptyset$ ($S_{J_0} = S$), setting $v_s := a_s$ ($s \in S_{J_0}$) and $v_w := c_w$ ($w \in V(J_0) \setminus S_{J_0}$) produces an open-interval realization of (J_0, S_{J_0}) around v_0 satisfying* (E-Non) *and* (E-Edge) *for every $w \geq 1$.*

(2) (Two-polarity pendant-P_3 open-interval lifting.) *Let $H_* = G_0 + (y - w_2 - x_2)$ ($h := |H_*| \geq 4$) have degree-1 leaf x_2 adjacent to degree-2 vertex w_2 ($N_{H_*}(w_2) = \{y, x_2\}$), let F' be $\overline{H_*}$-free, let $\eta_{\mathrm{low}} \in (0, 1/8]$, and let $\mathcal{B}' = (B'_1, \ldots, B'_\ell)$ ($|B'_i| = w_* \geq 2$) have pattern graph J' with $|Q_{J'}(B'_p, B'_q)| \leq \frac{\eta_{\mathrm{low}}^2}{64h^2} w_*^2$ for all $p \neq q$. Then for every $v_0 \in V(F') \setminus V(\mathcal{B}')$, $J'[\{i \in [\ell] : \eta_{\mathrm{low}} w_* <$*

$|N_{F'}(v_0) \cap B'_i| < \frac{1}{2}w_*\}]$ *is* $\overline{H_* - x_2}$*-free, and* $J'[\{i \in [\ell] : \frac{1}{2}w_* \le |N_{F'}(v_0) \cap B'_i| < (1-\eta_{\mathrm{low}})w_*\}]$ *is* $K_2[1 \leftarrow K_1, 2 \leftarrow \overline{G_0}]$*-free.*

Proof. In (1), for each $s \in S$, $|N_G(v_0) \cap B_{j_s}| \ge 1$ allows picking any $a_s \in N_G(v_0) \cap B_{j_s}$; for each $p \in U_{\mathrm{cut}}$, connectivity of $\overline{G}[B_{j_p}]$ across the non-empty cut $(B_{j_p} \setminus N_G(v_0)) \sqcup (N_G(v_0) \cap B_{j_p})$ yields an edge $c_p a_p \in E(\overline{G}[B_{j_p}])$ across the cut, and 100% completeness in G of all edges of J_0 incident to U_{cut} forces zero $\overline{G}$-edges between distinct blocks of U_{cut} and between U_{cut} and S. Ordering W_{seq} linearly, at each $u \in W_{\mathrm{seq}}$ at most $2|V(J_0)|$ vertices $\mathcal{X}_{<u} \subseteq \mathcal{X}_{\mathrm{all}} \setminus \{v_0\}$ precede u, so $|N_G(v_0) \cap B_{j_u}| + \sum_{x \in \mathcal{X}_{<u}:\, \mathrm{blk}(x)u \notin E(J_0)} |N_G(x) \cap B_{j_u}| < \frac{1}{2}w + 2|V(J_0)|\delta_{\mathrm{cross}} w \le w = |B_{j_u}|$ for every $w \ge 1$, allowing a valid choice $c_u \in (B_{j_u} \setminus N_G(v_0)) \setminus \bigcup_{x \in \mathcal{X}_{<u}:\, \mathrm{blk}(x)u \notin E(J_0)} N_G(x)$.

In (2), for any $(h-1)$-tuple $(j_1, \dots, j_{h-1})$, Markov's inequality deletes at most $\frac{\eta_{\mathrm{low}}}{16h} w_*$ vertices in B'_{j_q} having $\ge \frac{\eta_{\mathrm{low}}}{4h} w_*$ wrong neighbors in each fixed future block B'_{j_m} $(q < m \le h-1)$, removing $\le (h-2)\frac{\eta_{\mathrm{low}}}{16h} w_* < \frac{1}{16}\eta_{\mathrm{low}} w_*$ vertices in total per block. At step m, avoiding this set and the $< \frac{\eta_{\mathrm{low}}}{4h} w_*$ wrong neighbors of each previously chosen a_q $(q < m)$ removes $< (\frac{1}{16} + \frac{1}{4})\eta_{\mathrm{low}} w_* < \eta_{\mathrm{low}} w_*$ vertices, leaving a candidate in $B'_{j_{w_2}} \setminus N_{F'}(v_0)$ for w_2 and in $N_{F'}(v_0) \cap B'_{j_u}$ for $u \ne w_2$ (with $x_2 \mapsto v_0$) on the lower interval, and in $B'_{j_u} \setminus N_{F'}(v_0)$ for $u \in \{x_2, y\}$ and in $N_{F'}(v_0) \cap B'_{j_u}$ for $u \in V(G_0) \setminus \{y\}$ (with $w_2 \mapsto v_0$) on the upper interval. □

Proposition 3.4 (Master Open-Interval Witness Reduction for Thresholded Wonderfulness)**.** *Let* H_* *be a finite graph. Suppose* $J_0 \in \mathcal{EH}$ *is a non-empty graph with Erdős–Hajnal exponent* $c_{J_0} \in (0, 1/2]$ *(or more generally* $\{H_*, J_0\} \in \mathcal{EH}$ *with pair exponent* $c_{J_0} \in (0, 1/2]$ *and* $\overline{H_*} \subseteq_{\mathrm{ind}} G$ *excluded by* $|H_*| \le |V(J_0)|$*) and vertex partition* $V(J_0) = S_{J_0} \sqcup (V(J_0) \setminus S_{J_0})$ *(or* $V(J_0) = S \sqcup U_{\mathrm{cut}} \sqcup W_{\mathrm{seq}}$ *with* U_{cut} *a clique complete to* S *in* J_0*) such that every open-interval realization* $\{v_u : u \in V(J_0)\}$ *of* (J_0, S_{J_0}) *around* v_0 *(respectively, every cut-edge realization* $\mathcal{X}_{\mathrm{all}}$ *from Lemma 3.3(1)) in any graph* G *contains an induced copy of* H_* *in* $\overline{G}[\{v_0\} \cup \{v_u : u \in V(J_0)\}]$ *(respectively,* $\overline{G}[\mathcal{X}_{\mathrm{all}}]$*). Then* $\{H_*\}$ *is thresholded wonderful (Definition 3.1(3)) with explicit constants*

$$a_{J_0} := \max\Big(10,\ \lceil 6/c_{J_0} \rceil + 2,\ \lceil \log_2(8\,|V(J_0)|) \rceil + 6\Big) \quad \text{and} \quad y_{\mathrm{thr}}(J_0) := 2^{-8a_{J_0}} \in \big(0, \tfrac{1}{4}\big].$$

Proof. Let $y \in (0, y_{\mathrm{thr}}(J_0)]$, let G be any $\overline{H_*}$-free graph, and let $\mathcal{B} = (B_1, \dots, B_\ell)$ be an equal-sized anticonnected complete-or-(two-sided)-δ_{cross}-sparse (ℓ, w)-blockade in G with $\ell \ge y^{-a_{J_0}}$, $w \ge 1$, $\delta_{\mathrm{cross}} \in (0, \min(y^{a_{J_0}}, \frac{1}{8\ell^2})]$, and pattern graph J on $[\ell]$. Since $4|V(J_0)|\delta_{\mathrm{cross}} \le 4|V(J_0)|y_{\mathrm{thr}}(J_0) < 1$, Lemma 3.3(1) implies that if $J[I_0(v_0)]$ contained an induced copy of J_0 for some $v_0 \in V(G) \setminus V(\mathcal{B})$, then $\overline{G}$ would contain an induced copy of H_*, contradicting that G is $\overline{H_*}$-free. Thus $J[I_0(v_0)]$ is J_0-free (and similarly $\overline{H_*}$-free whenever $|H_*|\delta_{\mathrm{cross}} \le \frac{1}{2}$) for every $v_0 \in V(G) \setminus V(\mathcal{B})$.

If some $v_0 \in V(G) \setminus V(\mathcal{B})$ satisfies $|I_0(v_0)| \ge y\ell$, then since $J_0 \in \mathcal{EH}$ has exponent c_{J_0} and $(y\ell)^{c_{J_0}} \ge y^{-(a_{J_0}-1)c_{J_0}} \ge y^{-6} > \lceil 2y^{-4} \rceil =: m_{\mathrm{hom}}$, $J[I_0(v_0)]$ contains a clique or stable set $I_{\mathrm{hom}} \subseteq I_0(v_0)$ of size m_{hom}. Setting $U := G[\bigsqcup_{p \in I_{\mathrm{hom}}} B_p]$ $(|U| = m_{\mathrm{hom}} w \ge w)$: if I_{hom} is a clique in J, every cross-pair in U is 100% complete in G, so $\Delta(\overline{G}[U]) \le w - 1 < \frac{2}{m_{\mathrm{hom}}}|U| \le y^4|U|$; if I_{hom} is stable in J, every cross-pair in U is two-sided δ_{cross}-sparse in G with $(m_{\mathrm{hom}} - 1)\delta_{\mathrm{cross}} < 3y^{-4}y^{a_{J_0}} < 1$, so $\Delta(G[U]) \le (w-1) + (m_{\mathrm{hom}} - 1)\delta_{\mathrm{cross}} w < 2w \le y^4|U|$. Hence U satisfies (W1). Otherwise $|I_0(v)| < y\ell$ for every $v \in V(G) \setminus V(\mathcal{B})$, and averaging $\sum_{i=1}^{\ell} |\{v \notin V(\mathcal{B}) : i \in I_0(v)\}| = \sum_{v \notin V(\mathcal{B})} |I_0(v)| < y\ell|G|$ over $i \in [\ell]$ yields $i^* \in [\ell]$ satisfying (W2). □

Finally, we prove the Symmetric Two-Stage Master Theorem combining leaf-reducibility, coupled generalized niceness, and thresholded wonderfulness via one-shot viral co-sparse-to-100%-complete purification.

Theorem 3.5 (Symmetric Two-Stage Master Theorem; adapted from [16, Lemma 1.12 and Section 3]). *Every finite non-empty family $\mathcal{F}$ of non-empty graphs that is leaf-reducible, coupled generalized nice, and thresholded wonderful has the Erdős–Hajnal property, satisfies the polynomial Rödl property, and is viral (along with $\overline{\mathcal{F}}$). In particular, if H is a finite non-empty graph whose singleton family $\{H\}$ is leaf-reducible, coupled generalized nice, and thresholded wonderful, then both H and $\overline{H}$ belong to $\mathcal{EH}$, satisfy the polynomial Rödl property, and are viral.*

Proof. Let $H_* \in \mathcal{F}$ have $v_0 \in V(H_*)$ with $\deg_{H_*}(v_0) \le 1$ and $\mathcal{F}^- := (\mathcal{F} \setminus \{H_*\}) \cup \{H_* - v_0\} \in \mathcal{EH}$. If $|H_*| \le 2$, then $H_* \in \{K_1, K_2, \overline{K_2}\}$ yields $\mathcal{F} \in \mathcal{EH}$ immediately; if $|H_*| \ge 3$ and $\deg_{H_*}(v_0) = 0$, writing $H_* = \overline{K_2}[1 \leftarrow H_* - v_0, 2 \leftarrow K_1]$ gives $\mathcal{F} \in \mathcal{EH}$ by Proposition 2.4(1). Assume therefore $\deg_{H_*}(v_0) = 1$ with unique neighbor $u_0 \in V(H_* - v_0)$ and $h := |H_*| \ge 3$, and let $d' \ge 4$ be a viral exponent for $\overline{\mathcal{F}^-}$ (Theorem 2.3(b)). Let $c_1, \dots, c_9 > 0$ be the constants of Definition 3.1(2) (increasing c_2, c_9 if needed so $c_2 \ge \max(120, 4d'h+8)$, $c_6 \ge 1$, $c_7 \ge 2$, $c_4 \le 1$, and $3c_2 \ge c_7$), and let $a_{\mathrm{thr}} \ge 7$ and $y_{\mathrm{thr}} \in (0, 1/4]$ be the constants of Definition 3.1(3). Set $a_{\mathrm{leaf}} := 4d'h+6$, $b_{\mathrm{leaf}} := 3h \ge 9$, $a_5 := 4+b_{\mathrm{leaf}}$, $a := \max(a_{\mathrm{thr}}, 20, 4h, \lceil 4d'h+10 \rceil)$, $c_0 := \min(2^{-8a_5}, y_{\mathrm{thr}}, 2^{-120})$, $\lambda_0 := c_0^4$, $\delta_2 := \delta(\overline{\mathcal{F}}, \lambda_0) > 0$, $a_{\mathrm{peel}} := ac_1+ac_8+4d'h+4a+12$, $b_1 := \frac{11}{10}$, $b_3 := 2ac_3(3c_2+1)+a_{\mathrm{peel}}+a_{\mathrm{leaf}}+10$, $b_2 := \max(20b_3, \lceil \frac{\log(1/\delta_2)}{\log(1/\lambda_0)} \rceil + 1)$, $K_0 := \lceil \log_2(1/c_0) \rceil + 1$, and $C_1 := (5b_2+1)K_0 + ac_9 + a_{\mathrm{peel}} + b_2 + b_3 + c_5 + 20$. We prove that every non-empty $\overline{\mathcal{F}}$-free graph G satisfies $\kappa(G) \ge |G|^c$ with $c := \frac{c_4}{84C_1^2} > 0$.

Step 1 (One-shot viral co-sparse-to-100%-complete purification on (GN1)*).* Let F_t be any λ-restricted $\overline{\mathcal{F}}$-free graph ($\lambda \in (0, c_0^4]$) with $w_y := \lfloor y^{a_{\mathrm{peel}}} |F_t| \rfloor \ge 2$ ($y := \lambda/c_0 \in (0, c_0^3]$), and set $\varepsilon_{\mathrm{GN}} := y^a$, $\ell := \lceil y^{-a} \rceil$, and coupled sparsity $x_{\mathrm{sp}} := \varepsilon_{\mathrm{GN}}^{3c_2} = y^{3ac_2} \in [\varepsilon_{\mathrm{GN}}^{3c_2}, \varepsilon_{\mathrm{GN}}^{c_7}]$. Note that any $F' \subseteq F_t$ with $|F'| \ge 2^{-4a_5}|F_t|$ satisfies $|F'| \ge c_0|F_t|$. If F_t is $(c_0 y)$-sparse ($\Delta(F') \le y|F'|$) and Definition 3.1(2) on F' at $(\varepsilon_{\mathrm{GN}}, x_{\mathrm{sp}})$ gives (GN1), we obtain a weakly x_{sp}-sparse-or-co-sparse (ℓ, m_1)-blockade $(A_1, \dots, A_\ell)$ with pattern graph J and x_{sp}-independent width $m_1 = \lfloor y^{ac_1} |F'| \rfloor \ge 4\lceil y^{-(4d'h+2)} \rceil \ell^2 w_y =: W_0$. Sequential averaging shrinks each A_i to exact size W_0 with $|Q_J(A_i'', A_j'')| \le \ell^2 x_{\mathrm{sp}} W_0^2$, and Markov pruning deletes vertices with $> 2\ell^3 x_{\mathrm{sp}} W_0$ wrong neighbors in any block $j \ne i$ and sub-samples $(A_1', \dots, A_\ell')$ of exact size $W_{\mathrm{mid}} := \frac{1}{2} W_0$ with pointwise wrong degree $\le 4\ell^3 x_{\mathrm{sp}} W_{\mathrm{mid}}$.

Applying virality of $\overline{\mathcal{F}^-}$ at scale y^4 inside each $F'[A_i']$ either finds a y^4-restricted subgraph of order $\ge y^{4d'} |A_i'| \ge w_y$, or produces inside every A_i' an induced copy Φ_i ($|\Phi_i| = h-2$) of $\overline{H_* - v_0} \setminus \{u_0\}$ and an extension core $D_i \subseteq A_i' \setminus \Phi_i$ of size $> y^{4d'(h-1)} |A_i'| \ge 4\ell^2 w_y$ whose vertices complete $\Phi_i \cup \{u\} \cong \overline{H_* - v_0}$. Since $c_2 \ge 4d'h+8$, we have $\ell h \cdot 4\ell^3 x_{\mathrm{sp}} W_{\mathrm{mid}} = 8\ell^6 h \lceil y^{-(4d'h+2)} \rceil y^{3ac_2} w_y \le y^{3a} w_y$. Deleting from each D_j all non-neighbors in $\bigcup_{i : ij \in E(J)} \Phi_i$ removes at most $y^{3a} w_y < w_y$ vertices, leaving $|D_j'| > 2\ell^2 w_y$ such that every edge $ij \in E(J)$ is 100% complete between D_i' and D_j' (any non-edge $uv \notin E(F')$ with $u \in D_i', v \in D_j'$ would induce $\Phi_i \cup \{u, v\} \cong \overline{H_*}$ in F'). Applying Lemma 3.2(1) with parameter ℓ to each $F'[D_j']$ either produces in some $F'[D_{j_0}']$ a complete $(\ell, 2w_y)$-blockade (whose equalized block union of order $\ell w_y \ge w_y$ is y^a-co-sparse, hence y^4-restricted), or yields in every D_j' an anticonnected component of size $\ge |D_j'|/\ell \ge 2\ell w_y \ge w_y$; pruning spanning trees in $\overline{F'}$ to exact order w_y yields an equal-sized anticonnected (ℓ, w_y)-blockade $\mathcal{B} = (B_1, \dots, B_\ell)$ whose edges in J are 100% complete and whose non-edges have $|N_{F'}(u) \cap B_j| \le |N_{F'}(u) \cap A_j'| \le 4\ell^3 x_{\mathrm{sp}} W_{\mathrm{mid}} \le y^{3a} w_y = y^{3a} |B_j|$ in both directions.

Step 2 (Open-interval pure trigger pair and zero-defect termination). When $\overline{F_t}$ is $(c_0 y)$-sparse, virality of $\overline{\mathcal{F}^-}$ at scale y^4 inside a $\lceil y|F'| \rceil$-subset either finds a y^4-restricted subgraph of order $\ge w_y$ or gives a 100% complete pair (X, Y) ($|X| = w_y$, $|Y| \ge (1 - a_5 y)|F'|$). When F_t is $(c_0 y)$-sparse, applying Definition 3.1(3) to the purified blockade $\mathcal{B}$ from Step 1 either gives (W1) (y^4-restricted of order $\ge w_y$) or gives (W2), an index $i^* \in [\ell]$ with $|Z_{\mathrm{mid}}(i^*)| \le y|F'|$; since $\Delta(F') \le y|F'|$ forces

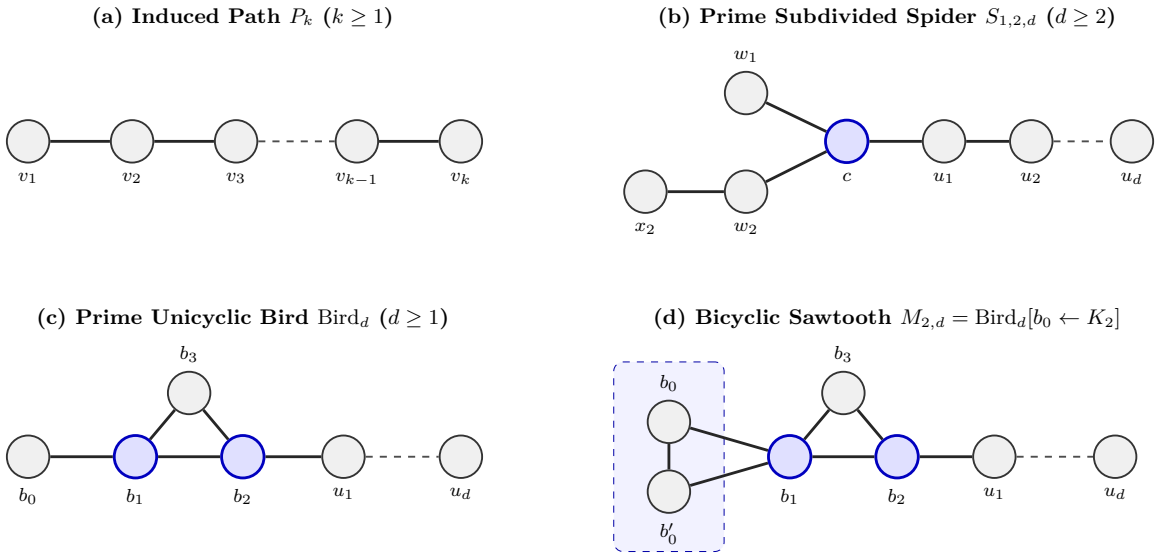


Figure 1: The infinite families resolved in Theorem 1.1: (a) paths P_k, (b) subdivided spiders $S_{1,2,d}$, (c) unicyclic birds Bird_d, and (d) bicyclic sawtooths $M_{2,d}$.

$|Z_{\mathrm{high}}(i^*)| \leq 2y|F'|$, $(B_{i^*}, V(F') \setminus (V(\mathcal{B}) \cup Z_{\mathrm{mid}}(i^*) \cup Z_{\mathrm{high}}(i^*)))$ is 100% anticomplete with $|B_{i^*}| = w_y$ and remainder $\geq (1-4y)|F'|$.

Finally, for $1 \leq |G| < 2^{84C_1^2}$, $|G|^c < 2^{c_4} \leq 2$ gives $\kappa(G) \geq |G|^c$ trivially. For $|G| \geq 2^{84C_1^2}$, set $\varepsilon := \frac{1}{2}|G|^{-1/(21C_1^2)}$ and $x := |G|^{-1/(3C_1)}$. Running multi-scale descent (Lemma 3.2(3) at $b_1 = 11/10$) from λ_0 to x on every $F \subseteq G$ with $|F| \geq \frac{1}{2}\varepsilon^{2C_1}|G|$ (along which $y^{a_{\mathrm{peel}}}|F_t| \geq x^{C_1}|F| \geq |G|^{1/3} > 2$) either reaches an x-restricted subgraph of order $\geq x^{-1}$, hits (GN2), or produces a 100% pairwise-pure cograph $(k_F, |F|/k_F^{C_1})$-blockade via (GN3) or Lemma 3.2(1) ($x_{\mathrm{def}} = 0$). If some such F has arity $k_F \geq \lceil \varepsilon^{-2} \rceil > (\lceil \varepsilon^{-1} \rceil - 1)^2$, Seinsche's theorem on one vertex per block gives $\kappa(G) \geq \lceil \varepsilon^{-1} \rceil \geq |G|^c$; otherwise $2 \leq k_F < \lceil \varepsilon^{-2} \rceil$ holds on every F, so Lemma 3.2(2a) yields a 2ε-restricted subgraph of order $\geq \varepsilon^{5C_1}|G| \geq (2\varepsilon)^{-1}$ and $\kappa(G) \geq (4\varepsilon)^{-1} \geq |G|^c$. Complementation and Theorem 2.3(b) complete the proof. □

4 Recursive Gallai Modules and Stage-1 Generalized Niceness

In this section we establish the Stage-1 comb and co-comb dichotomy and deduce coupled generalized niceness (Definition 3.1) uniformly across all finite paths P_k ($k \geq 6$), all subdivided spiders $S_{1,2,d}$ ($d \geq 2$), and all unicyclic birds Bird_d ($d \geq 2$) illustrated in Figure 1. Consider an (ℓ, w)-comb or (ℓ, w)-co-comb $((a_i, B_i) : i \in [\ell])$ in a graph F equipped with per-handle s-vertex external apex stems $\{\mathbf{v}^{(i)} = (v_1^{(i)}, \ldots, v_s^{(i)}) : i \in [\ell]\}$ as in Definition 2.2. Recall the unified handle-stem pair principle: for any ordered pair of distinct block indices (i, j) in $[\ell]$, setting

$$(\hat{\mathbf{v}}^{(i,j)}, \hat{a}_{i,j}) := \begin{cases} (\mathbf{v}^{(i)}, a_i) & \text{if } ((a_q, B_q) : q \in [\ell]) \text{ is a comb,} \\ (\mathbf{v}^{(j)}, a_j) & \text{if } ((a_q, B_q) : q \in [\ell]) \text{ is a co-comb,} \end{cases}$$

produces an $(s+1)$-tuple $(\hat{v}_1^{(i,j)}, \ldots, \hat{v}_s^{(i,j)}, \hat{a}_{i,j})$ that induces P_{s+1} in $\overline{F}$, has $\{\hat{v}_1^{(i,j)}, \ldots, \hat{v}_s^{(i,j)}\}$ complete in F to $B_i \cup B_j$, and has $\hat{a}_{i,j}$ complete in F to B_i and anticomplete in F to B_j.

When F is $\overline{P_{s+5}}$-free, any vertex $z_0 \in B_j$ $(j \neq i)$ not complete in F to a subset $M_0 \subseteq B_i$ satisfies the property that every anticonnected component of $F[N_F(z_0) \cap M_0]$ is a proper module of $F[M_0]$. When F is only $\overline{P_{s+4+r}}$-free for $r \geq 2$, however, vertices in $M_0 \setminus N_F(z_0)$ may be mixed on anticonnected components of $F[N_F(z_0) \cap M_0]$. We overcome this obstruction for all $r \geq 1$ simultaneously by iterating non-neighborhood peeling $(M_0 \supseteq M_1 \supseteq \cdots \supseteq M_{r-1})$ inside nested Gallai modular cores while growing an active-leaf split tree $\mathcal{T}_i$ inside each tooth B_i governed by the subadditive root potential $\Phi_{D_r}(\mathcal{T}_i) = \sum_{z \in \mathcal{L}(\mathcal{T}_i)} |W_z|^{1/D_r}$.

4.1 Gallai modular cores and active-leaf split trees

By Gallai's modular decomposition theorem [13], whenever a graph H on at least two vertices has both H and $\overline{H}$ connected, its maximal proper modules $Q_1, \ldots, Q_t$ $(t \geq 4)$ form a pairwise-pure partition of $V(H)$ whose quotient graph $H/\mathcal{Q}$ on $[t]$ is prime, and every proper module of H is contained in a unique Q_j. Moreover, if $M \subseteq W$ is a module of $F[W]$ and $N \subseteq M$ is a module of $F[M]$, then N is an exact module of $F[W]$. Our first lemma decouples Gallai's decomposition at an arbitrary arity cap p_{cap} and tracks active-leaf splits under Φ_D.

Lemma 4.1 (Decoupled Gallai core and active-leaf split-tree dichotomy)**.** *Let F be a graph, let $\ell \geq 16$ and $p_{\mathrm{cap}} \geq \ell$ be integers, and let $\theta \in (0, \frac{1}{64\ell^2}]$ and $\eta_0 \in (0,1)$.*

(1) (Single-step decoupled core extraction.) *For every subset $W \subseteq V(F)$ with $|W| \geq 64p_{\mathrm{cap}}^2\theta^{-1}$, running the descending module chain $W = M^{(0)} \supsetneq M^{(1)} \supsetneq \cdots \supsetneq M^{(T)}$—where as long as $|M^{(s)}| > \frac{1}{2}|W|$ and $m_{\max}(F[M^{(s)}]) \geq \frac{1}{64}|W|$, $M^{(s+1)}$ is chosen as a largest proper module of $F[M^{(s)}]$—produces an exact module $M^{(T)} \subseteq W$ of $F[W]$ satisfying at least one of three exhaustive outcomes:*

- (G-i) *$F[W]$ contains a pairwise-pure $(k, \frac{|W|}{64k^2})$-blockade $(U_1, \ldots, U_k)$ for some integer $k \in [2, p_{\mathrm{cap}} - 1]$ satisfying $\sum_{j=1}^{k} |U_j|^{1/8} \geq (\frac{k}{2})^{3/4}|W|^{1/8} \geq |W|^{1/8}$ and $\sum_{j=1}^{k} |U_j|^{1/16} \geq 2^{1/16}|W|^{1/16}$;*
- (G-ii) *$F[M^{(T)}]$ is connected and anticonnected with canonical Gallai partition $Q_1, \ldots, Q_t$ $(|Q_1| \geq \cdots \geq |Q_t|$, $t \geq p_{\mathrm{cap}})$ satisfying $|Q_{p_{\mathrm{cap}}}| \geq \theta|W|$, so $(Q_1, \ldots, Q_{p_{\mathrm{cap}}})$ is a pairwise-pure $(p_{\mathrm{cap}}, \theta|W|)$-blockade of exact length p_{cap}; or*
- (G-iii) *$|M^{(T)}| > \frac{1}{2}|W|$ and there exists a* Gallai core *$M^* \subseteq M^{(T)}$ of size $|M^*| > \frac{15}{64}|W| > 2^{-3}|W|$ such that every proper module X of $F[M^{(T)}]$ intersecting M^* satisfies $X \subseteq M^*$ and $|X| < \theta|W| < 5\theta|M^*|$.*

(2) (Iterated core extraction with stopping threshold $L_{\max}$.) *For every $L_{\max} \in [2, \ell]$ and every $B \subseteq V(F)$ with $|B| \geq 64p_{\mathrm{cap}}^2\theta^{-1}(4\ell)^{16}$, repeatedly applying (1) to a largest leaf of a partition tree $\mathcal{T}_0$ rooted at B while $|\mathcal{L}(\mathcal{T}_0)| < L_{\max}$ yields either:*

- (GC1) *a leaf $W \subseteq B$ $(|W| > L_{\max}^{-8}|B|)$, an exact module $M_T \subseteq W$ of $F[W]$, and a Gallai core $M^* \subseteq M_T$ with $|M^*| > 2^{-3}L_{\max}^{-8}|B|$ in which every proper module X of $F[M_T]$ meeting M^* satisfies $X \subseteq M^*$ and $|X| < 5\theta|M^*|$;*
- (GC2) *a pairwise-pure $(k, \frac{|B|}{64k^{10}})$-blockade in $F[B]$ for some integer $k \in [L_{\max}, p_{\mathrm{cap}} - 1]$; or*
- (GC3) *a pairwise-pure $(p_{\mathrm{cap}}, \min(\theta, \frac{1}{64p_{\mathrm{cap}}^2})\frac{|B|}{L_{\max}^8})$-blockade in $F[B]$ of exact length p_{cap}.*

(3) (Active-leaf split-tree potential lemma.) *Let $r_{\mathrm{tree}} \geq 1$ and $D = 8m \geq 64r_{\mathrm{tree}}$ be integers, set $E_D := \frac{5}{8}D = 5m$, let $L_{\max} \in [2, \ell]$, and define $\gamma_{D,L_{\max}} := L_{\max}^{-D}$ and $\beta_{D,L_{\max}} := 2^{-60r_{\mathrm{tree}}}L_{\max}^{-2D}$. Let $\mathcal{T}$ be a rooted tree of subsets of B with root B, where splits are applied only to* active leaves *$W_z \in \mathcal{L}(\mathcal{T})$ satisfying $|W_z| \geq \gamma_{D,L_{\max}}|B|$ into pairwise disjoint children $(W_{z,1}, \ldots, W_{z,k})$ $(2 \leq k < L_{\max})$ that are pairwise weakly η_0-sparse-or-co-sparse in F and satisfy either:*

- (Type I) $64 \le k < L_{\max}$ *(or* $4 \le k < L_{\max}$ *when* $D \ge 88$*) and* $|W_{z,q}| \ge 2^{-60} k^{-E_D} |W_z|$ *for all* $q \in [k]$*; or*
- (Type II) $2 \le k < L_{\max}$ *and* $|W_{z,q}| \ge 2^{-60 r_{\mathrm{tree}}} |W_z|$ *for all* $q \in [k]$.

Then the potential $\Phi_D(\mathcal{T}) := \sum_{z \in \mathcal{L}(\mathcal{T})} |W_z|^{1/D}$ *increases by at least* $(2^{4/D} - 1)|W_z|^{1/D} > \frac{1}{2DL_{\max}} |B|^{1/D}$ *at every split. Whenever* $|\mathcal{L}(\mathcal{T})| < L_{\max}$*, a largest leaf* $W_z \in \mathcal{L}(\mathcal{T})$ *satisfies* $|W_z| > \gamma_{D,L_{\max}} |B|$ *and is active; and at the first step where* $|\mathcal{L}(\mathcal{T})| \ge L_{\max}$*, the first* $L_{\max}$ *leaves* $L_1, \dots, L_{L_{\max}}$ *of* $\mathcal{T}$ *form a weakly* $(\eta_0 \beta_{D,L_{\max}}^{-2})$*-sparse-or-co-sparse* $\left(L_{\max}, \frac{|B|}{2^{60 r_{\mathrm{tree}}} L_{\max}^{2D}}\right)$*-blockade in* $F[B]$.

Proof. For (1), by module transitivity every $M^{(s)}$ $(0 \le s \le T)$ is an exact module of $F[W]$. If the chain halts with $|M^{(T)}| \le \frac{1}{2}|W|$, then $T \ge 1$, $|M^{(T-1)}| > \frac{1}{2}|W|$, and $|M^{(T)}| \ge \frac{1}{64}|W|$. Since $M^{(T)}$ is a module of $F[W]$ with $|W \setminus M^{(T)}| \ge \frac{1}{2}|W|$, the larger pure half $U_1 \subseteq W \setminus M^{(T)}$ satisfies $|U_1| \ge \frac{1}{4}|W| = \frac{|W|}{64 \cdot 2^2}$ and forms with $U_2 := M^{(T)}$ a pure 2-block blockade satisfying (G-i). Suppose instead that $|M^{(T)}| > \frac{1}{2}|W|$, so $m_{\max}(F[M^{(T)}]) < \frac{1}{64}|W|$. If $F[M^{(T)}]$ or $\overline{F}[M^{(T)}]$ is disconnected, every component has size below $\frac{1}{64}|W|$; taking a minimal prefix union U_1 of components of size at least $\frac{1}{4}|M^{(T)}|$ yields $\frac{1}{4}|M^{(T)}| \le |U_1| < \frac{1}{2}|M^{(T)}|$, so $(U_1, M^{(T)} \setminus U_1)$ is a pure 2-block blockade of width exceeding $\frac{1}{8}|W|$, giving (G-i). Otherwise both $F[M^{(T)}]$ and $\overline{F}[M^{(T)}]$ are connected, and Gallai's theorem [13] partitions $M^{(T)} = \bigsqcup_{j=1}^{t} Q_j$ $(t \ge 4,\ |Q_1| \ge \cdots \ge |Q_t|)$ into pairwise-pure maximal proper modules with $|Q_1| < \frac{1}{64}|W|$. If $|Q_k| \ge \frac{|W|}{64k^2}$ for some $k \in [2, \min(t, p_{\mathrm{cap}} - 1)]$, then $(Q_1, \dots, Q_k)$ satisfies (G-i) since $\sum_{j=1}^{k} |Q_j|^{1/8} \ge k(2^{-6}k^{-2})^{1/8}|W|^{1/8} = (k/2)^{3/4}|W|^{1/8} \ge |W|^{1/8}$; if $t \ge p_{\mathrm{cap}}$ and $|Q_{p_{\mathrm{cap}}}| \ge \theta|W|$, then $(Q_1, \dots, Q_{p_{\mathrm{cap}}})$ gives (G-ii). Otherwise $|Q_j| < \frac{|W|}{64j^2}$ for $j < p_{\mathrm{cap}}$ and $|Q_j| < \theta|W|$ for $j \ge p_{\mathrm{cap}}$; setting $I^* := \{j \in [t] : |Q_j| < \theta|W|\}$ and $M^* := \bigsqcup_{j \in I^*} Q_j$ gives

$$|M^{(T)} \setminus M^*| < \frac{|W|}{64} \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{384}|W| < \frac{1}{32}|W|,$$

so $|M^*| > \frac{15}{64}|W|$ and every proper module X of $F[M^{(T)}]$ meeting M^* lies inside some Q_j $(j \in I^*)$, giving $X \subseteq M^*$ and $|X| \le |Q_j| < \theta|W| < 5\theta|M^*|$ as in (G-iii).

For (2), applying (1) to a largest leaf W_z of $\mathcal{T}_0$ while $|\mathcal{L}(\mathcal{T}_0)| < L_{\max}$ preserves $\sum_{L \in \mathcal{L}(\mathcal{T}_0)} |L|^{1/8} \ge |B|^{1/8}$ and increases $\sum_{L \in \mathcal{L}(\mathcal{T}_0)} |L|^{1/16}$ by at least $(2^{1/16} - 1)|W_z|^{1/16}$ whenever a split of arity $k \in [2, L_{\max} - 1]$ occurs; since $|W_z| > L_{\max}^{-8}|B|$ prior to every split, every child created has size at least $\frac{|W_z|}{64k^2} > \frac{|B|}{64 L_{\max}^{10}}$, yielding (GC1), (GC2), or (GC3).

For (3), on any active leaf W_z $(|W_z| \ge L_{\max}^{-D}|B|)$, a Type I split satisfies $k^{3/8} 2^{-60/D} \ge (2^6)^{3/8} 2^{-60/D} = 2^{\frac{9}{4} - \frac{60}{D}} \ge 2^{4/D}$ for $k \ge 64$ and $D \ge 64$ (and $k^{3/8} 2^{-60/D} \ge 2^{\frac{3}{4} - \frac{60}{D}} \ge 2^{4/D}$ for $k \ge 4$ and $D \ge 88$), so

$$\sum_{q=1}^{k} |W_{z,q}|^{1/D} \ge k^{3/8} 2^{-60/D} |W_z|^{1/D} \ge 2^{4/D} |W_z|^{1/D},$$

while a Type II split $(k \ge 2)$ satisfies $\sum_{q=1}^{k} |W_{z,q}|^{1/D} \ge 2^{1 - 60 r_{\mathrm{tree}}/D} |W_z|^{1/D} \ge 2^{4/D} |W_z|^{1/D}$ since $D \ge 64 r_{\mathrm{tree}}$. Because only active leaves $(|W_z| \ge L_{\max}^{-D}|B|)$ are ever split, every child leaf L ever created satisfies $|L| \ge 2^{-60 r_{\mathrm{tree}}} L_{\max}^{-2D} |B| = \beta_{D,L_{\max}} |B|$. For any two leaves L_a, L_b with lowest common ancestor z ($L_a \subseteq W_{z,q}$, $L_b \subseteq W_{z,q'}$, $q < q'$), the number of wrong cross-pairs between L_a and L_b is at most $\eta_0 |W_{z,q}||W_{z,q'}| \le \eta_0 \beta_{D,L_{\max}}^{-2} |L_a||L_b|$, completing the proof. □

4.2 Iterated non-neighborhood peeling and path generalized niceness

Next we record how iterated non-neighborhood peeling ($M_0 \supseteq M_1 \supseteq \cdots \supseteq M_r$) inside a comb or co-comb forces modularity of anticonnected neighborhood components in $\overline{P_{s+5+r}}$-free graphs, and how extreme attachments promote into sub-combs or sub-co-combs with longer external apex stems.

Lemma 4.2 (Depth-r neighborhood modularity, dyadic stem promotion, and double counting)**.** *Let $s \geq 1$ and $r \geq 0$ be integers, and let $((a_i, B_i) : i \in [\ell])$ ($\ell \geq 16$) be an (ℓ, w)-comb or (ℓ, w)-co-comb in a graph F equipped with per-handle s-vertex external apex stems $\{\mathbf{v}^{(i)} : i \in [\ell]\}$.*

(1) (Depth-r neighborhood modularity.) *Suppose F is $\overline{P_{s+5+r}}$-free. Fix distinct $i, j \in [\ell]$, an external vertex $z_0 \in B_j$, and $M_0 \subseteq B_i$; for $1 \leq d \leq r$, recursively let $M_d \subseteq N_F(z_{d-1}) \cap M_{d-1}$ be non-empty and let $z_d \in M_{d-1} \setminus N_F(z_{d-1})$. If z_r is not complete in F to M_r, then every anticonnected component C_q of $F[N_F(z_r) \cap M_r]$ is a proper module of $F[M_r]$.*

(2) (Dyadic sub-comb/co-comb stem promotion.) *For any $i \in [\ell]$, any non-empty $M_i \subseteq B_i$, any $\tau \in (0, 2^{-20}]$, and external pools $S_i, D_i \subseteq \bigcup_{j \neq i} B_j$ where every $u \in S_i$ satisfies $|N_F(u) \cap M_i| < 2^{-20}|M_i|$ and every $u \in D_i$ satisfies $|M_i \setminus N_F(u)| < 2^{-20}|M_i|$, either there exists a subset $M_i' \subseteq M_i$ of size $|M_i'| > \frac{7}{8}|M_i|$ such that $|N_F(u) \cap M_i'| \leq \tau|M_i|$ for all $u \in S_i$ and $|M_i' \setminus N_F(u)| \leq \tau|M_i|$ for all $u \in D_i$, or dyadic bucket pruning extracts a $(k_{\mathrm{comb}}, w_{\mathrm{sub}})$-comb or co-comb inside M_i with length $512 < k_{\mathrm{comb}} < \tau^{-1}$, width $w_{\mathrm{sub}} > \max(\frac{|M_i|}{16k_{\mathrm{comb}}^2}, \frac{\tau}{2}|M_i|)$, and per-handle $(s+1)$-vertex external apex stems. Likewise, at any internal depth $d \geq 1$ with non-empty M_d and a pool $A \subseteq M_{d-1} \setminus N_F(z_{d-1})$ of cutters each having fewer than $2^{-20}|M_d|$ neighbors (respectively, non-neighbors) in M_d, either A is 2τ-sparse (respectively, 2τ-co-sparse) to a subset $R_{\mathrm{end}} \subseteq M_d$ of size $|R_{\mathrm{end}}| > \frac{1}{2}|M_d|$, or dyadic pruning extracts a $(k_{\mathrm{comb}}, w_{\mathrm{int}})$-comb or co-comb inside M_d with $512 < k_{\mathrm{comb}} < \tau^{-1}$, width $w_{\mathrm{int}} > \max(\frac{|M_d|}{16k_{\mathrm{comb}}^2}, \frac{\tau}{2}|M_d|)$, and shared $(s+1+d)$-vertex external apex stem $(\hat{v}_1^{(i,j)}, \ldots, \hat{v}_s^{(i,j)}, \hat{a}_{i,j}, z_0, \ldots, z_{d-1})$.*

(3) (Bipartite double-counting dichotomy.) *If $(M_1', \ldots, M_\ell')$ is a blockade in F such that for all $i \neq j$, every $u \in M_j'$ satisfies $\min(|N_F(u) \cap M_i'|, |M_i' \setminus N_F(u)|) \leq \frac{\eta_{\mathrm{out}}}{4\ell}|M_i'|$ ($\eta_{\mathrm{out}} > 0$), then $(M_1', \ldots, M_\ell')$ is weakly η_{out}-sparse-or-co-sparse in F.*

Proof. For (1), vertices of $(N_F(z_r) \cap M_r) \setminus C_q$ are complete in F to C_q. If some cutter $z_{r+1} \in M_r \setminus N_F(z_r)$ were mixed on C_q, connectivity of $\overline{F}[C_q]$ would yield an edge $x_2x_3 \in E(\overline{F}[C_q])$ with $z_{r+1}x_2 \in E(\overline{F})$ and $z_{r+1}x_3 \notin E(\overline{F})$. Using the unified handle-stem pair $(\hat{\mathbf{v}}^{(i,j)}, \hat{a}_{i,j})$, each vertex in the sequence

$$(\hat{v}_1^{(i,j)} - \cdots - \hat{v}_s^{(i,j)} - \hat{a}_{i,j} - z_0 - z_1 - \cdots - z_r - z_{r+1} - x_2 - x_3)$$

is non-adjacent in $\overline{F}$ to all non-consecutive later vertices (since $M_d \subseteq N_F(z_{d-1}) \cap M_{d-1}$ for $1 \leq d \leq r$ and $C_q \subseteq N_F(z_r) \cap M_r$), yielding an induced P_{s+5+r} in $\overline{F}$, a contradiction.

For (2), greedy peeling of maximum-degree neighborhoods $D_a := N_F(u_a) \cap A^{(a)}$ ($u_a \in S_i$) stops with $|A_i^S| > \frac{15}{16}|M_i|$ unless it removes at least $2^{-4}|M_i|$ vertices in $m < \tau^{-1}$ disjoint increments of size in $(\tau|M_i|, 2^{-20}|M_i|)$. Since $\sum_{t \geq 20}(t+1)^{-2} < 2^{-4}$, some dyadic bucket $J_t := \{a \in [m] : 2^{-(t+1)}|M_i| < |D_a| \leq 2^{-t}|M_i|\}$ ($t \geq 20$) satisfies $\sum_{a \in J_t}|D_a| > (t+1)^{-2}|M_i|$. Processing J_t in reverse order and retaining $a \in L$ with $B_a' := D_a \setminus \bigcup_{q \in L, q > a} N_F(u_q)$ whenever $|B_a'| \geq \frac{1}{2}|D_a|$ covers at least half of $\bigcup_{a \in J_t} D_a$ by sets of size at most $2^{-t}|M_i|$, giving $k_{\mathrm{comb}} = |L| > \frac{2^{t-1}}{(t+1)^2} > 2^{t/2-1} \geq 512$ and $|B_q'| > \max(\frac{|M_i|}{16k_{\mathrm{comb}}^2}, \frac{\tau}{2}|M_i|)$, with per-handle $(s+1)$-stems $(\hat{v}_1^{(i,j(q))}, \ldots, \hat{v}_s^{(i,j(q))}, \hat{a}_{i,j(q)})$ ($u_q \in B_{j(q)}$); repeating on D_i in $\overline{F}$ (or from A into M_d) completes (2).

For (3), partition $M'_j = S_{j,i} \sqcup D_{j,i}$ where $S_{j,i} := \{u \in M'_j : |N_F(u) \cap M'_i| \le \frac{\eta_{\text{out}}}{4\ell}|M'_i|\}$ and $D_{j,i} := M'_j \setminus S_{j,i}$. Counting edges between $D_{j,i}$ and $S_{i,j}$ gives

$$|D_{j,i}|\Big(|S_{i,j}| - \frac{\eta_{\text{out}}}{4\ell}|M'_i|\Big) \le |E_F(D_{j,i}, S_{i,j})| \le |S_{i,j}| \cdot \frac{\eta_{\text{out}}}{4\ell}|M'_j|,$$

which prevents $\min(|D_{j,i}|/|M'_j|, |S_{i,j}|/|M'_i|) > \frac{\eta_{\text{out}}}{2\ell}$ and yields $\min(|E_F(M'_i, M'_j)|, |E_{\overline{F}}(M'_i, M'_j)|) \le \frac{3\eta_{\text{out}}}{4\ell}|M'_i||M'_j| \le \eta_{\text{out}}|M'_i||M'_j|$. □

We now prove the depth-r comb/co-comb dichotomy for all $r \ge 1$ by strong induction on r. Define the split-tree exponent sequence $D_1 := 8$, $D_2 := 64$, and $D_r := 256 \cdot 8^{r-3}$ for $r \ge 3$: setting $r_{\text{tree}}(r) := 1$ for $r \le 2$ and $r_{\text{tree}}(r) := r$ for $r \ge 3$, we have $D_r \ge 64 r_{\text{tree}}(r)$, $25r \le 60 r_{\text{tree}}(r)$, and $2D_{r-1} + 2 + 10r \le E_{D_r} := \frac{5}{8}D_r$ for every integer $r \ge 2$ ($38 \le 40$ at $r = 2$, $160 \le 160$ at $r = 3$, and $2D_{r-1} + 2 + 10r \le 5D_{r-1} = E_{D_r}$ for $r \ge 4$). Define also the decoupled cap exponent $E_r := 8r \cdot 16^r$ for all $r \ge 1$.

Theorem 4.3 (Inductive depth-r comb/co-comb dichotomy)**.** *For every integer $r \ge 1$, every integer $s \ge 1$, all integers $p_{\text{cap}} \ge \ell \ge 64$ and $L_{\max} \in [2, \ell]$, every $\eta_{\text{out}} \in (0, 2^{-16}]$, and every $\overline{P_{s+4+r}}$-free graph F containing an (ℓ, w)-comb or (ℓ, w)-co-comb $((a_i, B_i) : i \in [\ell])$ equipped with per-handle s-vertex external apex stems and tooth width $w \ge w_{\min}(r, \eta_{\text{out}}, p_{\text{cap}}) := 2^{400r^2}\eta_{\text{out}}^{-3r}p_{\text{cap}}^{E_r+2D_r}$, the induced subgraph $F[\bigsqcup_{i=1}^{\ell} B_i]$ contains either:*

(a) *a weakly η_{out}-sparse-or-co-sparse $(k, \frac{w}{2^{60r}k^{3D_r}})$-blockade (in fact of width $\ge \frac{w}{2^{60r}k^{2D_r}}$) for some integer $k \in [L_{\max}, p_{\text{cap}}]$ (whose width is independent of η_{out} and p_{cap}); or*

(b) *a weakly η_{out}-sparse-or-co-sparse $(p_{\text{cap}}, 2^{-200r^2}\eta_{\text{out}}^{r}p_{\text{cap}}^{-E_r}w)$-blockade of exact length p_{cap}.*

Proof. We proceed by strong induction on $r \ge 1$, proving the stronger split/cross width $\frac{w}{2^{60r}k^{2D_r}}$ in (a) at every depth $r \ge 1$. For the base case $r = 1$ ($\overline{P_{s+5}}$-free, $2D_1 = 16$), set $p_{\text{comp}} := \min(p_{\text{cap}}, \lceil 8\ell^2\eta_{\text{out}}^{-1}\rceil) \in [\ell, p_{\text{cap}}]$ and $\theta_1 := \frac{\eta_{\text{out}}}{64\ell p_{\text{comp}}^2}$. Applying Lemma 4.1(2) with $L_{\max}$ and θ_1 to each tooth B_i yields (a) or (b) directly unless every $i \in [\ell]$ produces an exact module $M_{T_i} \subseteq B_i$ and core $M_i^* \subseteq M_{T_i}$ ($|M_i^*| > \frac{15}{64}L_{\max}^{-8}w > 2^{-3}L_{\max}^{-8}w$) in which every proper module of $F[M_{T_i}]$ meeting M_i^* has size less than $\tau_i := 5\theta_1|M_i^*|$. If some $u \in M_j^*$ ($j \ne i$) is not complete in F to M_{T_i} and satisfies $|N_F(u) \cap M_i^*| \ge \frac{\eta_{\text{out}}}{2\ell}|M_i^*| > 2p_{\text{comp}}\tau_i$, Lemma 4.2(1) ($r_{\text{cut}} = 0$) implies that the anticonnected components $(C_q)_q$ of $F[N_F(u) \cap M_{T_i}]$ satisfy $|C_q \cap M_i^*| \le \tau_i$; grouping consecutive intersections into p_{comp} intervals of size at least τ_i forms a complete $(p_{\text{comp}}, \tau_i)$-blockade in $F[M_i^*]$, giving (b) if $p_{\text{comp}} = p_{\text{cap}}$ and giving (a) of width $\tau_i > \theta_1 L_{\max}^{-8}w \ge \frac{w}{64k^{10}} > \frac{w}{2^{60}k^{16}}$ if $k = p_{\text{comp}} < p_{\text{cap}}$ (since $8\ell^2 \le k\eta_{\text{out}}$ implies $\theta_1 = \frac{\eta_{\text{out}}}{64\ell k^2} \ge \frac{\ell}{8k^3} \ge \frac{L_{\max}}{8k^3}$). Otherwise every $u \in M_j^*$ not complete to M_i^* satisfies $|N_F(u) \cap M_i^*| < \frac{\eta_{\text{out}}}{2\ell}|M_i^*|$; deleting for each non-complete pair (M_i^*, M_j^*) the $< \frac{\eta_{\text{out}}}{2\ell}|M_i^*|$ vertices complete to M_j^* leaves subsets $D_i \subseteq M_i^*$ ($|D_i| > 2^{-4}L_{\max}^{-8}w$) forming a complete-or-$\eta_{\text{out}}$-sparse blockade of length $\ell \in [L_{\max}, p_{\text{cap}}]$, establishing (a).

For the inductive step, let $r \ge 2$ and assume the theorem (with width $\frac{w}{2^{60r'}k^{2D_{r'}}}$ in (a)) holds for all $1 \le r' \le r-1$ and all $s' \ge 1$. Set $r_{\text{tree}} := r_{\text{tree}}(r)$, $\gamma := L_{\max}^{-D_r}$, $\beta := 2^{-60r_{\text{tree}}}L_{\max}^{-2D_r}$, $\eta_0 := \frac{1}{4}\beta^2\eta_{\text{out}}$, $p_{\text{comp},2} := \min(p_{\text{cap}}, \lceil 8\ell^2\eta_0^{-1}\rceil)$, and $\theta := \frac{\eta_0}{64\ell p_{\text{comp},2}^2}$. Maintain an active-leaf split tree $\mathcal{T}_i$ on each B_i via Lemma 4.1(3) at $D = D_r \ge 64r_{\text{tree}}$, stopping threshold $L_{\max}$, and sparsity η_0: as soon as $|\mathcal{L}(\mathcal{T}_i)| \ge L_{\max}$, the first $L_{\max}$ leaves of $\mathcal{T}_i$ satisfy (a). While $|\mathcal{L}(\mathcal{T}_i)| < L_{\max}$ for all $i \in [\ell]$, each largest leaf $W_{z_i} \in \mathcal{L}(\mathcal{T}_i)$ satisfies $|W_{z_i}| > L_{\max}^{-D_r}w$ and is active. Applying Lemma 4.1(1) with θ to each W_{z_i} yields (b), yields (a), splits W_{z_i} in $\mathcal{T}_i$, or extracts a Level-1 Gallai core $M_i \subseteq W_{z_i}$ ($|M_i| > 2^{-3}|W_{z_i}|$).

Case 1: Some external vertex $z_0 \in M_j$ ($j \neq i$) satisfies $2^{-20}|M_i| \le |N_F(z_0) \cap M_i| \le (1-2^{-20})|M_i|$. Recursively across depth $d = 1, \dots, r-1$, given a Level-d core $N^{(d-1)*} \subseteq W_{z_i}$ ($N^{(0)*} := M_i$, $|N^{(d-1)*}| > 2^{-23d}|W_{z_i}|$) and a cutter z_{d-1} with $N^{(d)} := N_F(z_{d-1}) \cap N^{(d-1)*}$ and $A_{d-1} := N^{(d-1)*} \setminus N^{(d)}$ satisfying $\min(|N^{(d)}|, |A_{d-1}|) \ge 2^{-20}|N^{(d-1)*}|$, applying Lemma 4.1(1) with θ to $F[N^{(d)}]$ either concludes via (a)/(b)/split or produces an exact module $N_T^{(d)} \subseteq N^{(d)}$ and Level-$(d+1)$ core $N^{(d)*} \subseteq N_T^{(d)}$ ($|N^{(d)*}| > 2^{-23d-3}|W_{z_i}|$):

- At $d = r-1$, Lemma 4.2(1) ($s+5+(r-1) = s+4+r$) ensures that for every cutter $z_{r-1} \in A_{r-2}$ not complete to $N_T^{(r-1)}$, the anticonnected components of $F[N_F(z_{r-1}) \cap N_T^{(r-1)}]$ are proper modules of $F[N_T^{(r-1)}]$; grouping components if $|N_F(z_{r-1}) \cap N^{(r-1)*}| \ge 10p_{\mathrm{comp},2}\theta|N^{(r-1)*}|$ gives (a) or (b), and otherwise the larger half $A_{r-2,*} \subseteq A_{r-2}$ is complete or η_0-sparse to $N^{(r-1)*}$ of block size $> 2^{-25r}|W_{z_i}| \ge 2^{-60r_{\mathrm{tree}}}|W_{z_i}|$, giving (a) directly if $L_{\max} = 2$ and a Type II split $(N^{(r-1)*}, A_{r-2,*})$ of W_{z_i} if $L_{\max} > 2$.
- If at some depth $1 \le d \le r-2$ no balanced cutter $z_d \in A_{d-1}$ exists, Lemma 4.2(2) at $\tau = \frac{\eta_0}{2}$ on the larger extreme half $A_{d-1,*} \subseteq A_{d-1}$ either yields a 2-block pair $(R_{\mathrm{end}}, A_{d-1,*})$ of block size $> 2^{-25r}|W_{z_i}|$ (giving (a) if $L_{\max} = 2$ and a Type II split of W_{z_i} if $L_{\max} > 2$) or extracts a $(k_{\mathrm{comb}}, w_{\mathrm{int}})$-comb or co-comb inside $N^{(d)*}$ ($k_{\mathrm{comb}} > 512$) with shared $(s+1+d)$-vertex stem. Applying the induction hypothesis at $r' := r-1-d \in [1, r-2]$ to the first $k' := \min(k_{\mathrm{comb}}, p_{\mathrm{cap}}) \ge 512$ pairs with stopping threshold $L'_{\max} := k'$ and sparsity η_0: when $k_{\mathrm{comb}} \le p_{\mathrm{cap}}$, outcome (a) has length $k \in [k', p_{\mathrm{cap}}]$ and width

$$\frac{w_{\mathrm{int}}}{2^{60r'}k^{2D_{r'}}} > \frac{|W_{z_i}|}{2^{60r'+23d+7}k^{2D_{r'}+2}} \ge \frac{|W_{z_i}|}{k^{2D_{r'}+2+10r}} \ge 2^{-60}k^{-E_{D_r}}|W_{z_i}|$$

  (using $k \ge k' > 64 = 2^6$ so $2^{60r'+23d+7} \le 2^{60r} \le k^{10r}$, together with $2D_{r'} + 2 + 10r \le E_{D_r}$ for all $r \ge 2$), yielding a Type I split of W_{z_i} if $k < L_{\max}$ and giving (a) of width $> \frac{w}{2^{60r}k^{2D_r}}$ if $k \ge L_{\max}$; whereas when $k_{\mathrm{comb}} > p_{\mathrm{cap}}$ (or outcome (b) occurs), $L'_{\max} = p_{\mathrm{cap}} \ge \ell \ge 64$ forces exact length $k = p_{\mathrm{cap}}$ and width exceeding $2^{-200r^2}\eta_{\mathrm{out}}^r p_{\mathrm{cap}}^{-E_r} w$ (absorbing $2^{-(120r+2)r'}$ into $p_{\mathrm{cap}}^{-30r^2}$ via $p_{\mathrm{cap}} \ge 64 = 2^6$), giving (b).

Case 2: Every cross-block attachment is $< 2^{-20}|M_i|$ or $> (1-2^{-20})|M_i|$. Applying Lemma 4.2(2) at $\tau = \frac{\eta_{\mathrm{out}}}{8\ell}$ across all $i \in [\ell]$ either yields subsets $M'_i \subseteq M_i$ ($|M'_i| > \frac{7}{8}|M_i| > 2^{-4}L_{\max}^{-D_r}w$) whose blockade $(M'_1, \dots, M'_\ell)$ satisfies (a) by Lemma 4.2(3), or promotes some M_i into a $(k_{\mathrm{comb}}, w_{\mathrm{sub}})$-comb or co-comb with per-handle $(s+1)$-stems; invoking the induction hypothesis at $r' = r-1$ ($d = 0$, $2D_{r-1} + 2 + 10r \le E_{D_r}$) with $k' = L'_{\max} := \min(k_{\mathrm{comb}}, p_{\mathrm{cap}}) \ge 512$ concludes identically. □

Corollary 4.4 (Stage-1 coupled generalized niceness for all finite paths)**.** *For every integer $k \ge 6$, both $\{P_k\}$ and $\{\overline{P_k}\}$ are unconditionally coupled generalized nice.*

Proof. Set $r := k-5 \ge 1$, $s := 1$, $L_{\max} := \ell$, $\eta_{\mathrm{out}} := x$, $p_{\mathrm{cap}} := \lceil x^{-2} \rceil$, and $C_{\mathrm{comb}}(k) := 3E_r + 4D_r + 100$. In any $\overline{P_k}$-free graph F containing an (ℓ, w)-comb ($w \ge |F|/\ell^{18}$) with external apex v, if $w < w_{\min}(r, x, p_{\mathrm{cap}})$ then $|F| < p_{\mathrm{cap}}^{C_{\mathrm{comb}}(k)}$ so singleton blocks give a pure $(k', |F|/(k')^{C_{\mathrm{comb}}(k)})$-blockade; otherwise Theorem 4.3 produces a weakly x-sparse-or-co-sparse $(k', |F|/(k')^{C_{\mathrm{comb}}(k)})$-blockade of length $k' \in [\ell, p_{\mathrm{cap}}]$. By Lemma 3.2 and complement symmetry of Definition 3.1(2), both $\{P_k\}$ and $\{\overline{P_k}\}$ are coupled generalized nice. □

4.3 Non-neighborhood modularity for subdivided spiders and unicyclic birds

For subdivided spiders $S_{1,2,d}$ and unicyclic birds Bird_d, replacing the neighborhood $N_F(z_r) \cap M_r$ at the bottom of the cutter chain by the *non-neighborhood* $M_r \setminus N_F(z_r)$ yields exact component-wise

modularity in F and $\overline{F}$, respectively.

Proposition 4.5 (Connected non-neighborhood modularity for spiders)**.** *Let $s \geq 1$ and $r \geq 0$ be integers, set $d := s + 1 + r \geq 2$, and let F be an $\overline{S_{1,2,d}}$-free graph containing an (ℓ, w)-comb or (ℓ, w)-co-comb $((a_i, B_i) : i \in [\ell])$ with per-handle s-vertex external apex stems. Fix distinct $i, j \in [\ell]$, $z_0 \in B_j$, $M_0 \subseteq B_i$, and for $1 \leq q \leq r$ let $M_q \subseteq N_F(z_{q-1}) \cap M_{q-1}$ and $z_q \in M_{q-1} \setminus N_F(z_{q-1})$. If $N_F(z_r) \cap M_r \neq \emptyset$, then every connected component K_p of $F[M_r \setminus N_F(z_r)]$ is a proper module of $F[M_r]$.*

Proof. Since $N_F(z_r) \cap M_r \neq \emptyset$, each component K_p of $F[M_r \setminus N_F(z_r)]$ satisfies $|K_p| < |M_r|$ and is anticomplete in F to $(M_r \setminus N_F(z_r)) \setminus K_p$. If some vertex $x_2 \in N_F(z_r) \cap M_r$ were mixed on K_p in F, connectivity of $F[K_p]$ would yield an edge $w_1w_2 \in E(F[K_p])$ ($w_1w_2 \notin E(\overline{F})$) with $w_1x_2 \in E(F)$ ($w_1x_2 \notin E(\overline{F})$) and $w_2x_2 \notin E(F)$ ($w_2x_2 \in E(\overline{F})$). With $(\hat{\mathbf{v}}^{(i,j)}, \hat{a}_{i,j})$ denoting the unified handle-stem pair, the $(d+1)$-vertex stem-cutter path

$$P_{\text{stem}} := (\hat{v}_1^{(i,j)} - \cdots - \hat{v}_s^{(i,j)} - \hat{a}_{i,j} - z_0 - z_1 - \cdots - z_r)$$

in $\overline{F}$ has length d and its first d vertices are anticomplete in $\overline{F}$ to $\{w_1, w_2, x_2\}$, while z_r is adjacent in $\overline{F}$ to $\{w_1, w_2\}$ and non-adjacent to x_2. Hence $\overline{F}[V(P_{\text{stem}}) \cup \{w_1, w_2, x_2\}]$ has degree-3 center z_r with arms $(z_r - w_1)$, $(z_r - w_2 - x_2)$, and P_{stem}, isomorphic to $S_{1,2,d}$, a contradiction. □

Proposition 4.6 (Anticonnected non-neighborhood modularity for birds)**.** *Under the exact hypotheses and notation of Proposition 4.5, if F is $\overline{\text{Bird}_d}$-free and $N_F(z_r) \cap M_r \neq \emptyset$, then every anticonnected component L_p of $F[M_r \setminus N_F(z_r)]$ is a proper module of $F[M_r]$.*

Proof. Every vertex of $(M_r \setminus N_F(z_r)) \setminus L_p$ is complete in F to L_p. If some vertex $x_2 \in N_F(z_r) \cap M_r$ were mixed on L_p in $\overline{F}$, connectivity of $\overline{F}[L_p]$ would produce an edge $w_1w_2 \in E(\overline{F}[L_p])$ such that $w_1x_2 \notin E(\overline{F})$ and $w_2x_2 \in E(\overline{F})$. In $\overline{F}[V(P_{\text{stem}}) \cup \{w_1, w_2, x_2\}]$, identifying $(b_0, b_1, b_2, b_3) = (x_2, w_2, z_r, w_1)$ shows that $\{w_2, z_r, w_1\}$ induces a triangle with $N_{\overline{F}}(w_1) \cap (V(P_{\text{stem}}) \cup \{w_2, x_2\}) = \{w_2, z_r\}$, $(w_2 - x_2)$ is the pendant edge at w_2, and P_{stem} is the d-edge path attached at z_r, forming an induced copy of Bird_d, a contradiction. □

Corollary 4.7 (Stage-1 coupled generalized niceness for spiders and birds)**.** *For every integer $d \geq 2$ and every integer $t \geq 2$, the pairs $\{P_t, S_{1,2,d}\}$ and $\{P_t, \text{Bird}_d\}$ are unconditionally coupled generalized nice; furthermore, whenever $P_{d+2}, P_{d+3} \in \mathcal{EH}$ (hence unconditionally across all $d \geq 2$ by Section 5, and at $d = 1$ for $\text{Bird}_1 = \text{Bull}$ via [7]), the singleton families $\{S_{1,2,d}\}$ and $\{\text{Bird}_d\}$ are leaf-reducible and $\{S_{1,2,d}\}$, $\{\overline{S_{1,2,d}}\}$, $\{\text{Bird}_d\}$, and $\{\overline{\text{Bird}_d}\}$ are coupled generalized nice.*

Proof. At any depth $r \geq 1$ and stem size $s \geq 1$ ($d = s + r$), replacing Lemma 4.2(1) in the proof of Theorem 4.3 by Proposition 4.5 (for $\overline{S_{1,2,s+r}}$-free graphs) and Proposition 4.6 (for $\overline{\text{Bird}_{s+r}}$-free graphs) decomposes $F[M_{r-1} \setminus N_F(z_{r-1})]$ into pairwise-anticomplete (or pairwise-complete) proper modules of $F[M_{r-1}]$ whenever $N_F(z_{r-1}) \cap M_{r-1} \neq \emptyset$. Either greedy interval grouping extracts a pure $(p_{\text{comp}}, 5\theta|N^*|)$-blockade inside $N^* \setminus N_F(z_{r-1})$, or every external vertex $u \in M_j^*$ at $r = 1$ (and every cutter $z_{r-1} \in A_{r-2}$ at $d_{\text{cut}} = r - 1$) satisfies either $N_F(\cdot) \cap N^* = \emptyset$ (100% anticomplete in F) or $|N^* \setminus N_F(\cdot)| < 10p_{\text{comp}}\theta|N^*|$ ($10p_{\text{comp}}\theta$-co-sparse in F). Deleting the $< \frac{\eta_{\text{out}}}{2\ell}|M_i^*|$ anticomplete vertices on non-anticomplete pairs at $r = 1$ yields an anticomplete-or-η_{out}-co-sparse ℓ-blockade, while taking the larger half $A_{r-2,*} \subseteq A_{r-2}$ at $r \geq 2$ gives an anticomplete or η_0-co-sparse pair $(N^{(r-1)*}, A_{r-2,*})$, recovering outcomes (a) and (b) of Theorem 4.3 for $H_{\text{fam}} \in \{S_{1,2,\cdot}, \text{Bird}_\cdot\}$ ($d = 1+r$ at $s = 1$). Since $\deg_{S_{1,2,d}}(w_1) = 1$ with $S_{1,2,d} - w_1 \cong P_{d+3}$ and $\deg_{\text{Bird}_d}(b_0) = 1$ with $\text{Bird}_d - b_0 \cong P_{d+2}[v_1 \leftarrow K_2]$, leaf-reducibility and coupled generalized niceness of $\{S_{1,2,d}\}$, $\{\overline{S_{1,2,d}}\}$, $\{\text{Bird}_d\}$, and $\{\overline{\text{Bird}_d}\}$ follow from Lemma 3.2(4) and complement symmetry. □

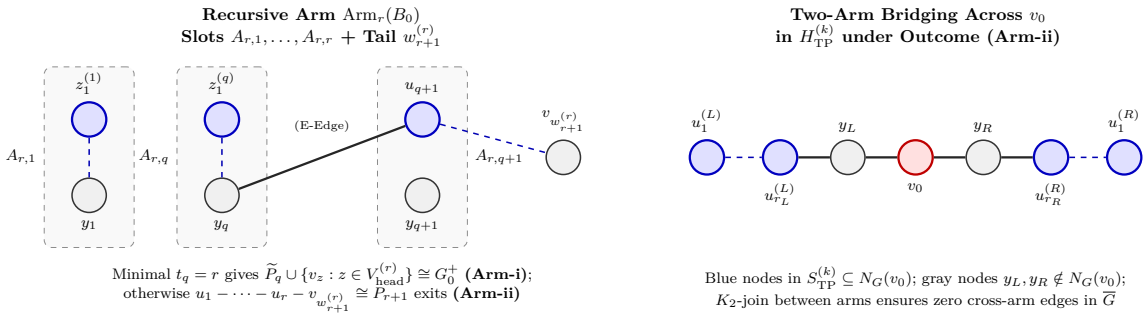


Figure 2: Left: Backward greedy routing inside $\mathrm{Arm}_r(B_0)$, yielding either an internal copy of G_0^+ **(Arm-i)** or an induced $(r+1)$-vertex exit path **(Arm-ii)**. Right: Bridging two **(Arm-ii)** exit paths across v_0 in $\overline{G}$.

5 Terminal-Path Pure-Substitution Arms and Stage-2 Wonderfulness

To complete the second stage of iterative sparsification across all infinite families in Theorem 1.1, we show that whenever G is $\overline{H_*}$-free and $\mathcal{B} = (B_1, \dots, B_\ell)$ is an equal-sized anticonnected complete-or-two-sided-δ_{cross}-sparse (ℓ, w)-blockade in G, the open-interval attachment subgraph $J[I_0(v_0)]$ on $I_0(v_0) := \{i \in [\ell] : 0 < |N_G(v_0) \cap B_i| < \frac{1}{2}|B_i|\}$ cannot contain an explicit pure-substitution witness graph $J_0 \in \mathcal{EH}$. By Lemma 3.3 and Proposition 3.4, in any embedding of (J_0, S_{J_0}) into $I_0(v_0)$ around an external vertex $v_0 \in V(G) \setminus V(\mathcal{B})$, picking any $v_u \in N_G(v_0) \cap B_{j_u}$ for $u \in S_{J_0}$ and sequential $v_u \in (B_{j_u} \setminus N_G(v_0)) \setminus \bigcup_{x \prec u:\, xu \notin E(J_0)} N_G(v_x)$ for $u \in V(J_0) \setminus S_{J_0}$ ensures:

- *(E-Non)* for every $xu \in E(J_0)$, we have $v_x v_u \notin E(\overline{G})$ ($v_0 v_u \notin E(\overline{G})$ for $u \in S_{J_0}$); and
- *(E-Edge)* for every distinct $x, u \in V(J_0)$ with $xu \notin E(J_0)$ and $\{x, u\} \not\subseteq S_{J_0}$, we have $v_x v_u \in E(\overline{G})$ ($v_0 v_u \in E(\overline{G})$ for $u \notin S_{J_0}$).

5.1 The recursive terminal-path pure-substitution arm

Our central Stage-2 tool is a recursive substitution operator $\mathrm{Arm}_r(B_0) \in \mathcal{J}(\{B_0, \overline{K_2}\})$ along the terminal path of any base graph $G_0 = \overline{B_0}$ (Figure 2).

Lemma 5.1 (Recursive terminal-path pure-substitution arm dichotomy)**.** *Let G_0 be a finite simple graph whose vertex set partitions as $V(G_0) = \{w_1, \dots, w_L\} \sqcup V_{\text{head}}$ for an integer $L \geq 2$ such that $(w_1 - w_2 - \cdots - w_L)$ is an induced path in G_0 and every $z \in V_{\text{head}}$ satisfies $N_{G_0}(z) \cap \{w_1, \dots, w_L\} \subseteq \{w_L\}$. Let $G_0^+ := G_0 + (w_0 - w_1)$ be obtained from G_0 by attaching a pendant edge $w_0 w_1$ at w_1, and set $B_0 := \overline{G_0}$. For $1 \leq r \leq L-1$, define $(\mathrm{Arm}_r(B_0), S_r(B_0))$ recursively with $\mathrm{Arm}_r(B_0) \in \mathcal{J}(\{B_0, \overline{K_2}\})$ by setting $(\mathrm{Arm}_1(B_0), S_1(B_0)) := (\{x_1, y_1\}_{\overline{K_2}}, \{x_1\})$ and, for $2 \leq r \leq L-1$, taking a copy $(B^{(r)}, w_1^{(r)}, \dots, w_L^{(r)}, V_{\text{head}}^{(r)}) \cong (B_0, w_1, \dots, w_L, V_{\text{head}})$ and substituting disjoint copies $(A_{r,1}, S_{r,1}), \dots, (A_{r,r}, S_{r,r}) \cong (\mathrm{Arm}_{r-1}(B_0), S_{r-1}(B_0))$ for $w_1^{(r)}, \dots, w_r^{(r)}$:*

$$\mathrm{Arm}_r(B_0) \;:=\; B^{(r)}\Big[w_1^{(r)} \leftarrow \mathrm{Arm}_{r-1}(B_0),\; \dots,\; w_r^{(r)} \leftarrow \mathrm{Arm}_{r-1}(B_0)\Big],$$

$$S_r(B_0) \;:=\; \bigsqcup_{m=1}^{r} S_{r,m}.$$

For any graph G and distinct vertices $\{v_u : u \in V(\mathrm{Arm}_r(B_0))\}$ satisfying (E-Non) *and* (E-Edge) *for $(\mathrm{Arm}_r(B_0), S_r(B_0))$, at least one of the following holds in $\overline{G}$:*

(Arm-i) *$\overline{G}[\{v_u : u \in V(\mathrm{Arm}_r(B_0))\}]$ contains an induced subgraph isomorphic to G_0^+; or*

(Arm-ii) *$\overline{G}[\{v_u : u \in V(\mathrm{Arm}_r(B_0))\}]$ contains an induced $(r+1)$-vertex path $(z_1 - \cdots - z_r - y)$ with $z_1, \ldots, z_r \in \{v_u : u \in S_r(B_0)\}$ and $y \in \{v_u : u \in V(\mathrm{Arm}_r(B_0)) \setminus S_r(B_0)\}$.*

Proof. We induct on $r \in \{1, \ldots, L-1\}$. For $r = 1$, $x_1 y_1 \notin E(\mathrm{Arm}_1(B_0))$ with $y_1 \notin S_1(B_0)$, so (E-Edge) gives $v_{x_1} v_{y_1} \in E(\overline{G})$ and $(v_{x_1} - v_{y_1})$ satisfies (Arm-ii).

For $2 \le r \le L-1$, each induced slot $(A_{r,m}, S_{r,m})$ $(1 \le m \le r$, with $S_{r,m} = V(A_{r,m}) \cap S_r(B_0))$ inherits (E-Non) and (E-Edge). By the induction hypothesis, either some slot $A_{r,m}$ realizes (Arm-i) (immediately giving (Arm-i) for $\mathrm{Arm}_r(B_0)$), or every slot $1 \le m \le r$ yields an induced r-vertex path $(z_1^{(m)} - \cdots - z_{r-1}^{(m)} - z_r^{(m)})$ in $\overline{G}$ with $z_1^{(m)}, \ldots, z_{r-1}^{(m)} \in \{v_u : u \in S_{r,m}\}$ and $z_r^{(m)} =: y_m \in \{v_u : u \in V(A_{r,m}) \setminus S_{r,m}\} \subseteq \{v_u : u \notin S_r(B_0)\}$. Set $u_r := z_1^{(r)}$; since $r \le L-1$ leaves $w_{r+1}^{(r)}, \ldots, w_L^{(r)}$ and $V_{\mathrm{head}}^{(r)}$ un-substituted (hence outside $S_r(B_0)$), (E-Edge) gives $u_r v_{w_{r+1}^{(r)}} \in E(\overline{G})$. Descending across $q = r-1, \ldots, 1$, since $z_r^{(q)} = y_q \notin \{v_u : u \in S_r(B_0)\}$ and $w_q^{(r)} w_{q+1}^{(r)} \notin E(B^{(r)})$, (E-Edge) gives $z_r^{(q)} u_{q+1} \in E(\overline{G})$. Let $t_q \in \{1, \ldots, r\}$ be the smallest index with $z_{t_q}^{(q)} u_{q+1} \in E(\overline{G})$:

- If $t_q = r$, then $z_1^{(q)}, \ldots, z_{r-1}^{(q)}$ are non-adjacent in $\overline{G}$ to u_{q+1}. Consider the $(L+1)$-vertex sequence

$$\widetilde{P}_q \;:=\; (z_{r-q}^{(q)},\, \ldots,\, z_r^{(q)},\, u_{q+1},\, y_{q+2},\, \ldots,\, y_r,\, v_{w_{r+1}^{(r)}},\, \ldots,\, v_{w_L^{(r)}}),$$

  whose last vertex is always $v_{w_L^{(r)}} \notin \{v_u : u \in S_r(B_0)\}$ since $L - r \ge 1$. When $q = r-1$, $\{y_{q+2}, \ldots, y_r\} = \emptyset$ and $u_{q+1} = u_r$ is adjacent in $\overline{G}$ directly to $v_{w_{r+1}^{(r)}}$; when $1 \le q \le r-2$, $\{y_{q+2}, \ldots, y_r\}$ is non-empty and lies outside $\{v_u : u \in S_r(B_0)\}$, so (E-Edge) gives $u_{q+1} y_{q+2}, \ldots, y_r v_{w_{r+1}^{(r)}} \in E(\overline{G})$. By (E-Non) across non-adjacent slots of G_0 and between $\{w_q^{(r)}, \ldots, w_{L-1}^{(r)}\}$ and $V_{\mathrm{head}}^{(r)}$, together with (E-Edge) and (E-Non) on $\{v_{w_L^{(r)}}\} \cup \{v_z : z \in V_{\mathrm{head}}^{(r)}\}$ (all vertices of which lie outside $\{v_u : u \in S_r(B_0)\}$), mapping $(w_0, w_1, \ldots, w_L)$ along $\widetilde{P}_q$ and each $z \in V_{\mathrm{head}}$ to $v_{z^{(r)}}$ induces G_0^+ in $\overline{G}$, giving (Arm-i).
- If $t_q \le r-1$, set $u_q := z_{t_q}^{(q)} \in \{v_u : u \in S_{r,q}\}$.

If $t_q \le r-1$ for all $1 \le q \le r-1$, then $(u_1 - \cdots - u_r - v_{w_{r+1}^{(r)}})$ has zero chords in $\overline{G}$ by (E-Non) across non-consecutive slots of G_0 and satisfies (Arm-ii). □

5.2 Stage-2 thresholded wonderfulness for all paths, spiders, and birds

Proposition 5.2 (Stage-2 thresholded wonderfulness for all paths P_k, $k \ge 6$)**.** *Let $k \ge 6$ be an integer, assume $P_{k-1}, \overline{P_{k-1}} \in \mathcal{EH}$, and set $r_L := \lceil \frac{k-3}{2} \rceil \ge 2$ and $r_R := \lfloor \frac{k-3}{2} \rfloor \ge 1$ ($r_L + r_R = k-3$).*

(a) (Staircase pure-substitution witness.) *Define $(\mathrm{Arm}_r^{(k)}, S_r^{(k)})$ recursively on $1 \le r \le r_L$ from $(\mathrm{Arm}_1^{(k)}, S_1^{(k)}) := (\{x_1, y_1\}_{\overline{K_2}}, \{x_1\})$ by setting, for $2 \le r \le r_L$ and $B^{(r)} \cong \overline{P_{k-1}}$ ($\overline{B^{(r)}} = w_1^{(r)} - \cdots - w_{k-1}^{(r)}$) with disjoint slot copies $(A_{r,q}, S_{r,q}) \cong (\mathrm{Arm}_q^{(k)}, S_q^{(k)})$ ($1 \le q < r$) and $(A_{r,r}, S_{r,r}) \cong (\mathrm{Arm}_1^{(k)}, S_1^{(k)})$,*

$$\mathrm{Arm}_r^{(k)} \;:=\; B^{(r)}\Big[w_1^{(r)} \leftarrow A_{r,1},\; w_2^{(r)} \leftarrow A_{r,2},\; \ldots,\; w_{r-1}^{(r)} \leftarrow A_{r,r-1},\; w_r^{(r)} \leftarrow A_{r,r}\Big],$$

$$S_r^{(k)} \;:=\; \bigsqcup_{q=1}^{r} S_{r,q}.$$

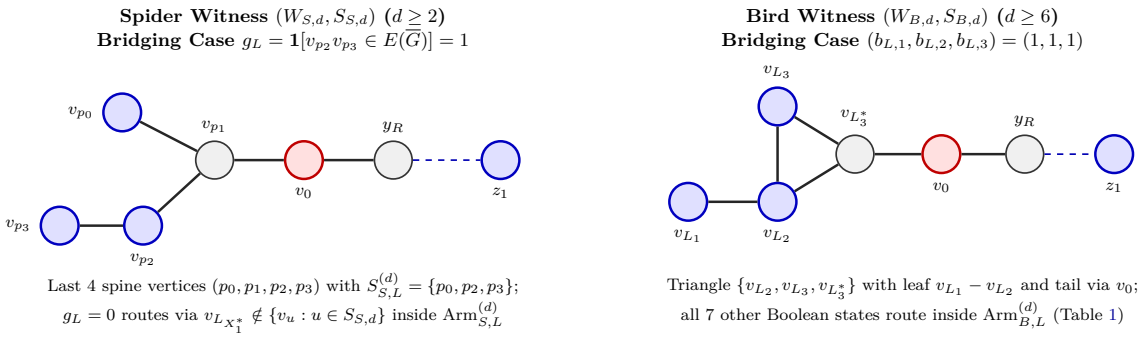


Figure 3: Two-arm pure-substitution Stage-2 witnesses $(W_{S,d}, S_{S,d})$ for $S_{1,2,d}$ (left) and $(W_{B,d}, S_{B,d})$ for Bird_d (right) bridging across the external vertex v_0 in $\overline{G}$.

Then $|V(\mathrm{Arm}^{(k)}_r)| = k\,2^{r-2} + 1$ *for all* $r \geq 2$ *and* $|S^{(k)}_r| = 2^{r-1}$ *for all* $r \geq 1$, *and* $H^{(k)} := K_2[1 \leftarrow \mathrm{Arm}^{(k)}_{r_L}, 2 \leftarrow \mathrm{Arm}^{(k)}_{r_R}] \in \mathcal{J}(\{\overline{P_{k-1}}, K_2, \overline{K_2}\}) \subseteq \mathcal{EH}$ *with* $S^{(k)} := S^{(k)}_{r_L} \sqcup S^{(k)}_{r_R}$ *certifies that* $\{P_k\}$ *is thresholded wonderful.*

(b) (Terminal-path pure-substitution witness.) *Taking* $G_0 := P_{k-1}$ *(*$L := k-1$, $V_{\mathrm{head}} := \emptyset$, $G_0^+ \cong P_k$*), the two-arm graph*

$$H^{(k)}_{\mathrm{TP}} := K_2[1 \leftarrow \mathrm{Arm}_{r_L}(\overline{P_{k-1}}), 2 \leftarrow \mathrm{Arm}_{r_R}(\overline{P_{k-1}})] \in \mathcal{J}(\{\overline{P_{k-1}}, K_2, \overline{K_2}\}) \subseteq \mathcal{EH}$$

with $S^{(k)}_{\mathrm{TP}} := S_{r_L}(\overline{P_{k-1}}) \sqcup S_{r_R}(\overline{P_{k-1}})$ *likewise certifies that* $\{P_k\}$ *is thresholded wonderful.*

Proof. Both $H^{(k)}$ and $H^{(k)}_{\mathrm{TP}}$ lie in $\mathcal{EH}$ by Alon–Pach–Solymosi substitution [1] from $\overline{P_{k-1}}, K_2, \overline{K_2} \in \mathcal{EH}$. In (a), strong induction on $r \geq 2$ gives

$$|V(\mathrm{Arm}^{(k)}_r)| = 2 + \sum_{q=2}^{r-1}(k\,2^{q-2}+1) + 2 + (k-1-r) = k\,2^{r-2}+1 \quad \text{and} \quad |S^{(k)}_r| = 1 + \sum_{q=2}^{r-1} 2^{q-1} + 1 = 2^{r-1};$$

backward selection on $(z^{(q)}_1 - \cdots - z^{(q)}_{d_q} - y_q)$ ($d_q = q$ for $q < r$ and $d_r = 1$) either stops at $t_q = q+1$ with the induced k-vertex path

$$\big(z^{(q)}_1, \ldots, z^{(q)}_q,\ y_q,\ u_{q+1},\ y_{q+2}, \ldots, y_r,\ v_{w^{(r)}_{r+1}}, \ldots, v_{w^{(r)}_{k-1}}\big) \cong P_k$$

in $\overline{G}$, or outputs $(u_1 - \cdots - u_r - v_{w^{(r)}_{r+1}})$. In (b), applying Lemma 5.1 to each arm yields the same dichotomy. Hence in both (a) and (b), either one arm already induces P_k in $\overline{G}$ or the two exit paths bridge across v_0 via the $(r_L+1) + 1 + (r_R+1) = k$-vertex induced path

$$\big(u^{(L)}_1 - \cdots - u^{(L)}_{r_L} - y_L - v_0 - y_R - u^{(R)}_{r_R} - \cdots - u^{(R)}_1\big) \cong P_k$$

(Figure 2). By Proposition 3.4, $\{P_k\}$ is thresholded wonderful. □

Remarkably, for subdivided spiders $S_{1,2,d}$, selecting the last four vertices of the $(d+2)$-vertex spine inside $\overline{\mathrm{Arm}^{(d)}_{S,L}} := S_{1,2,d-1}[u_{d-1} \leftarrow \{L_{X_1}, L_{X_1^*}\}_{K_2}]$ resolves *every* integer $d \geq 2$ uniformly with a single Boolean bit, starting directly from $S_{1,2,1} \cong B(2,2) = P_4[v_1 \leftarrow \overline{K_2}] \in \mathcal{J}(\{P_4, \overline{K_2}\}) \subseteq \mathcal{EH}$.

Table 1: Boolean routing of an induced Bird_6 inside $\overline{G}[V(W^{(6)}_{\mathrm{Bird}}) \cup \{v_0\}]$ across all 8 states of $(b_{X,1}, b_{X,2}, b_{X,3}) := (\mathbf{1}[v_{X_1}v_{X_2} \in E(\overline{G})], \mathbf{1}[v_{X_2}v_{X_3} \in E(\overline{G})], \mathbf{1}[v_{X_3}v_{X_4} \in E(\overline{G})]) \in \{0,1\}^3$ ($X \in \{L, R\}$).

Boolean state of arm $X \in \{L, R\}$	**Induced copy** $(b_0, b_1, b_2, b_3, u_1, \ldots, u_6) \cong \mathrm{Bird}_6$ **in** $\overline{G}$
$(b_{X,1}, b_{X,2}, b_{X,3}) \in \{(0,0,0), (0,0,1)\}$	$(v_{X_1}, v_{X_1^*}, v_{X_2}, v_{X_1'}, v_{X_3'}, v_{X_4}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8})$
$(b_{X,1}, b_{X,2}, b_{X,3}) = (0,1,0)$	$(v_{X_1}, v_{X_1^*}, v_{X_2}, v_{X_1'}, v_{X_3}, v_{X_4'}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8})$
$(b_{X,1}, b_{X,2}, b_{X,3}) = (0,1,1)$	$(v_{X_1}, v_{X_1^*}, v_{X_2}, v_{X_1'}, v_{X_3}, v_{X_4}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8})$
$(b_{X,1}, b_{X,2}, b_{X,3}) = (1,0,0)$	$(v_{X_1}, v_{X_2'}, v_{X_3^*}, v_{X_3}, v_{X_4}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8}, v_{X_9})$
$(b_{X,1}, b_{X,2}, b_{X,3}) = (1,0,1)$	$(v_{X_2}, v_{X_2^*}, v_{X_3}, v_{X_2'}, v_{X_4}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8}, v_{X_9})$
$(b_{X,1}, b_{X,2}, b_{X,3}) = (1,1,0)$	$(v_{X_1}, v_{X_2}, v_{X_3^*}, v_{X_3}, v_{X_4}, v_{X_5}, v_{X_6}, v_{X_7}, v_{X_8}, v_{X_9})$
$(b_{L,1}, b_{L,2}, b_{L,3}) = (b_{R,1}, b_{R,2}, b_{R,3}) = (1,1,1)$	$(v_{L_1}, v_{L_2}, v_{L_3^*}, v_{L_3}, v_0, v_{R_5}, v_{R_4}, v_{R_3}, v_{R_2}, v_{R_1})$

Proposition 5.3 (Stage-2 thresholded wonderfulness for all spiders $S_{1,2,d}$, $d \geq 2$)**.** *Let $d \geq 2$ be an integer, and assume $S_{1,2,d-1}, \overline{S_{1,2,d-1}} \in \mathcal{EH}$ (where $S_{1,2,1} \cong B(2,2) = P_4[v_1 \leftarrow \overline{K_2}] \in \mathcal{EH}$ for $d = 2$). Let $\overline{\mathrm{Arm}^{(d)}_{S,L}} := S_{1,2,d-1}[u_{d-1} \leftarrow \{L_{X_1}, L_{X_1^*}\}_{K_2}]$, let (p_0, p_1, p_2, p_3) denote the last four vertices of the $(d+2)$-vertex induced path $(L_{x_2} - L_{w_2} - L_c - L_{u_1} - \cdots - L_{u_{d-2}} - L_{X_1})$ in $\overline{\mathrm{Arm}^{(d)}_{S,L}}$—specifically $(L_{x_2}, L_{w_2}, L_c, L_{X_1})$ for $d = 2$, $(L_{w_2}, L_c, L_{u_1}, L_{X_1})$ for $d = 3$, $(L_c, L_{u_1}, L_{u_2}, L_{X_1})$ for $d = 4$, and $(L_{u_{d-4}}, L_{u_{d-3}}, L_{u_{d-2}}, L_{X_1})$ for all $d \geq 5$—and set $S^{(d)}_{S,L} := \{p_0, p_2, p_3\}$. Define the right arm $(\mathrm{Arm}^{(d)}_{S,R}, S^{(d)}_{S,R})$ to be $(\{R_y\}_{K_1}, \emptyset)$ for $d = 2$ and $(\mathrm{Arm}_{d-2}(\overline{S_{1,2,d-1}}), S_{d-2}(\overline{S_{1,2,d-1}}))$ (along the $(d-1)$-vertex terminal path $(u_{d-1} - \cdots - u_1)$ with $L = d - 1 \geq 2$, $V_{\mathrm{head}} = \{c, w_1, w_2, x_2\}$, and $S^+_{1,2,d-1} \cong S_{1,2,d}$) for $d \geq 3$. Then*

$$W_{S,d} \;:=\; K_2\Big[1 \leftarrow \mathrm{Arm}^{(d)}_{S,L},\; 2 \leftarrow \mathrm{Arm}^{(d)}_{S,R}\Big] \;\in\; \mathcal{J}(\{\overline{S_{1,2,d-1}}, K_2, \overline{K_2}\}) \subseteq \mathcal{EH}$$

with special set $S_{S,d} := S^{(d)}_{S,L} \sqcup S^{(d)}_{S,R}$ certifies unconditionally that $\{S_{1,2,d}\}$ is thresholded wonderful for every integer $d \geq 2$.

Proof. In $\overline{\mathrm{Arm}^{(d)}_{S,L}}$, $(p_0 - p_1 - p_2 - p_3)$ is an induced P_4, so inside $S^{(d)}_{S,L} = \{p_0, p_2, p_3\}$ both $\{p_0, p_2\}$ and $\{p_0, p_3\}$ belong to $E(\mathrm{Arm}^{(d)}_{S,L})$ ($v_{p_0}v_{p_2}, v_{p_0}v_{p_3} \notin E(\overline{G})$ by (E-Non)), leaving a single undetermined bit $g_L := \mathbf{1}[v_{p_2}v_{p_3} \in E(\overline{G})] \in \{0,1\}$. If $g_L = 0$, because $S^{(d)}_{S,L} \cap (V(S_{1,2,d-1}) \setminus \{u_{d-1}\}) = \{p_0, p_2\}$ consists of two vertices at distance 2 along the spine (hence $p_0p_2 \notin E(S_{1,2,d-1})$ for every $d \geq 2$) and $L_{X_1^*} \notin S_{S,d}$ is adjacent in $\overline{\mathrm{Arm}^{(d)}_{S,L}}$ solely to p_2 and $L_{X_1} = p_3$, every edge of $(v_{L_c}, v_{L_{w_1}}, v_{L_{w_2}}, v_{L_{x_2}}, v_{L_{u_1}}, \ldots, v_{L_{u_{d-2}}}, v_{L_{X_1^*}}, v_{L_{X_1}}) \cong S_{1,2,d}$ has at least one endpoint outside $\{v_u : u \in S_{S,d}\}$ (holding by (E-Edge)) while every non-edge holds by (E-Non) together with $v_{p_2}v_{p_3} \notin E(\overline{G})$. If $g_L = 1$, either $d \geq 3$ and $\mathrm{Arm}^{(d)}_{S,R}$ realizes (Arm-i) ($\cong S_{1,2,d}$) by Lemma 5.1, or $\mathrm{Arm}^{(d)}_{S,R}$ supplies an induced $(d-1)$-vertex exit path $(z_1 - \cdots - z_{d-2} - y_R)$ with $y_R \notin \{v_u : u \in S_{S,d}\}$ (namely (v_{R_y}) with $\{z_1, \ldots, z_{d-2}\} = \emptyset$ at $d = 2$, or (Arm-ii) at $d \geq 3$), so $(v_{p_1}, v_{p_0}, v_{p_2}, v_{p_3}, v_0, y_R, z_{d-2}, \ldots, z_1) \cong S_{1,2,d}$ in $\overline{G}$ (Figure 3, left), certifying thresholded wonderfulness by Proposition 3.4. □

Proposition 5.4 (Stage-2 thresholded wonderfulness for all birds Bird_d, $d \geq 2$)**.** *Assume $P_k, \overline{P_k} \in \mathcal{EH}$ for all $k \geq 1$.*

(a) (Symmetric 34-vertex cograph-substituted witness for $1 \le d \le 6$.) *For* $X \in \{L, R\}$, *let* $\mathrm{Arm}_X^{\mathrm{Bird}}$ *be the* 17*-vertex graph obtained from* $\overline{P_9}$ *(*$\overline{B_0} = X_1 - \cdots - X_9$*) by substituting the* 3*-vertex cograph* $\overline{K_2}[1 \leftarrow \{X_m, X'_m\}_{K_2},\ 2 \leftarrow \{X^*_m\}_{K_1}]$ *at each* $m \in \{1,2,3,4\}$ *(retaining un-substituted vertices* $X_5, \ldots, X_9$*). Then* $W^{(6)}_{\mathrm{Bird}} := K_2[1 \leftarrow \mathrm{Arm}_L^{\mathrm{Bird}},\ 2 \leftarrow \mathrm{Arm}_R^{\mathrm{Bird}}] \in \mathcal{J}(\{\overline{P_9}, K_2, \overline{K_2}\}) \subseteq \mathcal{EH}$ *with special set* $S^{(6)}_{\mathrm{Bird}} := \{L_1, L_2, L_3, L_4, R_1, R_2, R_3, R_4\}$ *certifies that* $\{\mathrm{Bird}_d\}$ *is thresholded wonderful for all* $1 \le d \le 6$ *(including* $\mathrm{Bull} = \mathrm{Bird}_1$*).*

(b) (General two-arm witness for $d \ge 6$.) *Assuming* $\mathrm{Bird}_{d-1}, \overline{\mathrm{Bird}_{d-1}} \in \mathcal{EH}$ *(*$d \ge 6$*), let* $\mathrm{Arm}^{(d)}_{B,L} \in \mathcal{J}(\{\overline{P_{d+3}}, K_2, \overline{K_2}\})$ *substitute* $\overline{K_2}[1 \leftarrow \{L_m, L'_m\}_{K_2},\ 2 \leftarrow \{L^*_m\}_{K_1}]$ *at* $m \in \{1,2,3,4\}$ *of* $\overline{P_{d+3}}$ *(with un-substituted vertices* $L_5, \ldots, L_{d+3}$*), and form* $\mathrm{Arm}_{d-2}(\overline{\mathrm{Bird}_{d-1}})$ *along the* $(d-1)$*-vertex terminal path* $(u_{d-1} - \cdots - u_1)$ *(*$L = d-1 \ge 5$*,* $V_{\mathrm{head}} = \{b_0, b_1, b_2, b_3\}$*, so* $\mathrm{Bird}^+_{d-1} \cong \mathrm{Bird}_d$*). Then* $W_{B,d} := K_2[1 \leftarrow \mathrm{Arm}^{(d)}_{B,L},\ 2 \leftarrow \mathrm{Arm}_{d-2}(\overline{\mathrm{Bird}_{d-1}})] \in \mathcal{EH}$ *with* $S_{B,d} := \{L_1, L_2, L_3, L_4\} \sqcup S_{d-2}(\overline{\mathrm{Bird}_{d-1}})$ *certifies that* $\{\mathrm{Bird}_d\}$ *is thresholded wonderful for every* $d \ge 6$.

Proof. In $\overline{\mathrm{Arm}_X^{\mathrm{Bird}}} = P_9[m \leftarrow K_2[1 \leftarrow \{X_m, X'_m\}_{\overline{K_2}}, 2 \leftarrow \{X^*_m\}_{K_1}]]_{m=1}^4$, each module $\{X_m, X'_m, X^*_m\}$ $(1 \le m \le 4)$ satisfies $v_{X_m} v_{X'_m} \notin E(\overline{G})$ by (E-Non) and $v_{X_m} v_{X^*_m}, v_{X'_m} v_{X^*_m} \in E(\overline{G})$ by (E-Edge) since $X^*_m \notin S^{(6)}_{\mathrm{Bird}}$. Table 1 checks directly that every Boolean state embeds $\mathrm{Bird}_6 \supseteq_{\mathrm{ind}} \mathrm{Bird}_d$ $(1 \le d \le 6)$ in $\overline{G}$. For $d \ge 6$ in (b), appending $(v_{L_9}, \ldots, v_{L_{d+2}})$ when $b_{L,1} = 0$ and $(v_{L_{10}}, \ldots, v_{L_{d+3}})$ when $b_{L,1} = 1$ (empty at $d = 6$, and using $L_9, \ldots, L_{d+3} \notin S_{B,d}$ for $d \ge 7$) extends the internal states of Table 1 to Bird_d, while at $(b_{L,1}, b_{L,2}, b_{L,3}) = (1,1,1)$, Lemma 5.1 either embeds $\mathrm{Bird}^+_{d-1} \cong \mathrm{Bird}_d$ inside $\mathrm{Arm}_{d-2}(\overline{\mathrm{Bird}_{d-1}})$ or bridges $(v_{L_1}, v_{L_2}, v_{L^*_3}, v_{L_3}, v_0, y_R, z_{d-2}, \ldots, z_1) \cong \mathrm{Bird}_d$ across v_0 (Figure 3, right). □

6 Proof of the Main Theorems

We now assemble Sections 4 and 5 with the viral closures and two-stage reduction of Sections 2 and 3 to establish Theorems 1.1, 1.2, and 1.3 (Figure 1).

Theorem 6.1 (Twin-leaf-free lower bound and classification of small trees)**.** *Every twin-leaf-free tree* T *of order* $|T| \ge 3$ *and maximum degree* $d_c := \Delta(T) \ge 3$ *satisfies* $|T| \ge 1 + 1 + 2(d_c - 1) = 2d_c$*. Consequently, every non-empty tree of order at most* 6*, every non-empty forest whose connected components each have order at most* 6*, and* 10 *of the* 11 *non-isomorphic trees on* 7 *vertices (every* 7*-vertex tree except solely the subdivided claw* $S_{2,2,2}$*), together with all of their complements, belong to* $\mathcal{H}(\mathcal{B}^*_{\mathrm{new}}) \subseteq \mathcal{EH}$*, satisfy the polynomial Rödl property, and are viral.*

Proof. Let T be a twin-leaf-free tree (no two distinct leaves of T share a neighbor) containing a vertex $c \in V(T)$ of degree $d_c := \Delta(T) \ge 3$. Since T is connected and acyclic, deleting c partitions $V(T) \setminus \{c\} = \bigsqcup_{i=1}^{d_c} V(C_i)$ into d_c pairwise disjoint connected components $C_1, \ldots, C_{d_c}$ of $T - c$, each containing a unique neighbor $y_i \in N_T(c)$ (any path in $T - c$ between distinct neighbors of c would form a cycle with c in T). Because T is twin-leaf-free, at most one neighbor of c has degree 1 in T; ordering $N_T(c)$ so that $\deg_T(y_i) \ge 2$ for all $1 \le i \le d_c - 1$, each such non-leaf neighbor y_i has at least one neighbor $w_i \in N_T(y_i) \setminus \{c\}$ in C_i, so $|C_i| \ge |\{y_i, w_i\}| = 2$ for $1 \le i \le d_c - 1$ while $|C_{d_c}| \ge 1$. Summing vertex counts yields

$$|T| \;=\; 1 + \sum_{i=1}^{d_c} |C_i| \;\ge\; 1 + 1 + 2(d_c - 1) \;=\; 2d_c.$$

Observe next that if any tree T contains distinct twin leaves $u \neq v$ ($\deg_T(u) = \deg_T(v) = 1$) sharing their unique neighbor $p \in V(T)$, then $uv \notin E(T)$, $T - v$ is a tree of order $|T| - 1$ in which u and v were false twins ($N_T(u) = N_T(v) = \{p\}$), and compatibility of vertex substitution with complementation ($\overline{H_1[w \leftarrow H_2]} = \overline{H_1}[w \leftarrow \overline{H_2}]$) yields simultaneously

$$T \cong (T - v)[u \leftarrow \overline{K_2}] \qquad \text{and} \qquad \overline{T} \cong \overline{(T - v)[u \leftarrow \overline{K_2}]} = \overline{T - v}[u \leftarrow K_2].$$

Since both $\overline{K_2}$ and K_2 belong to $\mathcal{B}^*_{\text{new}} \subseteq \mathcal{J}(\mathcal{B}^*_{\text{new}})$, $T - v \in \mathcal{H}(\mathcal{B}^*_{\text{new}})$ implies $T, \overline{T} \in \mathcal{J}(\mathcal{B}^*_{\text{new}})$, hence $T \in \mathcal{H}(\mathcal{B}^*_{\text{new}})$. By strong induction on $|T|$ for $1 \leq |T| \leq 6$, deleting any twin leaf v reduces T to a twin-leaf-free tree: if $\Delta(T) \leq 2$, then $T \cong P_{|T|} \in \mathcal{B}^*_9 \subseteq \mathcal{H}(\mathcal{B}^*_{\text{new}})$; if $d_c := \Delta(T) \geq 3$, then $2d_c \leq |T| \leq 6$ forces $d_c = 3$, $|T| = 6$, and sorted component sizes $(|C_1|, |C_2|, |C_3|) = (1, 2, 2)$ of $T - c$ ($C_1 \cong K_1$ and $C_2, C_3 \cong K_2$), giving $T \cong S_{1,2,2} \in \mathcal{B}^*_{\text{new}} \subseteq \mathcal{H}(\mathcal{B}^*_{\text{new}})$.

For $|T| = 7$, if T has twin leaves $u \neq v$, then $|T - v| = 6$ so $T - v \in \mathcal{H}(\mathcal{B}^*_{\text{new}})$ by the order-≤ 6 case above and hence $T \cong (T - v)[u \leftarrow \overline{K_2}] \in \mathcal{H}(\mathcal{B}^*_{\text{new}})$. Suppose instead that T is twin-leaf-free of order $|T| = 7$. If $\Delta(T) \leq 2$, then $T \cong P_7 \in \mathcal{B}^*_9 \subseteq \mathcal{H}(\mathcal{B}^*_{\text{new}})$. If $d_c := \Delta(T) \geq 3$, then $2d_c \leq |T| = 7$ forces $d_c = 3$, and $1 + |C_1| + |C_2| + |C_3| = 7$ with $|C_1| \geq 1$ and $|C_2|, |C_3| \geq 2$ forces $(|C_1|, |C_2|, |C_3|) \in \{(1, 2, 3), (2, 2, 2)\}$: when $(|C_1|, |C_2|, |C_3|) = (2, 2, 2)$, $C_1, C_2, C_3 \cong K_2$ so $T \cong S_{2,2,2}$; when $(|C_1|, |C_2|, |C_3|) = (1, 2, 3)$, $C_1 \cong K_1$, $C_2 \cong K_2$, and $C_3 \cong P_3$, where the unique neighbor $y_3 \in V(C_3) \cap N_T(c)$ must have $\deg_{C_3}(y_3) = 1$ (if $\deg_{C_3}(y_3) = 2$, the two ends of C_3 would be twin leaves of T sharing the neighbor y_3, contradicting twin-leaf-freeness of T), so $T \cong S_{1,2,3} \in \mathcal{B}^*_{\text{new}} \subseteq \mathcal{H}(\mathcal{B}^*_{\text{new}})$.

Finally, for any disconnected forest $F = C_1 \sqcup \cdots \sqcup C_m$ ($m \geq 2$) whose components $C_1, \ldots, C_m$ have order at most 6, writing $F = C_1 \sqcup F'$ with $F' := \bigsqcup_{q=2}^{m} C_q$ and inducting on m via

$$\begin{aligned} F &= C_1 \sqcup F' \cong \overline{K_2}[1 \leftarrow C_1,\, 2 \leftarrow F'] \in \mathcal{J}(\mathcal{B}^*_{\text{new}}), \\ \overline{F} &= \overline{C_1 \sqcup F'} \cong K_2[1 \leftarrow \overline{C_1},\, 2 \leftarrow \overline{F'}] \in \mathcal{J}(\mathcal{B}^*_{\text{new}}) \end{aligned}$$

shows $F \in \mathcal{H}(\mathcal{B}^*_{\text{new}})$. Once $\mathcal{B}^*_{\text{new}} \subseteq \mathcal{EH}$ is established in Steps 1–2 below, Theorem 2.3(b) and Proposition 2.4(3) yield $\mathcal{H}(\mathcal{B}^*_{\text{new}}) \subseteq \mathcal{EH}$, the polynomial Rödl property, and virality for all such trees, forests, and their complements. □

Proof of Theorems 1.1, 1.2, and 1.3. We proceed in strict non-circular dependency order directly from graphs of order at most 5 ($|H| \leq 4$ [23], Bull $\in \mathcal{EH}$ [7], $C_5 \in \mathcal{EH}$ [8, 9], and $P_5 \in \mathcal{EH}$ [21]), noting first that Theorem 4.3 establishes the depth-r Stage-1 comb/co-comb dichotomy unconditionally for all $r \geq 1$:

1. *All paths $P_k, \overline{P_k}$ ($k \geq 1$), linear/co-linear forests, brooms, and double brooms.* For $1 \leq k \leq 4$, P_k is a cograph ($P_k \in \mathcal{EH}$ [23]), and $P_5 \in \mathcal{EH}$ holds by Nguyen, Scott, and Seymour [21]. By strong induction on $k \geq 6$: assuming $P_{k-1}, \overline{P_{k-1}} \in \mathcal{EH}$, the singleton $\{P_k\}$ is leaf-reducible ($P_k - v_k \cong P_{k-1} \in \mathcal{EH}$), both $\{P_k\}$ and $\{\overline{P_k}\}$ are unconditionally coupled generalized nice by Corollary 4.4, and $\{P_k\}$ is Stage-2 thresholded wonderful by Proposition 5.2; hence Theorem 3.5 and Theorem 2.3(b) establish $P_k, \overline{P_k} \in \mathcal{EH}$, the polynomial Rödl property, and virality for all $k \geq 1$. Every non-empty linear forest $F = \bigsqcup_{q=1}^{m} P_{k_q}$ embeds as an induced subgraph into $P_{\sum_{q=1}^{m} k_q + m - 1} \in \mathcal{EH}$, and since $\overline{K_s} \subseteq_{\text{ind}} P_{2s} \in \mathcal{EH}$ ($s \geq 1$), vertex substitution [1, 11] (Proposition 2.4(1)) into $P_{d+2}, P_{d+3} \in \mathcal{EH}$ yields all brooms $B(s, d) := P_{d+2}[v_1 \leftarrow \overline{K_s}]$ ($s, d \geq 1$, including $S_{1,2,1} \cong B(2, 2)$) and double brooms $\mathrm{DB}(a, d, b) := P_{d+3}[v_1 \leftarrow \overline{K_a}, v_{d+3} \leftarrow \overline{K_b}]$ ($a, d, b \geq 1$), together with their complements, polynomial Rödl property, and virality. This proves Theorem 1.1(1).

2. *All spiders $S_{1,2,d}$ ($d \geq 2$), leaf-substituted spiders, all birds* Bird_d *($d \geq 1$), bicyclic sawtooths $M_{2,d}$ ($d \geq 1$), and small trees.* Since $P_{d+2}, P_{d+3} \in \mathcal{EH}$ and $P_{d+2}[v_1 \leftarrow K_2] \in \mathcal{EH}$ for all $d \geq 2$ by Step 1, $\{S_{1,2,d}\}$ ($S_{1,2,d} - w_1 \cong P_{d+3} \in \mathcal{EH}$) and $\{\mathrm{Bird}_d\}$ ($\mathrm{Bird}_d - b_0 \cong P_{d+2}[v_1 \leftarrow K_2] \in \mathcal{EH}$) are unconditionally leaf-reducible, and $\{S_{1,2,d}\}, \{\overline{S_{1,2,d}}\}, \{\mathrm{Bird}_d\}, \{\overline{\mathrm{Bird}_d}\}$ are unconditionally coupled generalized nice for every $d \geq 2$ by Corollary 4.7. Starting from $S_{1,2,1} \cong B(2,2) \in \mathcal{EH}$ (Step 1), strong induction on $d \geq 2$ via Proposition 5.3 and Theorem 3.5 establishes Stage-2 thresholded wonderfulness and $S_{1,2,d}, \overline{S_{1,2,d}} \in \mathcal{EH}$ for every $d \geq 2$. Starting from $\mathrm{Bird}_1 = \mathrm{Bull} \in \mathcal{EH}$ [7] and $P_{d+3} \in \mathcal{EH}$ ($d \geq 6$), Proposition 5.4(a)–(b), Theorem 3.5, and strong induction on $d \geq 2$ establish Stage-2 thresholded wonderfulness for all $d \geq 2$ and $\mathrm{Bird}_d, \overline{\mathrm{Bird}_d} \in \mathcal{EH}$ for every $d \geq 1$. Substituting $\overline{K_a}, \overline{K_b}, \overline{K_c} \in \mathcal{EH}$ at w_1, x_2, u_d of $S_{1,2,d}$ and substituting K_2 at b_0 of Bird_d yield all leaf-substituted spiders $S_{1,2,d}[w_1 \leftarrow \overline{K_a}, x_2 \leftarrow \overline{K_b}, u_d \leftarrow \overline{K_c}]$ ($d \geq 2$) and bicyclic sawtooths $M_{2,d}$ ($d \geq 1$) along with their complements, polynomial Rödl property, and virality. This proves $\mathcal{B}^*_{\mathrm{new}} \subseteq \mathcal{B}^*_\infty \subseteq \mathcal{EH}$, so Theorem 6.1 concludes the proof of Theorems 1.1 and 1.2.
3. *Strict closure hierarchy, opposite-chirality closure and forest pairs, and hooked-claw pairs.* Since $K_1, K_2, \overline{K_2}, P_5, \overline{P_5} \in \mathcal{B}^*_9$ and $\mathrm{Bull} \in \mathcal{H}(\{K_1, K_2\}) \subseteq \mathcal{H}(\mathcal{B}^*_9)$ (deleting the degree-1 vertices b_0, u_1 of Bull leaves $K_3 = K_2[1 \leftarrow K_2]$, and $\overline{\mathrm{Bull}} \cong \mathrm{Bull}$), while every generator of $\mathcal{B}^*_9 \subseteq \mathcal{B}^*_{\mathrm{new}} \subseteq \mathcal{B}^*_\infty$ lies in $\mathcal{EH}$ by Steps 1–2, we have $\mathcal{H}(\mathcal{B}^*_0) \subseteq \mathcal{H}(\mathcal{B}^*_9) \subseteq \mathcal{H}(\mathcal{B}^*_{\mathrm{new}}) \subseteq \mathcal{H}(\mathcal{B}^*_\infty) \subseteq \mathcal{EH}$. Since P_6 has order $6 > 5$, $S_{1,2,2}$ (6 vertices, 5 edges, $\Delta = 3$) is non-isomorphic to $\mathrm{Bird}_2, \overline{\mathrm{Bird}_2}, P_6, \overline{P_6}$, and $S_{1,2,6}$ (10 vertices, 9 edges) is non-isomorphic to Bird_6 (10 edges) and $\overline{\mathrm{Bird}_6}$ (35 edges), we have $P_6 \notin \mathcal{B}^+_0$, $S_{1,2,2} \notin \mathcal{B}^*_9$, and $S_{1,2,6} \notin \mathcal{B}^*_{\mathrm{new}}$, so Proposition 2.6(1)–(2) certifies $P_6 \in \mathcal{H}(\mathcal{B}^*_9) \setminus \mathcal{H}(\mathcal{B}^+_0)$, $S_{1,2,2} \in \mathcal{H}(\mathcal{B}^*_{\mathrm{new}}) \setminus \mathcal{H}(\mathcal{B}^*_9)$, and $S_{1,2,6} \in \mathcal{H}(\mathcal{B}^*_\infty) \setminus \mathcal{H}(\mathcal{B}^*_{\mathrm{new}})$. By Proposition 2.4(3), every opposite-chirality pair $\{H_1, \overline{H_2}\}$ ($H_1, H_2 \in \mathcal{J}(\mathcal{B}^*_\infty)$) belongs to $\mathcal{EH}$, satisfies the polynomial Rödl property, and is viral; since every non-empty finite forest F is either $K_1 \in \mathcal{B}^*_0$ or has $\delta(F) \leq 1$, induction on $|F|$ gives $F \in \mathcal{J}(\{K_1\}) \subseteq \mathcal{J}(\mathcal{B}^*_\infty)$ and resolves all opposite-chirality forest pairs $\{F_1, \overline{F_2}\}$ [15, 18]. Combining $P_t \in \mathcal{EH}$ ($t \geq 1$) with Proposition 2.5(2) resolves $\{P_t, \mathrm{Hook}_4(Q,S)\}$ and $\{\overline{P_t}, \overline{\mathrm{Hook}_4(Q,S)}\}$, proving Theorem 1.3(1)–(2).
4. *Infinite prime non-split viral extension towers.* Let G_0 be any prime non-split graph of order $n_0 := |G_0| \geq 6$ with $\Delta(G_0) \leq n_0 - 3$, leaf $x_0 \in V(G_0)$ ($\deg_{G_0}(x_0) = 1$), and $G_0, G_0 - x_0 \in \mathcal{H}(\mathcal{B}^*_\infty)$. Then $\mathcal{B}(G_0) := \{G_0, \overline{G_0}, G_0 - x_0, \overline{G_0 - x_0}, K_1, K_2, \overline{K_2}\} \subseteq \mathcal{J}(\mathcal{B}^*_\infty)$, so Theorem 2.7 yields $\{H^{(G_0)}_{n_0+r}\}_{r \geq 0} \subseteq \mathcal{H}(\mathcal{B}(G_0)) \subseteq \mathcal{H}(\mathcal{B}^*_\infty) \subseteq \mathcal{EH}$, where each $H^{(G_0)}_{n_0+r}$ has order $n_0 + r$ (hence $H^{(G_0)}_{n_0+r_1} \cong H^{(G_0)}_{n_0+r_2} \iff r_1 = r_2$), is prime ($m_{\max} = 1$) and non-split, and satisfies $\overline{H}^{(G_0)}_{n_0+r} \notin \mathcal{J}(\mathcal{B}^+_0) \supseteq \mathcal{J}(\{K_1, K_2, \overline{K_2}, C_5\})$. Finally, Proposition 2.6(1)–(3) verifies primality, non-splitness, $|G_0| \geq 6$, $\Delta(G_0) \leq |G_0| - 3$, and $\deg_{G_0}(x_0) = 1$ for (P_k, v_k) ($k \geq 6$), $(S_{1,2,d}, u_d)$ ($d \geq 2$), and (Bird_d, u_d) ($d \geq 2$), while $G_0 \in \mathcal{B}^*_\infty \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$ and:
   - for (P_k, v_k) ($k \geq 6$): $P_k - v_k \cong P_{k-1} \in \mathcal{B}^*_\infty \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$;
   - for $(S_{1,2,d}, u_d)$ ($d \geq 2$): $S_{1,2,2} - u_2 \cong B(2,2) = P_4[v_1 \leftarrow \overline{K_2}] \in \mathcal{H}(\mathcal{B}^*_9) \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$ for $d = 2$, and $S_{1,2,d} - u_d \cong S_{1,2,d-1} \in \mathcal{B}^*_\infty \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$ for $d \geq 3$;
   - for (Bird_d, u_d) ($d \geq 2$): $\mathrm{Bird}_2 - u_2 \cong \mathrm{Bird}_1 = \mathrm{Bull} \in \mathcal{H}(\mathcal{B}^*_{\mathrm{new}}) \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$ (since $\overline{\mathrm{Bull}} \cong \mathrm{Bull}$), and $\mathrm{Bird}_d - u_d \cong \mathrm{Bird}_{d-1} \in \mathcal{B}^*_\infty \subseteq \mathcal{H}(\mathcal{B}^*_\infty)$ for $d \geq 3$.

   This completes the proof of Theorem 1.3(3). □

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